

# Identifiability by Stationarity: Blind Gain–Object Separation in Self-Calibrating Ghost Imaging

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**Abstract**—For a fixed square linear inverse with known gain and matched acquisition budget, an ordered orthogonal scan minimizes conditioning-driven amplification of additive noise; with unknown slow multiplicative gain and unconstrained objects, the same deterministic chronology can create an exact gain–object ambiguity. In this paper, we resolve this paradox with an identifiability theory of blind gain–object separation, distinguishing algebraic identifiability, statistical identifiability, and estimation conditioning. Every square invertible design fails algebraically — for unconstrained objects, nonzero carrier entries, and admissible nonconstant log-gain perturbations, modulo the global scale; tall designs restore uniqueness generically at  $N \geq K + p$  frames for a  $p$ -dimensional drift class. Independently, a stationarity anchor (Theorem B) pins the relative gain statistically: once the object carrier is stationary in time, slow modulation of the record can only be gain. Randomized Hadamard/SRHT acquisition preserves orthogonal conditioning, and under Walsh-flatness or a successful random permutation of a sufficiently spread object it supplies the stationary or exchangeable carrier needed by the proposed windowed gain estimator; an exact finite-noise relative-MSE identity then yields a phase diagram with a computable flip boundary between ordered and randomized acquisition. In matched-budget simulations, raw ordered Hadamard leaves a non-decaying blind gain error near 0.96 where randomized arms reach 0.02, and SRHT recovers +5.45 dB under slow drift (descriptive values on the simulation grid). The proposed acquisition rule: randomize the carrier’s chronology — make the object stop pretending to be time.

**Index Terms**—ghost imaging; single-pixel imaging; blind calibration; identifiability; bilinear inverse problems

## I. INTRODUCTION

Single-pixel (ghost) imaging [1], [2], [3], [4] has become the method of choice for imaging through media that defeat conventional cameras [5], [6]: turbid water, fog, smoke, and scattering channels around corners [7], [8]. The instrument is disarmingly simple—a series of structured illumination patterns and a bucket detector [9], [10]—and the textbook design instinct is equally simple: use an ordered orthogonal basis. A Hadamard scan acquires exactly one transform coefficient per frame [11], [12], the measurement matrix is perfectly conditioned, and an exact inverse is waiting at the end of the acquisition. Against additive noise, this is provably the safe choice; nothing about the pattern set is left to chance.

Dynamic media break this instinct in a specific and, as will be argued, instructive way. A time-varying channel multiplies the bucket record by a slowly drifting gain: the  $n$ -th measurement is  $R_n = a_n B_n$ , where  $B_n$  is the object-dependent bucket coefficient and  $a_n > 0$  is an unknown per-frame gain with log-gain  $\ell_n = \log a_n$  confined to a slow-drift class  $S$

of dimension  $p$ . Recent experimental work on ghost imaging through dynamic and complex scattering media [7] and on optical wireless transmission around corners [8] shows that self-calibrating schemes built on *random* illumination with an empirical correction of dynamic scaling factors succeed where ordered scans fail. Read naively, this is a puzzle: the ordered Hadamard scan is the better-conditioned instrument, yet randomness—usually a concession, not a feature—wins. Therefore, it is desirable to determine when, and by what mechanism, a drifting gain becomes blindly separable at all.

The resolution is that under gain drift the ordering itself becomes the liability. An ordered scan sorts the object’s transform coefficients into a smooth sequence in acquisition time, so the object acquires a *temporal signature*: a slow gain trace and a slow coefficient trace cast the same shadow on the one-dimensional bucket record, and no estimator can tell them apart from that record alone. Hadamard is not defeated by randomness; it is defeated by chronology. It is shown that this confounding is not an artifact of any particular correction algorithm but a genuine algebraic ambiguity—for a square unconstrained design, *any* invertible pattern matrix can absorb a nonconstant gain into a perfectly valid alternative object (Theorem A). Conversely, random carriers escape it by a statistical mechanism: when the pattern sequence makes the bucket carrier’s distribution stationary in time, slow gain becomes the only non-stationary component of the record, and the relative gain becomes identifiable up to the unavoidable global scale (Theorem B). Self-calibration is thus a property one can *design into acquisition time*, and its price, rates, and failure boundaries can be computed in advance.

In this paper, we develop that observation into a quantitative identifiability theory. The contributions are:

- 1) **A three-notion identifiability framework** for the bilinear gain–object model  $R_n = a_n B_n$ , separating exact-algebraic identifiability, statistical (asymptotic) identifiability, and estimation conditioning. Within it, the square-design obstruction (Theorem A) is proved, and the tall-design thresholds (Theorem A’)—generic exact identifiability at  $N \geq K + p$  and uniform identifiability at  $N \geq 2K + p - 1$ —are established at generic-pair level for the physical Fourier low-pass drift family, with the uniform-over-drift reading resting on a labeled dimension-count argument plus an exact empirical rank transition (Secs. 4, 9). This is distinct from generic bilinear-injectivity results [13], [14], [15], [16], [17]: the operator here is a diagonal-row map, and the anchor is temporal, not structural.
- 2) **The stationarity anchor (Theorem B)** and its consistency condition ( $\star$ ): random carriers pin the relative gain statis-

tically, with a windowed log-domain estimator achieving the minimax rate over the drift class, including a photon-limited variant whose error scales as  $1/(W\bar{\lambda})$  for windows  $W$ .

- 3) **A synthesis, not a trade-off:** randomized Hadamard (SRHT) designs retain orthogonal conditioning while randomizing the temporal carrier — the stationary/exchangeable carrier being supplied under Walsh-flatness of  $T^2$  or a successful random permutation of a sufficiently spread object — with an exact Walsh-domain necessary-and-sufficient whitening condition separating the roles of sign flips and permutations.
- 4) **An exact finite-noise reconstruction bridge** (Theorem 1) expressing relative MSE for any linear reconstructor through a leverage functional  $B_L$ , covering residual gain, read noise, and Poisson noise without Gaussian approximation, and specializing to orthogonal, pairwise-Hadamard, and random/DGI pipelines [18], [19] (with the DGI constant  $C_0$  measured per implementation—no universal value exists).
- 5) **A phase diagram with a finite- $N$  flip boundary  $\rho^*$**  in the drift-decorrelation parameter  $\rho_{\text{pair}}$  (Prop 3), giving a pre-acquisition go/no-go design chart (Fig. 5) for choosing between ordered and randomized acquisition.

The scope is emphasized: all results are theoretical, supported by numerical evidence; no hardware validation is reported. Identifiability near  $N = K + p$  is a uniqueness statement, not a conditioning guarantee, and random square designs remain algebraically non-identifiable—their advantage is statistical. The organizing idea throughout is the distilled form of the paradox: one makes the gain observable not by measuring it directly, but by making the object stop pretending to be time.

The paper is organized as follows. Section 2 fixes the model and the three notions of identifiability; Fig. 1 places ordered-Hadamard, iid-random, tall-random, and SRHT designs on the resulting two-axis map of conditioning versus identifiability. Section 3 positions the work against bilinear inverse-problem theory. Sections 4 and 5 establish the algebraic obstruction and the stationarity anchor; Sec. 6 derives estimation rates including the low-photon regime. Section 7 develops the SRHT synthesis, Sec. 8 the reconstruction bridge and phase diagram. Section 9 presents the numerical experiments, and Sec. 10 collects design rules, scope, and outlook.

## II. PROBLEM SETUP AND THREE NOTIONS OF IDENTIFIABILITY

We consider a single-pixel (bucket-detection) acquisition of  $N$  frames. On frame  $n$  a programmed pattern  $m_n \in \mathbb{R}^K$  illuminates a static object  $T \in \mathbb{R}^K$ , producing the ideal bucket value  $B_n = m_n^\top T$ ; we call the sequence  $\{B_n\}$  the *object carrier*. The recorded signal is corrupted by a slowly drifting multiplicative gain (source power, detector responsivity, coupling efficiency) and can be described by

$$R_n = a_n B_n, \quad a_n = e^{\ell_n} > 0,$$

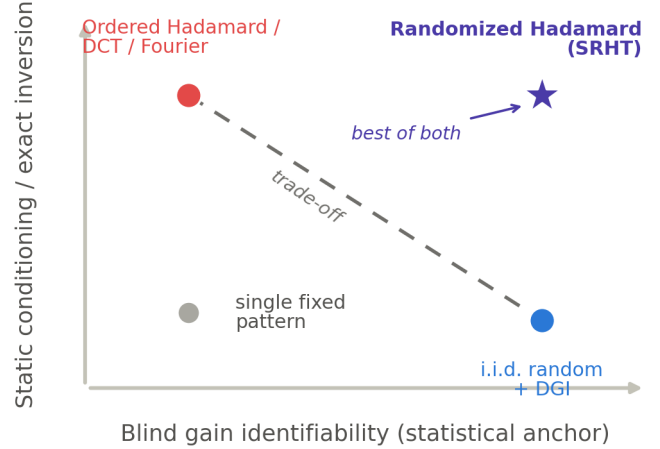


Fig. 1. Two-axis map of inversion conditioning against blind identifiability. Ordered Hadamard, i.i.d.-random (DGI), tall random, and SRHT acquisitions occupy visibly different corners: ordered Hadamard is well-conditioned yet blindly non-identifiable, while randomized and tall designs trade toward identifiability. The separation is explained by Secs. 4 and 5.

where  $a_n$  denotes the unknown per-frame gain and the log-gain trajectory  $\ell = (\ell_1, \dots, \ell_N)$  is confined to a *drift class*  $S \subset \mathbb{R}^N$ ,  $p = \dim S$ , which always contains the constants. The constant direction is a pure gauge:  $(a, T) \mapsto (ca, T/c)$  leaves every  $R_n$  invariant, so all statements below are up to a global scale. For physically slow drift we take  $S$  to be a low-pass (Fourier) space of normalized bandwidth  $\rho_{\text{bw}}$ , so that  $p \approx \rho_{\text{bw}}N$ ; this bandwidth parameter must not be confused with the adjacent-pair decorrelation  $\rho_{\text{pair}}$  entering the finite- $N$  flip boundary of Sec. 8. The blind problem — recover  $(a, T)$  from  $\{R_n\}$  alone — is bilinear, and asking whether it is “solvable” turns out to be three distinct questions. Separating them is itself a contribution of this paper.

### Conventions (fixed throughout).

- $K, T$  — number of pixels; object  $T \in \mathbb{R}^K$ , entrywise nonnegative unless stated otherwise.
- $N$  — number of acquired frames.
- $S$  — linear log-gain drift class; **always contains the constants**.
- $p = \dim S$  — drift dimension; the *nonconstant* drift directions number  $p - 1$ .
- global scale — **all** identifiability statements are modulo the gauge  $(a, T) \mapsto (ca, T/c)$ ; we fix the gauge by centering the log-gain ( $\bar{\ell} = 0$ ); reconstruction error metrics use least-squares scale alignment (Sec. 8).
- identifiability notions — *local-differential*, *generic-exact*, and *uniform-exact* identifiability are distinct properties with distinct thresholds (Theorem A’); none of them implies conditioning.
- $\rho_{\text{bw}}$  — normalized gain **bandwidth**:  $p \approx \rho_{\text{bw}}N$  (tall-design law, Sec. 4).
- $\rho_{\text{pair}}$  — **adjacent-pair** gain decorrelation, entering  $\text{Var}(\Delta) \approx 2s^2r(\rho_{\text{pair}})$  (flip boundary, Sec. 8); related to,

but not equal to,  $\rho_{\text{bw}}$ .

- $K_{\text{eff}} = S_1^2/S_2$  — excitation functional, with  $S_1 = \sum_j T_j$ ,  $S_2 = \sum_j T_j^2$ .
- $K_\infty = S_2/\|T\|_\infty^2$  — Lindeberg spikiness functional (Gaussian/log-domain approximations).
- $K_4 = S_2^2/\sum_j T_j^4$  — spectral-spread functional (permutation whitening, Sec. 7).
- $C_0$  — one-sample DGI correlator constant,  $C_0 = \mathbb{E}\|Z\|^2/S_2 - 1$ ; pipeline-specific, no universal value.
- $D_H$  — pairwise-Hadamard factor,  $\approx 1$  for non-DC coefficients; measured per pipeline.
- $B_L$  — leverage of a linear reconstructor  $L$ ,  $B_L = (1/S_2) \sum_n b_n^2 \|Le_n\|^2$  (Sec. 8).

Pattern entries  $m_n$  may be signed (Hadamard) or nonnegative (speckle) even though  $T$  is nonnegative; where signed coefficients appear as carriers, the bucket conventions of Sec. 5 specify how they are physically realized.

**Algebraic identifiability.** Does the noiseless record determine  $(a, T)$  up to gauge, or does some other pair  $(a', T')$  with  $\ell' - \ell \in S$  nonconstant reproduce the data exactly? This is a finite-sample, deterministic question about collisions  $MT = D_s MT'$ , and its answer depends on the design  $M$  and on the dimension count  $N$  versus  $K + p$ . Section 4 (Theorem A and the tall-design thresholds, Theorem A') shows that every square invertible design — however well-conditioned — fails it, and that overdetermination restores it generically at  $N \geq K + p$ .

**Statistical identifiability.** When the patterns are drawn from a random ensemble, a weaker but operationally decisive property becomes available: if the carrier  $B_n$  is *stationary* — its distribution does not depend on  $n$  — then any slow modulation of local averages over windows  $W$  is attributable to the gain alone, and the relative gain is identifiable in distribution even where exact algebraic separation is impossible. This stationarity anchor is developed in Sec. 5 (Theorem B) and is the mechanism the field's empirical successes have been implicitly exploiting.

**Estimation conditioning.** Uniqueness is not accuracy. Even when one of the first two notions holds, the finite-noise error of a concrete estimator is governed by conditioning quantities — the window size  $W$ , the residual-gain variance  $v$ , the leverage  $B_L$  of the reconstruction operator — quantified by the rate analysis of Sec. 6 and the reconstruction bridge of Sec. 8. In particular, identifiability at the threshold  $N = K + p$  says nothing about stability there.

We contend that much of the confusion surrounding blind gain correction stems from conflating these three notions: judging an acquisition scheme by the visual quality of a blindly corrected reconstruction entangles algebraic ambiguity, the presence or absence of a statistical anchor, and estimator conditioning into a single scalar, although the three failures call for entirely different remedies. Figure 1 disentangles them as a two-axis map — inversion conditioning against blind identifiability — on which ordered Hadamard, i.i.d.-random (DGI), tall random, and SRHT acquisitions occupy visibly different corners, a separation that Secs. 4 and 5 explain.

### III. RELATION TO PRIOR WORK

Gain-object separation is a bilinear inverse problem: writing the log-gain as  $\ell_n = \log a_n$  and lifting the unknowns to the rank-one array  $X = aT^\top$ , each bucket reads  $R_n = a_n B_n$  with  $B_n = m_n^\top T$ , a bilinear form in the object  $T$  and the gain trace  $a$ . This places our question inside a mature literature on bilinear identifiability, and it is worth stating precisely what that literature settles and what it leaves open for the diagonal-gain geometry considered here.

The sharpest injectivity counts for lifted bilinear maps are Kech–Krahmer's optimal conditions for generic identifiability [13], which for a rank-one measurement model require on the order of  $2(n_1 + n_2) - 4$  generic measurements. These counts do not transfer to our setting: the operator  $L_M(X)_n = a_n m_n^\top T$  is a *diagonal-row* map, not a generic rank-one measurement, so the object carrier and the gain enter through a structured — not generic — set of linear forms. The relevant identifiability question is instead the intersection geometry of the object subspace with its gain-modulated copy,  $U \cap D_s U$ . Choudhary–Mitra's scaling laws for bilinear inverse problems [16] are the closest in spirit to our Theorem A: their rank-two nullspace obstruction is exactly the mechanism by which a square unconstrained design absorbs a nonconstant gain into a redefined object. Li–Lee–Bresler's unified framework [14] supplies the subspace- and sparsity-constrained identifiability thresholds that our tall-design counts ( $N \geq K + p$  generic,  $N \geq 2K + p - 1$  uniform) specialize to for the diagonal-gain model. Ahmed–Recht–Romberg [15] establish convex recovery for blind deconvolution under subspace priors, and Kueng–Rauhut–Terstiege [17] give the low-rank-from-rank-one-measurement guarantees; both are the algorithmic back-drop against which our purely algebraic and statistical criteria should be read.

On the imaging side, the practice of single-pixel and computational ghost imaging has long weighed ordered Hadamard bases against random speckle patterns for their conditioning and noise properties, but without an identifiability account of slow multiplicative drift. Learned correction of dynamic scaling factors in ghost imaging [7] and learned encoding for optical wireless links through complex channels [8] address the same physical nuisance empirically; our analysis explains why their *static-correction ceiling* — the error floor left by any time-independent correction (one learned or fitted on static data) when applied to a drifting record — is the honest performance bound, and when a drifting gain becomes blindly separable at all. Blind self-calibration of unknown sensor gains and phases has a long history outside imaging — from array-processing eigenstructure methods [20] to blind calibration of sensor networks [21] and biconvex compressive self-calibration [22] — and Balzano–Nowak's blind sensor calibration similarly finds gains recoverable under mild over-sampling, which our tall-design threshold  $N \geq K + p$  makes exact for the diagonal bilinear model. On the acquisition-design side, pink-noise speckle illumination [23] and sparse Hadamard sequencing [24] optimize the pattern set directly, but target additive-noise robustness, whereas our mechanism concerns multiplicative-drift identifiability.

Against this background the contribution of the present work is specific. We do not add to generic bilinear injectivity, which is settled by the references above. What is new is the temporal-statistical mechanism: relative-gain identifiability secured by making the object carrier stationary (Theorem B), the associated nonparametric slow-gain calibration rate, the ordered-orthogonal confounding failure mode that this mechanism repairs, the SRHT synthesis that recovers orthogonal conditioning without sacrificing self-calibration, and the finite-noise reconstruction bridge that renders the phase diagram quantitative. The novelty is the stationarity anchor, not bilinear injectivity per se.

#### IV. THE ALGEBRAIC OBSTRUCTION AND WHEN OVERSAMPLING CURES IT

The paradox of Sec. 1 is not an estimator artifact. Write the noiseless bucket record as  $R_n = a_n B_n$  with  $B_n = m_n^T T$  the carrier coefficient of the object  $T \in \mathbb{R}^K$  under illumination row  $m_n$ , and  $\ell_n = \log a_n$  the log-gain. Stack  $N = K$  frames into an invertible square design  $M$ . Then the multiplicative drift is absorbed *exactly* into a relabelled object.

**Theorem A (square non-identifiability).** *For any invertible  $M \in \mathbb{R}^{K \times K}$  and any gain  $a = (a_n)$  nonconstant on the support of the carrier  $MT$ , there exists a distinct object  $T' \neq cT$  producing the identical bucket record:  $\text{diag}(a)MT = MT'$  with  $T' = M^{-1} \text{diag}(a)MT$ . Gain-object separation is unidentifiable up to the global-scale gauge  $a \mapsto ca$ ,  $T \mapsto T/c$ . (Proof: Appendix A.)*

The object is a barcode printed onto time and the drift is a smudge across it; when the barcode carries slow structure, the theorem says the smudge can be *reread as a different barcode* rather than dismissed as blur. This is a genuine orbit of the bilinear map, and it holds for *every* invertible design — Hadamard, Fourier, or otherwise. Two corollaries sharpen the trap.

**Corollary 1 (slowness does not help).** *Restricting  $\ell \in S$  to a slow drift class does not restore identifiability at  $N = K$ : for any nonconstant  $s \in S$ , the alternative pair  $(\ell - s, T' = M^{-1} \text{diag}(e^s)MT)$  explains the record exactly, and the alternative log-gain  $\ell - s$  is again an admissible slow drift because  $S$  is a linear space. A slowly varying gain thus masquerades as a slowly varying object.*

**Corollary 2 (positivity is not enough; known support zeros can be).** *Nonnegativity of  $T$  alone does not break the orbit. Known zeros in the object support can: each enforced zero removes a degree of freedom from  $T'$ , and  $K_0 \geq p-1$  generic zeros at known support locations restore local identifiability via a rank condition on the constrained collision map — one per nonconstant drift dimension,  $p-1 = \dim(S/\text{span}\{1\})$ . This is a local rank criterion, not a global guarantee. (Proof: Appendix A.)*

The first escape is overdetermination. Take a generic tall design  $M \in \mathbb{R}^{N \times K}$ ,  $N > K$ , with  $\ell \in S$ ,  $\dim S = p$ ,  $S \ni$  constants (the gauge). A collision  $MT = \text{diag}(e^s)MT'$  with  $s = \ell' - \ell$  is governed by the intersection  $\text{range}(M) \cap$

$\text{diag}(e^s) \text{range}(M)$ . Counting the projective incidence dimension of diagonal-gain collisions gives  $\Delta = (p-1) + 2K - 1 - \text{rank}[M, -\text{diag}(b)M]$ , with the rank equal to  $\min(N, 2K)$  on the dense stratum; generic-point identifiability requires  $\Delta < K-1$  and uniform identifiability  $\Delta < 0$ . This yields three thresholds.

**Theorem A' (tall-design thresholds).** *For generic tall  $M$  and transverse nonconstant gain family of dimension  $p$ :*

- *local differential identifiability:*  $N \geq K + p - 1$ ;
- *generic exact finite-sample identifiability:*  $N \geq K + p$ ;
- *uniform (every nonzero object):*  $N \geq 2K + p - 1$ .

(Proof: Appendix B.)

For the *physical* low-pass drift space these bounds hold at three distinct levels of rigor, which we keep separate. For  $S_{\text{LP}} = \text{span}\{1, \cos q\omega, \sin q\omega : q \leq m\}$ ,  $p = 2m + 1$ , a level-multiplicity lemma (a nonconstant degree- $2m$  trigonometric polynomial attains each value at most  $p-1$  times) bounds the collision rank, giving — for each *fixed* nonconstant  $s \in S_{\text{LP}}$  and generic  $M$  —  $\text{rank}[M, -\text{diag}(b)M] = \min(N, 2K)$  whenever  $N \geq K + p$ : the clean thresholds are thus *proven at generic-pair level* — for a generic pair  $(M, s)$  — for the Fourier low-pass family. The *uniform-over-drift* version (one generic  $M$  serving every nonconstant  $s \in S_{\text{LP}}$  simultaneously) rests on a dimension-count argument, labeled as such in Appendix B: under a free determinantal codimension estimate it covers  $N \leq 2K - p + 1$  and  $N \geq 2K + p - 1$ , leaving the gap window  $2K - p + 1 < N < 2K + p - 1$  at generic-pair level only. Empirically, the  $K = 64$  and  $K = 128$  threshold scans of Sec. 9 (Fig. 8) show an exact 0%  $\rightarrow$  100% rank transition at  $N = K + p - 1$  across all five drift dimensions tested at both object sizes (3250 cells in total, no exceptions), consistent with the uniform reading at the local level. (Proof: Appendix B.) For other conventions one checks the level count  $L_S \leq K$  (if  $N < 2K$ ) or  $L_S \leq N - K$  (if  $N \geq 2K$ ). Throughout, the middle threshold of Theorem A' asserts *generic exact finite-sample identifiability* — uniqueness of  $(a, T)$  up to gauge — not a conditioning bound and not an algorithmic recovery guarantee; stability near the thresholds is the separate third notion of Sec. 2, quantified in Secs. 6 and 8.

The picture is a dichotomy, and it sets up the rest of the paper. When  $N \geq K + p$ , overdetermination *alone* separates gain from object — no statistical mechanism is logically needed, though randomization still buys stable conditioning (the identifiability axis of Fig. 1). When  $K < N < K + p$ , tallness is insufficient even generically, and the stationarity anchor of Sec. 5 (or a support/sparsity prior) becomes *required*, not merely convenient. The two regimes meet through the drift bandwidth: a gain occupying a fraction  $\rho_{\text{bw}}$  of the frame band spans  $p \approx \rho_{\text{bw}} N$  low-pass modes, so  $N \geq K + p$  reads as

$$N \gtrsim \frac{K}{1 - \rho_{\text{bw}}}.$$

Slow drift ( $\rho_{\text{bw}} \ll 1$ ) is cured by modest oversampling; as  $\rho_{\text{bw}} \rightarrow 1$ ,  $p \rightarrow N$  and blind algebraic separation is impossible for *any*  $N$ , forcing the statistical route.

## V. STATISTICAL IDENTIFIABILITY: THE STATIONARITY ANCHOR

Section 4 offered one escape from the gain–object orbit: buy identifiability with surplus rows. This section develops a second, logically distinct escape that operates even at fixed  $N$  and, crucially, explains *why* randomized acquisition self-calibrates while ordered acquisition does not. The idea is best stated before the mathematics. A drifting gain is a boat carried by a slow current, and the bucket record is our only view of the water. If the waves change character with position — as the ordered Hadamard coefficient sequence does, sweeping through the object’s spectral envelope — we cannot tell the current from the coastline: a slow change in the record may be drift, or may be the scenery. But if the waves are *statistically identical everywhere*, any slow modulation of the record has exactly one available explanation. Local averaging then drops an anchor into the distribution itself: it pins the *relative* drift, while the absolute scale remains gauged away. This intuition, and nothing stronger, is what the results below formalize.

The log-domain record can be described by  $Y_n = \log R_n = \ell_n + \log B_n$ . The anchor requirement is a condition on the carrier alone:

**Condition  $(\star)$  (stationary carrier, quantitative form).** There is a scalar  $m_T$  — depending on the object but not on time, and absorbed by the global-scale gauge — such that the windowed carrier averages over windows  $W_n$  of length  $W$  satisfy

$$\max_n \left| W^{-1} \sum_{k \in W_n} \log B_k - m_T \right| \rightarrow 0$$

in probability as  $W \rightarrow \infty$ ,  $W/N \rightarrow 0$ .

The condition presupposes a positive carrier on the analyzed record — physically, a positivity/offset margin, or the soft-log  $\psi_\alpha$  of Sec. 6 in the photon-counting case.

The displayed criterion is deliberately *uniform* — a maximum over all window locations; pointwise consistency at a single  $n$  is strictly weaker, and upgrading it to the displayed uniform control over the  $\sim N$  window locations typically costs a  $\log N$  / high-probability refinement (the  $\delta \mapsto \delta/N$  step of Appendix C).

I.i.d. random illumination satisfies  $(\star)$  by construction. For a finite random row permutation of a deterministic pattern set, or a single fixed SRHT draw,  $(\star)$  is *not* automatic: a permuted record is exchangeable — which anchors windowed averages in distribution — but a particular draw can still fail for a spectrally concentrated object; the quantitative version of this distinction is the permutation bound of Sec. 7, and the permuted-arm rejections in Fig. 2 are its visible trace. For the i.i.d. case:

**Proposition 1 (random-basis carrier).** *For i.i.d. random patterns with pixel mean  $\mu_I$  and standard deviation  $\sigma_I$ , the carrier  $\{B_n\}$  is white and strictly stationary, and the object enters its first two moments only through the scalars  $S_1 = \sum_j T_j$  and  $S_2 = \sum_j T_j^2$ :  $\mathbb{E} B_n = \mu_I S_1$ ,  $\text{Var } B_n = \sigma_I^2 S_2$ , hence  $\text{CV}_B = (\sigma_I/\mu_I)/\sqrt{K_{\text{eff}}}$  with  $K_{\text{eff}} = S_1^2/S_2$ . Gaussian and log-domain approximations additionally require the Lindeberg-type spikiness condition  $K_\infty = S_2/\|T\|_\infty^2 \rightarrow \infty$ ;*

*the reduction of the object to two scalars holds at this Gaussian-approximation level, not exactly. Higher cumulants and the log-domain mean can depend on the object’s shape beyond  $(S_1, S_2)$ , but for an i.i.d. carrier every such term is time-independent and shifts only the gauge scalar  $m_T$ ; it does not affect relative-gain recovery. (Proof: Appendix C.)*

Figure 2 shows the two carriers side by side: the random trace is a flat, object-independent noise band, whereas the ordered Hadamard trace is the object’s coefficient envelope — a deterministic, strongly non-stationary signature imprinted on acquisition time. Under ordered scanning the object masquerades as a temporal trend, precisely the degree of freedom a slow gain occupies, and  $(\star)$  fails. We emphasize the correct logical status of the condition:

**Proposition 2  $(\star)$  is an estimator criterion, not a universal one).** *Condition  $(\star)$  is necessary and sufficient for consistency of the blind windowed (stationarity-anchor) gain correction. It is not an if-and-only-if condition for identifiability by all mechanisms: the overdetermined designs of Sec. 4 identify algebraically without any stationarity, and conversely  $(\star)$  confers no exact algebraic identifiability. (Proof: Appendix C.)*

**Bucket conventions.** Three bucket-signal conventions coexist in this paper, and the log-domain results apply to them differently. (i) *Signed coefficient model:*  $B_n$  is a signed transform coefficient (e.g. a bare Walsh–Hadamard coefficient); it can vanish or change sign, so  $\log B_n$  is undefined and no log-domain gain estimator is available without further structure. (ii) *Physical nonnegative offset model:* the optical bucket is a nonnegative intensity; signed patterns are realized with a bias/offset so that  $B_n \geq B_{\min} > 0$ . Theorem B and the estimators of Sec. 6 are stated in this model (assumption (iv) below). (iii)  $\pm$  *pairwise model:* each signed pattern is acquired as a complementary pair of nonnegative patterns whose buckets are differenced; the difference is signed again, but the drift now enters through adjacent-pair increments controlled by  $\rho_{\text{pair}}$  (Sec. 8). The raw signed arms of Sec. 9, which feed bare Walsh coefficients to the gain estimator, follow convention (i) and are *diagnostic only*: they visualize the chronology failure but are not a valid physical log-domain estimator without the offset of (ii) or the pairing of (iii).

The main statistical result is then:

**Theorem B (statistical relative-gain identifiability).** *Assume the log-domain model*

$$Y_n = \log R_n = \ell_n + m_T + z_n,$$

*where (i)  $m_T = \mathbb{E} \log B_n$  is a scalar depending on the object but not on time; (ii) the carrier fluctuations  $\{z_n\}$  are centered, strictly stationary, and weakly dependent with summable autocovariances,  $\sigma_{\text{abs}}^2 = \sum_h |\gamma(h)| < \infty$  where  $\gamma(h) = \text{Cov}(z_0, z_h)$  ( $\beta$ -mixing with summable coefficients is sufficient) — the signed long-run variance  $\sigma_{\text{LR}}^2 = \gamma(0) + 2 \sum_{h \geq 1} \gamma(h) (\leq \sigma_{\text{abs}}^2)$  then also exists and is the asymptotic variance of window means; the two constants play distinct roles below; (iii) the log-gain  $\ell \in S$  lies in a normalized-time Hölder- $\beta$  class,  $|\ell_j - \ell_k| \leq L(|j - k|/N)^\beta$ ; and (iv)  $R_n > 0$  on the analyzed record so the logarithm is defined (positivity/offset margin; the*

Noiseless bucket carriers (digit\_0, 8192 frames)

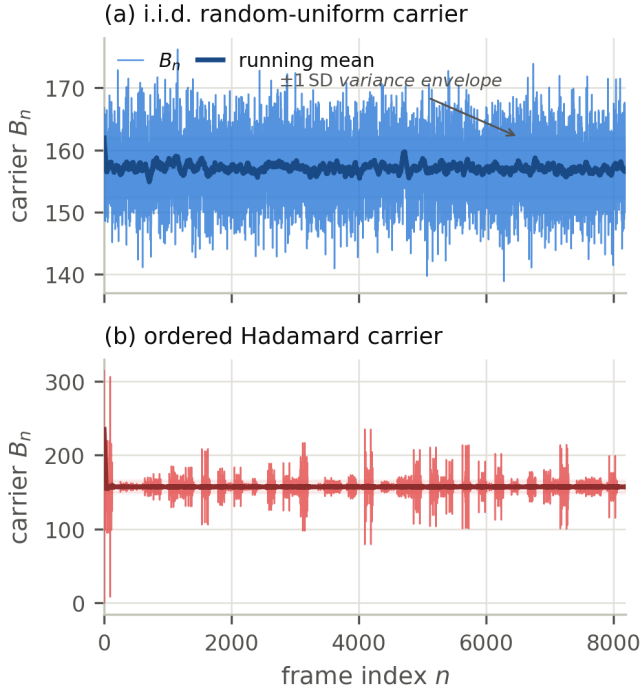


Fig. 2. Carrier stationarity. Noiseless bucket carriers  $B_n$  at 8192 frames per arm on  $64 \times 64$  objects, before any gain is applied, split into 8 chunks with a 128-frame transient cutoff, AGC window  $W = 64$ , five arms. The ordered Hadamard trace is the object’s coefficient envelope — a deterministic, strongly non-stationary temporal signature — whereas the random traces are flat, object-independent noise bands. The Brown–Forsythe variance-stationarity test at  $p < 10^{-3}$  rejects 7 of 10 objects for ordered Hadamard (against 0 of 10 for both i.i.d. random ensembles), while the permuted arms each reject 2 of 10, the theory’s own predicted residue of fixed-permutation aliasing. Every exhaustively paired arm has carrier  $CV_B = 1/\sqrt{K_{\text{eff}}}$  to a relative accuracy of about  $10^{-6}$ . Descriptive counts over the ten-object panel.

*soft-log of Sec. 6 covers the photon-counting case). Then, for windows satisfying both  $W \rightarrow \infty$  and  $W/N \rightarrow 0$ , windowed averages of  $Y_n$  estimate  $\ell_n + m_T$  consistently, with non-asymptotic stochastic error  $O_P(\sigma_{\text{abs}}/\sqrt{W})$  — the asymptotic variance of the window mean being  $\sigma_{\text{LR}}^2/W$  — and windowed drift bias  $O(L(W/N)^\beta)$ ; consequently the centered log-gain  $\ell_n - \bar{\ell}$  — the relative gain, up to one global scale — is identifiable and consistently estimable. The scalar  $m_T$ , and with it the absolute gauge of  $\ell$ , is not recoverable. (Proof: Appendix C.)*

Theorem B must be read with its scope intact. It is an asymptotic, high-probability statement about *relative-gain* recovery up to the global-scale gauge — never exact finite-sample identifiability of  $(a, T)$ . In particular it does not contradict Theorem A: a random square design remains algebraically non-identifiable; what randomness buys is statistical separability, a different currency. This is also what distinguishes the mechanism from the bilinear-inverse identifiability literature [14], [13], [16]: the anchor is a property of the *temporal distribution* of the carrier, not of subspace injectivity.

The anchor also explains why the self-calibrating gain corrections of Refs. [7], [8] can work at all. Under Proposition 1’s Gaussian approximation the bucket record given  $(a, \mu_B, \sigma_B)$  is a slow trend plus stationary residuals, and an SCGI-style Gaussian-likelihood fit of that trend is a maximum-likelihood realization of the windowed anchor estimator, inheriting Theorem B’s consistency guarantee (and its gauge).

Figure 3 is the core numerical evidence. Across a panel of objects, the blind gain-estimation error  $\|\hat{a} - a\|/\|a\|$  under random illumination tracks  $CV_B/\sqrt{W}$  object by object — to leading Gaussian/log-variance order, and within the tested object panel, the object enters only through  $K_{\text{eff}}$  — and once normalized by that floor the curves collapse across objects to within  $\pm 5\text{--}7\%$ ; under ordered Hadamard the curves scatter by object and do not decay with  $W$  — the object envelope is a bias no amount of averaging removes. The window  $W$  itself carries a bias–variance trade-off (drift curvature versus  $CV_B^2/W$ ), whose optimal resolution and minimax rate are the subject of Sec. 6.

## VI. ESTIMATION RATE, LOWER BOUNDS, AND LOW-PHOTON ROBUSTNESS

Section 5 establishes that relative gain is *identifiable* from a stationary carrier; here we show it is recovered *efficiently and predictably*. The baseline estimator is the plain windowed log of Theorem B: average  $Y_n = \log R_n$  over a window  $W$  of frames sharing a slowly varying gain. For photon-counting records we work with its robust generalization, the windowed *soft-log*: average  $\psi_\alpha(c) = \log(c + \alpha)$  and invert the calibration curve,  $\hat{\theta}_n = m_\alpha^{-1}(\text{mean}_W \psi_\alpha)$ , where  $m_\alpha(\theta) = \mathbb{E}[\psi_\alpha(C)]$  denotes the calibration curve for  $C \sim \text{Pois}(\Lambda_0 e^\theta B + d)$ . As  $\alpha \rightarrow 0$  at high mean count  $\bar{\lambda}$  the calibration curve becomes the identity and the soft-log reduces to the plain windowed log estimator; the rate theorem below is stated for the general (soft-log) estimator and covers both regimes. Because the log-gain  $\ell$  lies in a Hölder- $\beta$  class of smooth drift (notation warning:  $\beta$  is reserved for Hölder smoothness and  $\alpha$  for the soft-log offset throughout; the two symbols are unrelated and must not be conflated), the window trades bias against variance in the usual nonparametric way: wider  $W$  suppresses variance as  $O(1/W)$  but incurs an approximation bias set by the local smoothness  $\beta$  and the drift Lipschitz constant  $L_a$  (the per-frame form of Theorem B’s normalized-time constant,  $L_a = L N^{-\beta}$ ).

**Theorem C (windowed rate).** *For log-gain in a Hölder- $\beta$  class with constant  $L_a$ , the bias-optimized windowed estimator attains*

$$\text{MSE}^* \leq C \kappa_{\min}^{-2} [\kappa_{\max} L_a]^{2/(2\beta+1)} \sigma_{\alpha, \text{LR}}^{4\beta/(2\beta+1)},$$

*the standard nonparametric rate in the effective per-frame noise  $\sigma_{\alpha, \text{LR}}$ . (Proof: Appendix D.)*

This rate is not merely achievable but *optimal*. An Assouad two-point construction over the same smoothness class (a van Trees / Pinsker argument in the Sobolev variant) produces a matching lower bound, so no estimator improves on Theorem C by more than constants.

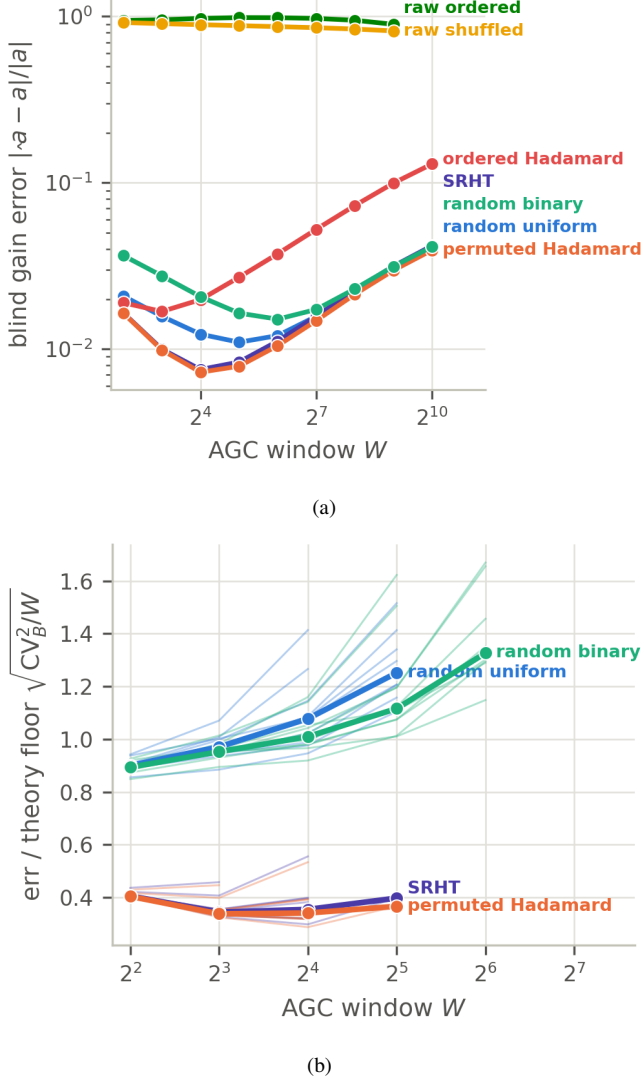


Fig. 3. Blind gain identifiability (seven arms;  $32 \times 32$  objects,  $K = 1024$ , 2048 frames). (a) Blind gain error  $\| \hat{a} - a \| / \| a \|$  versus window  $W$ . The raw ordered Hadamard arm sits at  $\approx 0.96$  and does not decay (0.95 at  $W = 4$ , still 0.90 at  $W = 512$ ); shuffling the raw frames improves it to  $\approx 0.86$  but it remains roughly  $40\times$  worse than the random and paired arms at  $\approx 0.02$ . Slow-drift variance-segment slopes of  $\log(\text{err})$  against  $\log W$  are  $-0.39$  (random binary),  $-0.60$  (SRHT), and  $-0.61$  (randomly permuted paired Hadamard) — all inside the  $W^{-1/2}$  band — while naturally ordered paired Hadamard is shallow at  $-0.24$ . (b) Normalizing by the theory floor  $\sqrt{CV_B^2/W}$  collapses the panel: paired arms at 0.40–0.41, random arms at 0.89–0.90, each bundle collapsing across all ten objects to within  $\pm 5$ –7%; best-window error correlates with  $K_{\text{eff}}$  at  $-0.86$  to  $-0.90$ . Descriptive values over the ten-object panel.

**Theorem D (minimax optimality).** *Over the Hölder- $\beta$  drift class the windowed estimator is minimax-rate optimal: the lower bound matches Theorem C at the level of rates. Sharp minimax constants remain open. (Proof: Appendix D.)*

The floor beneath both bounds is Fisher-information geometry. Working in the gauge quotient (global scale is unobservable, Sec. 5), the relevant object is the *quotient Fisher information*: the Fisher matrix restricted to directions transverse to the scale orbit. This makes the identifiability dichotomy quantitative — the Fisher matrix is singular exactly along the

null direction of Theorem A. Non-identifiability is not poor conditioning; it is a genuine flat direction of the likelihood, the ambiguity  $(a, T) \mapsto (as^{-1}, sT)$  made infinitesimal.

Which design constant controls what is a spikiness question, and three distinct functionals appear.  $K_{\text{eff}} = S_1^2/S_2$  (Prop 1) governs excitation and the pairwise-Hadamard residual-gain budget;  $K_\infty = S_2/\|T\|_\infty^2$  is the Lindeberg spikiness parameter gating the Gaussian and log-domain approximations (Prop 1) and the windowed-energy concentration;  $K_4 = S_2^2/\sum_j T_j^4$  enters the concentration guarantee for permutation whitening (Sec. 7, via  $\min(\epsilon^2 K_4, \epsilon K_\infty) \geq C \log(K/\delta)$ ). Conflating them mislabels the operative bottleneck: a design flat in  $K_{\text{eff}}$  can still be spiky in  $K_\infty$ .

Low photon counts do not break the log domain. The Poisson derivative identity  $\frac{d}{d\lambda} \mathbb{E}f(C) = \mathbb{E}[f(C+1) - f(C)]$  gives a sensitivity  $\kappa_\alpha(\lambda) \rightarrow 1$  at high  $\bar{\lambda}$  and  $\kappa_\alpha(\lambda) \rightarrow \lambda \log(1 + 1/\alpha)$  as  $\bar{\lambda} \rightarrow 0$ ; the offset  $\alpha$  absorbs empty bins so no frame ever contributes  $\log 0$ . Propagating through Theorem C, the photon-starved variance degrades gracefully as  $\sim 1/(W\bar{\lambda})$  — the Fisher-matched scaling, since the Poisson information for log-intensity is  $\lambda$  itself. Zero photons carry zero Fisher information; they cut precision but never diverge (Fig. 7).

The caveat is stated honestly. At  $\bar{\lambda} < 1$  the soft-log becomes shrinkage-bias-dominated:  $\psi_\alpha$  pulls the estimate toward  $\log \alpha$ , and the estimator sits *below* its unbiased (local) Fisher reference not because it beats the bound but because bias is present. The  $1/(W\bar{\lambda})$  law is the variance-limited regime above this crossover; below it, honest accounting reports the bias floor rather than the optimistic Fisher curve, and recovering the rate requires the offset-design or full Poisson-mixture MLE variants.

## VII. BASIS DESIGN: RANDOMIZED ORTHOGONAL ACQUISITION (SRHT)

A reader who has followed the stationarity anchor of Sec. 5 will raise a natural objection: the anchor was purchased with random illumination, yet the appeal of ordered Hadamard was its exact, perfectly conditioned inverse. Must one surrender orthogonal conditioning to buy self-calibration? The answer is no. The randomized Hadamard transform — an orthogonal Hadamard carrier composed with a random sign flip and a random pixel permutation, known in the sketching literature as the SRHT (subsampled randomized Hadamard transform) — keeps a full-rank, well-conditioned forward map while randomizing the *temporal handwriting* of the object. One shuffles the deck without bending the cards. We use the *full*  $N = K$  randomized transform throughout; “subsampled” is reserved for the  $N < K$  case, which additionally raises a bias/prior question outside our scope.

The mechanism is exact. Writing the per-coefficient excitation as  $Z_g$  under a random diagonal sign  $D$ , the residual gain-object coupling is governed entirely by the Walsh spectrum of the squared object. With  $w = T^2$ ,

$$\text{Cov}_D(Z_g, Z_h) = \hat{w}(g + h),$$

the Walsh transform of  $w$  evaluated at the dyadic sum of indices. Sign randomization alone therefore whitens the excitation — decorrelates every coefficient pair — **if and only**

if  $w$  is Walsh-flat, i.e.  $\hat{w}(k)$  is negligible for all  $k \neq 0$ . This is a condition on the object, not the design: a structured  $T$  concentrates  $\hat{w}$  off the origin and sign flips cannot remove it.

The random permutation supplies the missing genericity. It acts as a probabilistic Walsh-flattener: permuting pixels spreads the mass of  $\hat{w}$  across the spectrum, and the off-origin covariance is driven below tolerance  $\varepsilon$  with probability  $1 - \delta$  once the object is spectrally spread enough,

$$\min(\varepsilon^2 K_4, \varepsilon K_\infty) \geq C \log(K/\delta),$$

with  $K_4$  and  $K_\infty$  the spread functionals of Sec. 6. Both diverge as the object's energy delocalizes and collapse when a few pixels dominate; the bound quantifies precisely how much delocalization the permutation needs.

Permutation is a genuine second ingredient, not a substitute for the analysis. Under permutation alone,

$$\text{Var}_P Z_g = \frac{K S_2 - S_1^2}{K - 1} = S_2 \frac{K - K_{\text{eff}}}{K - 1},$$

so the residual carries an  $O(1/K_{\text{eff}})$  upper bound on variance — reassuring on average — but no matching lower bound. A flat object realizes the gap: it produces zero excitation, so permutation cannot guarantee two-sided control coefficient by coefficient. The whitening claim is one-sided unless the sign flip and the spectral-spread condition are both in force. A row-order shuffle of the acquisition sequence is a third, weaker operation: it makes the temporal carrier exchangeable — enough for the windowed anchor of Sec. 5 to hold in distribution — but it does not by itself guarantee two-sided per-coefficient excitation. Figure 6 isolates these contributions by ablation (none / permutation / sign / both), and the two-sided regime is exactly the “both” cell.

This yields a design rule rather than a single number.

#### Design rule (blind slow gain).

- (i) Do *not* acquire in ordered Hadamard when the nuisance is a blind, slowly drifting gain — the ordering hands the object a temporal signature (Sec. 4).
- (ii) Randomize the carrier: apply the SRHT sign flip and permutation.
- (iii) Verify Walsh-flatness of the realized (permuted)  $T^2$  — the covariance-whitening guarantee is the sign-randomization statement applied after the permutation — or rely on the permutation bound above with  $K_4, K_\infty$  estimated from the object class.
- (iv) Keep a positivity / offset margin so the log or soft-log stage of Sec. 6 stays away from its singularity.
- (v) Choose the averaging window  $W$  from the bias–variance trade of Sec. 6, not from conditioning.

The rule prescribes *both* randomizations because that is the configuration with a two-sided theorem guarantee under the stated spectral-spread condition. The ablation of Fig. 6 reports a different, empirical fact: in the tested panel, either randomization alone already recovered essentially the full advantage. The two statements must not be merged — the theorem guarantees whitening under its hypotheses, while the ablation shows that, for these objects and this drift, any tested

way of breaking the chronology sufficed; we do not promote the panel result to a universal claim.

The synthesis is that orthogonality and self-calibration are compatible: SRHT preserves the  $B_L = 1$  leverage of a full orthogonal inversion (Sec. 8) while — under Walsh-flatness of  $T^2$  or a successful permutation of a sufficiently spread object — making the object carrier statistically stationary in time, so that the only non-stationary structure left in the bucket record is the gain itself. The conditioning that made ordered Hadamard attractive is retained; only its chronology is discarded.

(Proof of the Walsh identity and permutation bound: Appendix E.)

### VIII. FROM IDENTIFIABILITY TO IMAGES: THE RECONSTRUCTION BRIDGE AND PHASE DIAGRAM

Sections 4–7 answer the question a statistician asks — *is the gain identifiable, and at what rate?* — but an imaging practitioner asks a different one: *how many gray levels do I lose?* This section supplies the bridge. The result is exact, holds at finite  $N$  and finite photon count, and reduces every acquisition strategy in this paper to three numbers that can be read off a single chart before any hardware is built.

The corrected bucket record can be described by  $z_n = (1 + \delta_n)b_n + \xi_n$  with  $b = AT$ , where  $\delta_n$  denotes the *residual* multiplicative gain after whatever blind correction the pipeline applied; let  $\hat{T} = Lz$  be any linear reconstruction.

**Theorem 1 (master finite-noise relMSE identity).** *Assume the residual gain  $\delta$  and the additive noise  $\xi$  are uncorrelated,  $C_{\delta\xi} := \text{Cov}(\delta, \xi) = 0$ . Then, without Gaussian or small-noise approximation,*

$$\frac{\mathbb{E}\|\hat{T} - T\|^2}{S_2} = \underbrace{\frac{\|(LA - I)T + L \text{diag}(b) m_\delta\|^2}{S_2}}_{\text{bias}} + \underbrace{\frac{\text{tr}(L \text{diag}(b) V_\delta \text{diag}(b) L^\top)}{S_2}}_{\text{residual gain}} + \underbrace{\frac{\text{tr}(L \Sigma_\xi L^\top)}{S_2}}_{\text{read + Poisson}}.$$

*If the residual gains are independent with  $\text{Var } \delta_n = v$ , the gain term collapses to  $v B_L$  with the **leverage**  $B_L = (1/S_2) \sum_n b_n^2 \|Le_n\|^2$ ; coherent (smooth) residuals require the full matrix form. Without the uncorrelatedness hypothesis the identity acquires exactly one additional term,  $(2/S_2) \text{tr}[L \text{diag}(b) C_{\delta\xi} L^\top]$ ; it vanishes under the stated hypothesis, but not automatically when the gain estimate  $\hat{a}$  is computed from the same noisy record that carries  $\xi$  (same-record dependence). (Proof: Appendix F.)*

The Poisson plug-in  $\Sigma_\xi = \text{diag}(\tau_{G,n}^2 + b_n^2/\Phi_n)$  is exact at all photon counts *after the post-calibration substitution*  $\hat{a}_n \rightarrow a_n$ ; before that substitution both noise variances carry the factor  $(a_n/\hat{a}_n)^2$  (Appendix F). With that qualifier, the identity — not an approximation to it — is what Fig. 4

tests. One metric convention matters when reading measured numbers: the identity is stated for the raw error  $\mathbb{E}\|\hat{T}-T\|^2/S_2$ , whereas the  $\text{relMSE}$  *measured* in Figs. 4–5 is computed after least-squares scale alignment of  $\hat{T}$  to  $T$  (the gauge convention of Sec. 2); alignment can pull  $\text{relMSE}/v$  below 1 for DC-dominated objects by absorbing the gain component co-linear with the object (Sec. 9). The two roles are kept separate in what follows: the *raw* (un-aligned) identity is verified directly by the injected coherent-residual experiment of Sec. 9 (E4, supplementary panels), while Fig. 4 reads the scale-aligned image metric. Specializing  $L$ :

- **Orthogonal / SRHT full inversion:**  $B_L = 1$ , so  $\text{relMSE} = v + (1/S_2) \sum_n (\tau_{G,n}^2 + b_n^2/\Phi_n)$ . The imager passes residual gain through at unit leverage — no amplification, no dilution.
- **Pairwise (differenced) Hadamard:** the gain term is  $R_{\text{pair}} = (K_{\text{eff}}/4) D_H(T) \text{Var}(\Delta)$ , with  $D_H \approx 1$  for non-DC coefficients and  $\text{Var}(\Delta) \approx 2s^2 r(\rho_{\text{pair}})$  the adjacent-pair gain increment, where  $r(\cdot)$  denotes the adjacent-pair decorrelation function of the drift —  $r(\rho) = 1 - e^{-\rho}$  for the OU drift of Sec. 9, with  $r(\rho) \approx \rho$  for small  $\rho$ . Differencing converts *slowness* of the drift directly into suppression.
- **Random / DGI:**  $\text{relMSE}_{\text{DGI}} = C_0(\text{pipeline})/N + (1 + C_0(\text{pipeline}))v/N + \text{noise terms}$ , where  $C_0(\text{pipeline}) = \mathbb{E}\|Z\|^2/S_2 - 1$  denotes the exact one-sample constant of the correlator ( $C_0 = K + \beta_4 - 2$  for symmetric zero-mean patterns). There is **no universal**  $C_0$  — hence the argument: it depends on the pattern statistics, background, and normalization, and must be measured from the pipeline at hand — for our SCGI-style testbed [7], [8] the measured value is used throughout Figs. 4 and 5.

This yields an honest corollary that the identifiability results alone would obscure: **at equal residual variance  $v$ , orthogonal inversion is the best conduit** — it has  $B_L = 1$  and no  $C_0(\text{pipeline})/N$  estimator floor. The randomized bases do not win because they transmit gain error more gracefully; they win because Theorem B and the window estimator of Sec. 6 let  $v$  *itself* be driven down blindly, at rate  $1/W$  per window, whereas an ordered scan offers no blind handle on  $v$  at all.

The competition between these closed forms produces a finite- $N$  flip boundary.

**Prop 3 (finite- $N$  flip boundary).**  $\rho^* = \inf\{\rho_{\text{pair}} : R_{\text{pair}}(\rho_{\text{pair}}) \geq R_{\text{rand}}(\rho_{\text{pair}}, N)\}$  satisfies, in the small-drift regime,

$$r(\rho^*) = \frac{2}{K_{\text{eff}} D_H s^2} [C_0(\text{pipeline})/N + (1 + C_0(\text{pipeline}))v_{\text{blind}}/N + R_{\text{DGI,noise}} - R_{\text{pair,noise}}],$$

whose leading order ( $r(\rho) = \rho$ ,  $D_H = 1$ , negligible noise and  $v_{\text{blind}}$ ) is the heuristic  $\rho^* \approx 2C_0(\text{pipeline})/(NK_{\text{eff}}s^2)$ . (Proof: Appendix F.)

**Remark (two decorrelation conventions — do not conflate).** Throughout,  $\rho_{\text{pair}}$  denotes *adjacent-pair* gain decorrelation, the variable entering  $\text{Var}(\Delta)$  and Prop 3;  $\rho_{\text{bw}}$  denotes the normalized gain *bandwidth*,  $p \approx \rho_{\text{bw}}N$ , the variable entering

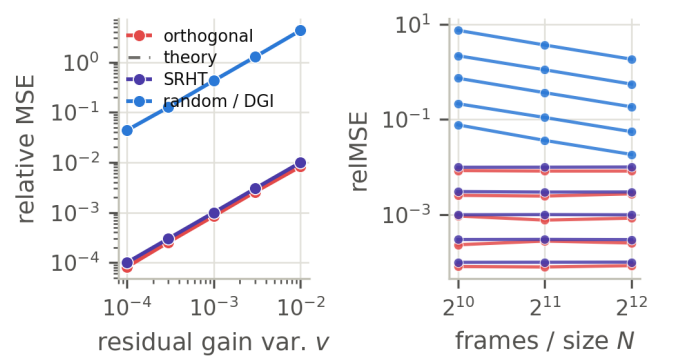


Fig. 4. Reconstruction bridge (Theorem 1 tested as an identity, on a balanced 8100-row design: three reconstructors,  $N \in \{1024, 2048, 4096\}$ , six injected residual-gain variances  $v$ , ten objects, fifteen seeds). Predicted versus measured scale-aligned  $\text{relMSE}$ . The deterministic inverses are flat in  $N$ , as  $B_L = 1$  demands: per- $v$  log-log slopes have maximal magnitude 0.074 (orthogonal) and 0.014 (SRHT), so there is no  $1/N$  reconstruction floor. The random/DGI arm decays as  $1/N$  at every  $v$  (per- $v$  slopes  $-1.031$  to  $-0.974$ ), with measured leverage  $\times N$  between  $7.38 \times 10^5$  and  $7.68 \times 10^5$  across the  $N$  sweep (a 4% drift). For the orthogonal inverse  $\text{relMSE}/v \approx 1.0$  for 8 of 10 objects; the two DC-dominated objects fall below the  $v$  line because the scale alignment absorbs the gain component co-linear with the object, an alignment effect rather than a failure of the identity.

the tall-design thresholds of Sec. 4 ( $N \gtrsim K/(1 - \rho_{\text{bw}})$ ). Both grow with drift speed but are not equal, and every axis in Fig. 5 is labeled by which one it carries.

Figure 4 shows the bridge validated term by term: predicted bias, gain, and noise contributions against measured  $\text{relMSE}$  across bases, drift speeds, and photon budgets — numerical evidence, on simulation, that Theorem 1 is an accounting identity rather than a bound. Figure 5 assembles the payoff: the phase diagram over (basis,  $\rho_{\text{pair}}$ , noise level  $\sigma$ ) with the  $\rho^*$  curve overlaid. Three regimes emerge. For **slow drift** ( $\rho_{\text{pair}} < \rho^*$ ), pairwise Hadamard is fine: differencing suppresses what little the gain moves, and orthogonal conditioning is kept for free. For **intermediate drift**, randomized carriers win — the stationarity anchor buys a small  $v_{\text{blind}}$  that no ordered scan can reach; the above-floor winner is the SRHT-paired pipeline, which keeps orthogonal conditioning, not bare i.i.d.-random/DGI, which pays its  $C_0(\text{pipeline})/N$  estimator floor in absolute error. For **fast drift** ( $\rho_{\text{bw}} \rightarrow 1$ ,  $p$  approaching  $N$ ), every *blind algebraic* strategy hits the noise floor together — consistent with the information-loss intuition behind Theorem A’s counting; this is a statement about blind separation, not about the physics — an externally calibrated or metered gain restores the problem to ordinary inversion. The chart is a pre-acquisition instrument: locate your drift spectrum and photon budget on it, and the go/no-go decision — and the basis — follow before the first frame is taken.

The measured constants and conventions behind Fig. 5 are collected in Table 1.

## IX. NUMERICAL EXPERIMENTS

The theory of Secs. 4–8 makes quantitative predictions — a stationarity dichotomy, a  $W^{-1/2}$  estimation law whose object

TABLE I

TABLE I (MEASURED CONSTANTS AND CONVENTIONS BEHIND FIG. 5). EVERY CONSTANT IS PIPELINE-SPECIFIC (NO UNIVERSAL VALUE EXISTS); CONFIDENCE INTERVALS BELOW USE 20 DRIFT SEEDS, BOOTSTRAPPED OVER OBJECT $\times$ SEED CELLS, WITH SLOPE FITS RESTRICTED TO  $W \leq 16$  ( $n = 200$  FITS/ARM).

| Quantity                                                                                                                         | Value in this pipeline                                                                                                                                                                                                                                                                                                                                                                                                        | Uncertainty                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| $C_0$ (pipeline), measured via the DGI leverage law (leverage $\times N = 1 + C_0$ , i.e. $C_0 = \text{leverage} \times N - 1$ ) | $C_0 = \text{leverage} \times N - 1 \approx 7.4\text{--}7.7 \times 10^5$ (leverage $\times N \in [7.38 \times 10^5, 7.68 \times 10^5]$ across the $N$ sweep, 4% drift)                                                                                                                                                                                                                                                        | not a resampled statistic (deterministic across the $N$ -sweep design); no bootstrap CI computed    |
| $D_H(T)$ (pairwise-Hadamard factor)                                                                                              | measured over the ten-object panel: min/median/max = 0.971/0.988/0.999 (per object via the analytic identity $D_H(T) = \frac{1}{K} \sum_k (1 - x_k^2)^2$ , $x_k = c_k/S_1$ )                                                                                                                                                                                                                                                  | deterministic per object (identity evaluated on the panel); no resampling needed                    |
| $K_{\text{eff}}$ (object-panel range)                                                                                            | 59.8–864.3                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                   |
| $v_{\text{blind}}$ (residual gain variance after blind correction)                                                               | best-window blind gain error at $\rho_{\text{pair}} = 10^{-3}$ , $s = 0.1$ : srht_paired 0.0074 [0.0073, 0.0076]; hadamard_random_paired 0.0071 [0.0070, 0.0073]; random_binary 0.0148 [0.0144, 0.0151]; $v_{\text{blind}} = \text{Var}(\delta)$ is estimated by the squared best-window scale-aligned gain error (an estimator convention, not an identity): (gain error) $^2 \approx 5 \times 10^{-5}$ – $2 \times 10^{-4}$ | 95% CIs shown inline (20-seed bootstrap; see caption)                                               |
| Drift / noise / flux conventions                                                                                                 | OU log-gain, $\rho \in [10^{-3}, 10]$ , drift s.d. 0.030–0.104; noise $\sigma \in [0.05, 0.5]$ ; equal frames and equal total flux per arm                                                                                                                                                                                                                                                                                    | —                                                                                                   |
| Above-floor gate                                                                                                                 | relMSE $< 0.5$ ; 28 of 44 cells above floor                                                                                                                                                                                                                                                                                                                                                                                   | cell count, not resampled; no bootstrap CI computed                                                 |
| Flip-boundary fit to Prop 3                                                                                                      | $R^2 = 0.989\text{--}0.9995$                                                                                                                                                                                                                                                                                                                                                                                                  | boundary fits reported with $R^2$ and point counts (Fig. 5); survival-style CIs left to future work |

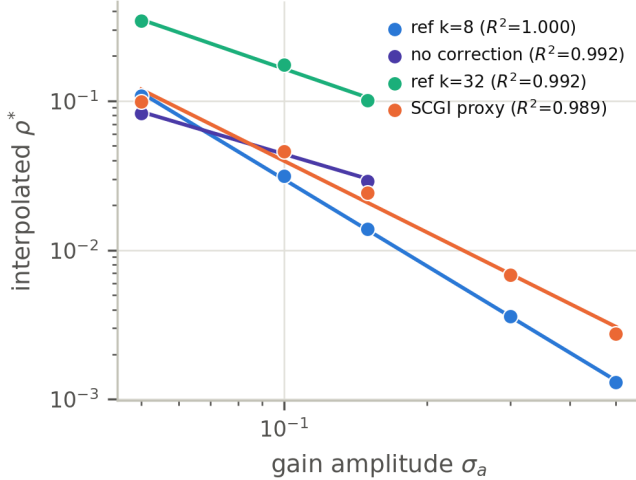


Fig. 5. Finite- $N$  flip boundary / phase diagram over (basis,  $\rho_{\text{pair}}$ , noise level  $\sigma$ ) with the  $\rho^*$  curve of Prop 3 overlaid. Interpolated  $\rho^*$ , the adjacent-pair decorrelation at which randomized carriers overtake pairwise Hadamard, is plotted against gain amplitude  $\sigma_a$  for the SRHT-paired arm under several correction references, with boundary fits reaching  $R^2 = 0.989\text{--}0.9995$  against the closed form of Prop 3. SRHT-paired acquisition with pairwise correction wins all 28 above-floor cells of the 44-cell winner map (above-floor gate relMSE  $< 0.5$ ; the remaining 16 cells are labeled noise floor). Three regimes: slow drift ( $\rho_{\text{pair}} < \rho^*$ ) favors pairwise Hadamard; intermediate drift favors randomized (SRHT-paired) carriers; fast drift ( $\rho_{\text{bw}} \rightarrow 1$ ,  $p$  approaching  $N$ ) drives every blind strategy to the noise floor together.

dependence is confined to  $K_{\text{eff}}$ , an exact relMSE identity, a flip boundary, and a photon-limited Fisher scaling — and this section tests each of them in a series of simulations. A closing

subsection integrates five verification experiments completed after the main panel: the tall-design rank threshold (Fig. 8), permutation-whitening power, the coherent-residual validation of Theorem 1, the  $C_0$  audit, and oracle-to-blind baselines (supplementary panels).

**Protocol.** All results are simulations; no hardware data enter, consistent with the scope of Sec. 1. Comparisons are made under matched acquisition budgets — equal frame counts and equal total flux for every arm — so that no basis is granted extra photons or extra rows. The drift is an Ornstein–Uhlenbeck log-gain trace with per-frame decorrelation rate  $\rho$ , the knob that sets the adjacent-pair decorrelation  $\rho_{\text{pair}}$  of Sec. 8; within each seed a single trace is shared by every measurement arm, so the bases face the identical nuisance realization rather than their own luck, while the traces differ across seeds (drift standard deviation 0.030–0.104). Blind correction uses a single estimator for all arms — the sliding-window mean automatic gain control (AGC) realization of the proposed windowed estimator of Sec. 6 — with no per-arm tuning and no denoising or post-processing anywhere in the pipeline. The object panel holds ten objects spanning  $K_{\text{eff}} = S_1^2/S_2$  from 60 to 864, each run at five drift seeds (fifteen for the reconstruction bridge), with errors computed against ground truth on the object support mask after the least-squares scale alignment of Sec. 8. Concretely: the object panel comprises ten shared objects from a fixed generator — two synthetic digits, the letters A and L, a vertical stripe (width  $\approx \text{image}/8$ ), an annulus (radii  $\approx 0.22/0.34$  of the image span), and four procedural natural patches — each normalized to  $[0, 1]$  and spanning  $K_{\text{eff}} = 59.8\text{--}864.3$ ; every experiment below shares base seed 20240708. Per-figure enumerables:

- **Fig. 2 (carrier stationarity):** 8192 frames per arm at  $64 \times$

64, split into 8 chunks with a 128-frame transient cutoff, AGC window  $W = 64$ , five arms.

- **Fig. 3 (blind gain identifiability):**  $32 \times 32$  objects ( $K = 1024$ ), 2048 frames, seven arms,  $\rho_{\text{pair}} \in \{10^{-3}, 10^{-2}\}$ , drift scale  $s = 0.1$ ,  $W$  swept over a log-spaced grid from 4 to 1024; the final uncertainty replication (Sec. 9.2, Table 1) uses 20 drift seeds per (object,  $\rho$ ) pair — 24,400 rows — while the  $R^2$  exploration above used 5 seeds.
- **Fig. 4 (reconstruction bridge):** injected  $v \in \{0, 10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$ ,  $N \in \{1024, 2048, 4096\}$ , 15 seeds, three reconstruction arms.
- **Fig. 7 (low-photon robustness):** mean photon budgets  $\bar{\lambda} \in \{0.25, \dots, 128\}$  on a  $\times 2$  grid, soft-log  $\alpha = 0.5$ ,  $W = 64$ ,  $\rho_{\text{pair}} = 10^{-3}$ ,  $s = 0.1$ , 5 seeds, with all three estimators compared on the same paired Poisson draw at each budget.
- **Fig. 8 + supplementary panels (threshold, whitening power, coherent residuals,  $C_0$ , baselines):** rank/solver threshold scan at  $K = 64$  (20 seeds, 1300 cells) and replicated at  $K = 128$  (30 seeds, 1950 cells),  $p \in \{3, 5, 9, 17, 33\}$ , margins  $N - K - p \in [-8, +16]$  (lower end truncated at  $N > K$  for small  $p$ ); whitening power with 32 pixel permutations  $\times$  10 objects under the Fig. 2 protocol; E4 coherent-residual injection with five covariance structures at  $v = 10^{-3}$ ,  $K = N = 1024$ ;  $C_0$  audit with four pattern ensembles  $\times$  five objects,  $2 \times 10^4$  single-frame draws each; oracle/metered/blind bars on four objects  $\times$  five gain treatments  $\times$  five seeds.
- **Pattern conventions:** i.i.d. uniform  $[0, 1]$ ; i.i.d. binary Bernoulli(0.5); paired Hadamard as complementary  $(1 \pm H)/2$  frames (2 frames per coefficient); SRHT as  $H \cdot \text{diag}(\pm 1)$  with an optional row permutation, paired the same way; raw signed Walsh arms are diagnostic-only (Sec. 5 bucket convention (i)) and not a valid physical estimator without offset or pairing.
- **Metrics:** blind gain error is the scale-aligned relative error (both the estimated and true gain traces normalized to unit mean, with the optimal scalar alignment of Sec. 8); mask-based contrast uses GT-mask CNR at threshold 0.5; reconstruction error is the scale-aligned relMSE of Sec. 8. The OU drift trace is generated with `normalize_mean=1`.

**Carrier stationarity (Fig. 2).** This experiment applies a *variance-stationarity diagnostic* to the noiseless carrier  $\{B_n\}$ , before any gain is applied, at 8192 frames per arm on  $64 \times 64$  objects; the Brown–Forsythe test checks chunk-variance equality — a consequence of the stationarity that  $(*)$  requires, not a direct test of  $(*)$ ’s windowed log-carrier means. As shown in Fig. 2, the Brown–Forsythe variance-stationarity test at  $p < 10^{-3}$  rejects 7 of 10 objects for ordered Hadamard, with  $p$  reaching  $10^{-20}$ , against 0 of 10 for both i.i.d. random ensembles. (These are uncorrected per-test  $p$ -values; under a multiplicity correction across the object  $\times$  arm panel — e.g. Bonferroni — the strongest rejections clear the adjusted threshold by many orders of magnitude, while rejections near the  $p < 10^{-3}$  gate should be read with that correction in mind.) The permuted arms (randomly permuted paired Hadamard and SRHT) each reject 2 of 10: a *fixed* permutation does not Walsh-flatten every object’s  $T^2$  spectrum

— the permutation whitening of Sec. 7 is a probabilistic statement over permutations, and an object with concentrated spectral energy can alias against a particular draw. This residue is the theory’s own prediction, not an anomaly. The Parseval invariant closes the loop: every exhaustively paired arm has carrier  $\text{CV}_B = 1/\sqrt{K_{\text{eff}}}$  to a relative accuracy of about  $10^{-6}$ , and the random arms match  $(\sigma_I/\mu_I)/\sqrt{K_{\text{eff}}}$  within  $\pm 2\%$  — so the ordered pathology lives entirely in the temporal variance structure, not in the marginal spread of the carrier. The rejection proportions are descriptive counts over this ten-object panel (no resampled intervals).

**Blind gain identifiability (Fig. 3).** This is the core evidence for Theorem B, run over seven arms including the raw, non-paired Hadamard arms that feed bare signed Walsh coefficients to the estimator — convention (i) of the bucket conventions of Sec. 5, a *diagnostic-only* configuration that is not a valid physical log-domain estimator without offset or pairing. As shown in Fig. 3, the chronology failure of Theorem A is made visible: the raw ordered arm’s blind gain error is  $\approx 0.96$  and does not decay with the window (0.95 at  $W = 4$ , still 0.90 at  $W = 512$ ). Shuffling the raw frames improves this to  $\approx 0.86$  — permutation helps even here — but the arm remains roughly  $40\times$  worse than the random and paired arms at  $\approx 0.02$ , because without pairing the near-zero-mean carrier leaves the AGC ill-posed. Within the variance-obeying arms, the slow-drift variance-segment slopes of  $\log(\text{err})$  against  $\log W$  are  $-0.39$  (random binary),  $-0.60$  (SRHT), and  $-0.61$  (randomly permuted paired Hadamard) — all inside the  $W^{-1/2}$  band — while naturally ordered paired Hadamard is shallow at  $-0.24$ : differencing cancels the drift within pairs but leaves the ordered envelope. Normalizing by the theory floor  $\sqrt{\text{CV}_B^2/W}$  collapses the panel: the paired arms sit at 0.40–0.41 and the random arms at 0.89–0.90, each bundle collapsing across all ten objects to within  $\pm 5\text{--}7\%$ , and the best-window error correlates with  $K_{\text{eff}}$  at  $-0.86$  to  $-0.90$  (descriptive correlations over the ten-object panel; no resampled intervals) — consistent, within this tested panel, with the object entering through  $K_{\text{eff}}$  alone, as Proposition 1 predicts to leading Gaussian/log-variance order. The shuffled-paired arm performing like the random arms bears out the permutation analysis of Sec. 7 end to end on this panel: pairing cancels the slow drift within pairs, and the permutation restores the stationarity that the ordering destroyed.

The 20-seed uncertainty replication (24,400 rows total) confirms these slopes under the fixed  $W \leq 16$  variance-segment convention: median log–log slope with 95% bootstrap CI is  $-0.41$   $[-0.42, -0.41]$  for random binary,  $-0.39$   $[-0.40, -0.37]$  for random uniform,  $-0.58$   $[-0.60, -0.54]$  for SRHT-paired, and  $-0.62$   $[-0.64, -0.61]$  for randomly permuted paired Hadamard, while the naturally ordered paired arm sits at  $+0.03$   $[-0.01, +0.04]$  — a CI straddling zero, i.e. no detectable decay — and the raw ordered arm’s error is 0.958  $[0.956, 0.959]$ , essentially flat. The corresponding best-window blind gain errors at this replication’s operating point ( $\rho_{\text{pair}} = 10^{-3}$ ,  $s = 0.1$ ) are reported in Table 1. It is worth noting that this replication’s slope fits use the fixed  $W \leq 16$  variance-segment convention, whereas the exploratory slopes

and  $R^2$  figures above used an adaptive segment; the two conventions agree on every qualitative conclusion.

**The reconstruction bridge (Fig. 4).** This experiment tests Theorem 1 as an identity rather than a bound, on a fully balanced 8100-row design: three reconstructors, three transform sizes  $N \in \{1024, 2048, 4096\}$ , six injected residual-gain variances  $v$ , ten objects, fifteen seeds. It can be observed that the deterministic inverses are flat in  $N$ , as  $B_L = 1$  demands: the per- $v$  log-log slopes have maximal magnitude 0.074 (orthogonal) and 0.014 (SRHT) — no  $1/N$  reconstruction floor. The random/DGI arm decays as  $1/N$  at every  $v$  (per- $v$  slopes  $-1.031$  to  $-0.974$ ), and its leverage law is quantitative: the measured relMSE agrees with leverage  $\times v$  within  $[0.91, 1.06]$ , with leverage  $\times N$  between  $7.38 \times 10^5$  and  $7.68 \times 10^5$  across the  $N$  sweep (a 4% drift). For the orthogonal inverse, relMSE/ $v \approx 1.0$  for 8 of 10 objects; the two DC-dominated objects — natural\_patch at 0.35 and stripe at 0.54 — fall below the  $v$  line because the scale alignment of the metric absorbs the gain component co-linear with the object, an alignment effect rather than a failure of the identity. Fig. 4’s numbers are accordingly read in the *scale-aligned image metric*; the raw-MSE identity-verification role is carried by the injected-residual E4 comparison below.

**Phase diagram and ablation (Figs. 5 and 6).** The phase diagram sweeps the drift rate  $\rho \in [10^{-3}, 10]$  and noise level  $\sigma \in [0.05, 0.5]$  at equal frames, with an above-floor gate relMSE  $< 0.5$  separating cells where reconstruction is meaningful from cells at the noise floor — the gate excludes reconstructions whose error energy is at least half the object energy, a conservative meaningfulness cut; cells failing it are reported as noise floor rather than assigned a winner. It is numerically demonstrated that SRHT-paired acquisition with pairwise correction wins all 28 above-floor cells of the 44-cell winner map; the remaining 16 cells are honestly labeled noise floor rather than assigned a winner, and the flip-boundary fits reach  $R^2 = 0.989\text{--}0.9995$  against the closed form of Prop 3 (winner counts and boundary locations are descriptive results of the specified grid; survival-style intervals are left to future work, Table 1). The ablation isolates the mechanism at the single above-floor drift point  $\rho = 10^{-3}$ : SRHT gains +5.45 dB over ordered Hadamard, while sign randomization alone gains +5.47 dB and permutation alone +5.39 dB — each ingredient by itself recovers essentially the full advantage, and combining them is not additive; these dB gains are descriptive values at this single grid point (no resampled intervals). What matters is that the chronology is randomized, not the particular form of the randomization. At the largest simulated adjacent-pair decorrelation rates ( $\rho \geq 1$ ), all blind variants empirically approach the noise floor within  $\pm 0.09$  dB; this is consistent with the information-loss intuition behind the  $p \rightarrow N$  algebraic limit of Theorem A’, but it is not a direct test of the low-pass dimension threshold unless gain bandwidth ( $\rho_{\text{bw}}$ ) and adjacent-pair decorrelation ( $\rho_{\text{pair}}$ ) are controlled separately.

**Low-photon robustness (Fig. 7).** This experiment tests the soft-log analysis of Sec. 6 under Poisson counting, on identical random-basis carriers and shared gain traces. As shown in Fig. 7, the soft-log stays finite at every photon budget and, at mean counts  $\bar{\lambda} \leq 1$ , outperforms the naive clipped log-

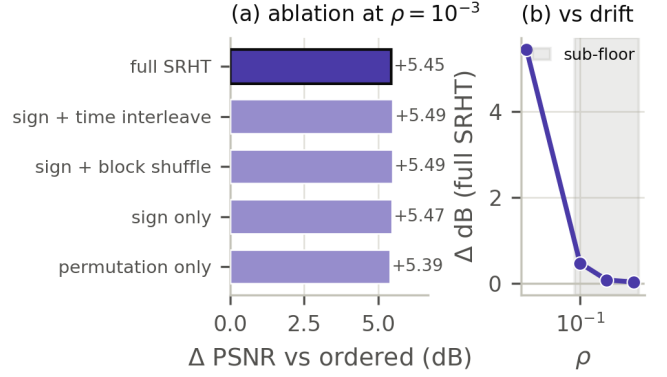


Fig. 6. Ablation isolating the randomization mechanism at the single above-floor drift point  $\rho = 10^{-3}$  (none / permutation / sign / both). SRHT gains +5.45 dB over ordered Hadamard, while sign randomization alone gains +5.47 dB and permutation alone +5.39 dB — each ingredient by itself recovers essentially the full advantage, and combining them is not additive. What matters is that the chronology is randomized, not the particular form of the randomization. Descriptive values at this single grid point (no resampled intervals).

arithm by  $23\text{--}46\times$  (the Anscombe transform, separately, by  $19.5\text{--}30\times$ ). In the variance-limited regime the Fisher-matched  $1/(W\bar{\lambda})$  law holds: the fitted log-log slope is  $-0.910$  over  $\bar{\lambda} \in [2, 32]$ , shallowing to  $-0.730$  over  $[1, 16]$  as the fit window approaches the  $\bar{\lambda} \sim 1$  bias-to-variance transition. Below  $\bar{\lambda} \sim 1$  the estimator is shrinkage-bias-dominated, and its MSE sits below the unbiased (local) Fisher reference — because it is biased, not because it beats the bound, exactly the caveat of Sec. 6. (A per-budget decomposition of that MSE into bias and variance contributions below the crossover requires tracking the windowed estimate against the known gain trace per draw and is deferred to a follow-up run; the sub-Fisher MSE together with the calibration-curve shrinkage of Sec. 6 already identifies the regime as bias-dominated.) At the high-photon end the floor is drift-limited rather than photon-limited: the gain MSE of  $2.78 \times 10^{-4}$  at  $\rho = 10^{-3}$  falls to  $1.73 \times 10^{-4}$  when the drift slows to  $\rho = 10^{-4}$ .

**Threshold, whitening-power, and accountability verifications (Fig. 8 and supplementary panels).** A series of five further experiments closes the remaining theory-evidence gaps. Unless a bootstrap interval is stated, every number in this subsection is a descriptive result of its specified grid, with no resampled intervals.

**Tall-design threshold scan (Fig. 8).** A direct test of Theorem A’'s counting at two object sizes:  $K = 64$  (20 seeds per cell, 1300 cells) and  $K = 128$  (30 seeds per cell, 1950 cells), each over five low-pass drift dimensions  $p \in \{3, 5, 9, 17, 33\}$  with frame counts swept through the wall at margins  $N - K - p \in [-8, +16]$ . As shown in Fig. 8, at both sizes the local rank test on the collision Jacobian transitions from 0% to 100% pass exactly at margin  $N - K - p = -1$  for every  $p$  (3250 cells, no exceptions) — verifying the local differential rank threshold  $N = K + p - 1$  of Theorem A’; being a rank statement, it verifies the local level and does not by itself test generic-exact uniqueness at  $N = K + p$ . The

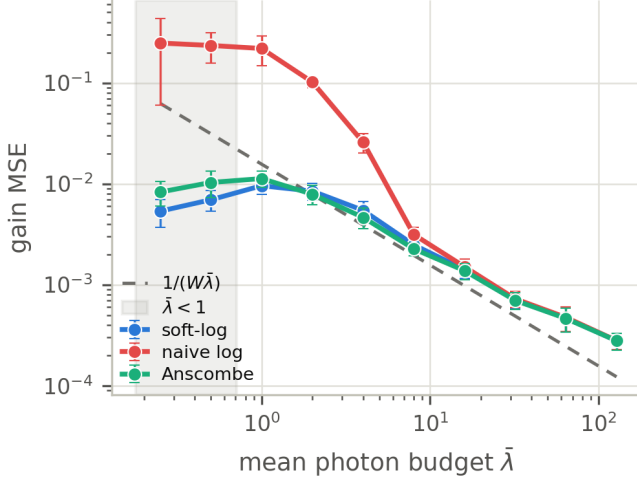


Fig. 7. Low-photon robustness under Poisson counting, on identical random-basis carriers and shared gain traces (soft-log  $\alpha = 0.5$ ,  $W = 64$ ,  $\rho_{\text{pair}} = 10^{-3}$ ,  $s = 0.1$ ). Gain MSE versus mean photon budget  $\bar{\lambda}$ . The soft-log stays finite at every budget and, at  $\bar{\lambda} \leq 1$ , outperforms the naive clipped logarithm by  $23\text{--}46\times$  (and the Anscombe transform by  $19.5\text{--}30\times$ ). In the variance-limited regime the Fisher-matched  $1/(W\bar{\lambda})$  law holds: fitted log-log slope  $-0.910$  over  $\bar{\lambda} \in [2, 32]$ , shallowing to  $-0.730$  over  $[1, 16]$ . Below  $\bar{\lambda} \sim 1$  the estimator is shrinkage-bias-dominated and its MSE sits below the unbiased (local) Fisher reference. At the high-photon end the floor is drift-limited: gain MSE  $2.78 \times 10^{-4}$  at  $\rho = 10^{-3}$  falls to  $1.73 \times 10^{-4}$  at  $\rho = 10^{-4}$ .

companion solver panel (blind alternating recovery, 8 restarts  $\times$  150 iterations) starts at the same wall but climbs slowly — about 2% success at margin 0 and only  $\sim 50\%$  at margin +16 for  $K = 64$ , slower still at  $K = 128$  ( $\sim 41\%$  at margin +16) — an *algorithmic-recovery* observation, not a uniqueness test: uniqueness and recovery separate exactly as the third notion of Sec. 2 anticipates, with the gap widening as the dimension grows (see also Sec. 10). This exact transition is the empirical support cited for the low-pass three-level hierarchy of Sec. 4 and Appendix B.

*Multi-permutation whitening power (supplementary panels).* Theorem E.3’s permutation whitening is probabilistic — over permutations, not per draw — and this experiment measures that distinction under the Fig. 2 protocol: 32 random pixel permutations  $\times$  10 objects, i.e. 320 permuted draws. It can be observed that the Brown–Forsythe rejection rate at  $p < 10^{-3}$  is 5.9% over the 320 permuted draws, against 70% (7 of 10 objects) for the ordered reference arm. The natural per-draw certificate candidate fails: the maximum non-DC Walsh coefficient of the permuted  $T^2$  does *not* predict which draws reject (Spearman 0.092,  $p = 0.10$ ) — an honest null, consistent with the probabilistic (not per-draw-certifiable) nature of permutation whitening (Sec. 7).

*Coherent-residual validation of Theorem 1 (E4, supplementary panels).* Five injected residual-covariance structures — i.i.d., AR(1), sinusoidal, low-pass OU, blockwise — at fixed  $v = 10^{-3}$ , across three reconstructors and ten objects (150 cells), measured in the raw-residual (identity-verification) metric rather than the scale-aligned image metric of Fig. 4. The full matrix law of Theorem 1 tracks the measured relMSE

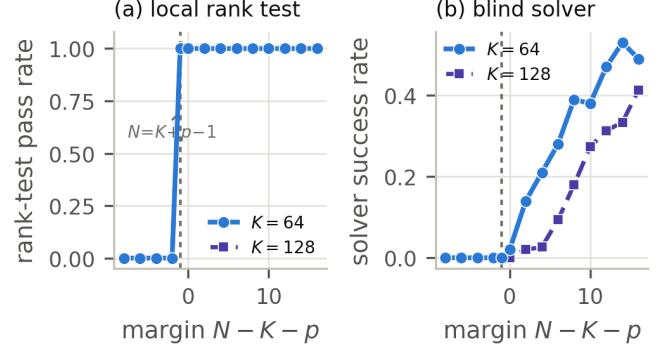


Fig. 8. Tall-design threshold scan (a direct test of Theorem A’*s* counting) at  $K = 64$  (20 seeds/cell, 1300 cells) and  $K = 128$  (30 seeds/cell, 1950 cells), over five low-pass drift dimensions  $p \in \{3, 5, 9, 17, 33\}$  with margins  $N - K - p \in [-8, +16]$ , all  $p$  pooled. (a) The local rank test on the collision Jacobian transitions from 0% to 100% pass *exactly* at margin  $N - K - p = -1$  for every  $p$  at both sizes (3250 cells, no exceptions), verifying the local differential threshold  $N = K + p - 1$ . (b) The companion blind alternating solver (8 restarts  $\times$  150 iterations) starts at the same wall but climbs slowly — about 2% success at margin 0 and only  $\sim 50\%$  at margin +16 for  $K = 64$  ( $\sim 41\%$  for  $K = 128$ ) — an *algorithmic-recovery* observation, not a uniqueness test: uniqueness and recovery separate as Sec. 2’s third notion anticipates.

within 4.2% across all 150 cells. The scalar  $v B_L$  law is exact for the complete orthogonal and SRHT inversions — for these,  $LL^T \propto I/N$  analytically, so the gain term feels only  $\text{diag}(V_\delta)$  — but fails by up to  $3.65\times$  for random/DGI under low-pass-coherent residuals: a direct verification of Theorem 1’s coherent clause. It is worth noting that E4 validates the matrix law under *injected* residuals; it does not probe the same-record  $\delta\text{--}\xi$  dependence flagged after Theorem 1.

*C<sub>0</sub> audit (supplementary panels).* The one-sample constant measured from  $2 \times 10^4$  pattern draws per ensemble agrees with the moment formula of Appendix F to 0.35% mean (2.01% worst) relative error across four pattern ensembles  $\times$  five objects. The strong ensemble dependence —  $O(K)$  for signed Rademacher patterns versus  $O(K_{\text{eff}}K)$  for the nonnegative offset ensembles — confirms that  $C_0$  is pipeline-specific, as Theorem 1’s DGI specialization insists.

*Oracle, metered, and blind baselines (supplementary panels).* On four held-out objects with OU drift ( $\rho = 10^{-3}$ ,  $s = 0.1$ ), the accountability ordering static  $\geq$  oracle  $\geq$  blind-AGC  $\geq$  no-correction holds in CNR for 4 of 4 objects, with an oracle–blind CNR gap of 0.035–0.147: at this drift point the blind windowed estimator concedes little to a gain oracle, while forgoing correction altogether costs the most.

## X. DISCUSSION: SIGNIFICANCE, SCOPE, AND OUTLOOK

The results of Sections 2–9 are, at bottom, a set of pre-acquisition design tools: quantities one can evaluate on the illumination basis and the expected drift *before* photons are collected, and which predict whether blind gain–object separation will succeed. They are collected here as four deliverables, followed by their limits and the experimental program they invite.

**A go/no-go identifiability predictor.** The tall-design thresholds convert “will this acquisition self-calibrate?” into arithmetic. For a physical low-pass drift class  $S$  of dimension  $p = \dim S$ , local differential identifiability holds at  $N \geq K + p - 1$ , generic exact finite-sample identifiability at  $N \geq K + p$ , and uniform (every-object) identifiability at  $N \geq 2K + p - 1$  (Theorem A' — proven at generic-pair level for the physical low-pass drift family, with the uniform-over-drift reading resting on a labeled dimension-count argument plus the exact empirical rank transition of Fig. 8). Written through the bandwidth relation  $p \approx \rho_{\text{bw}}N$ , the generic threshold becomes  $N \gtrsim K/(1 - \rho_{\text{bw}})$ : slow drift needs only modest oversampling, while  $\rho_{\text{bw}} \rightarrow 1$  ( $p \rightarrow N$ ) makes blind algebraic separation impossible for any  $N$ , forcing reliance on the statistical anchor of Theorem B — which itself degrades once the gain decorrelates within the averaging window (the  $\rho \geq 1$  convergence of Fig. 6). This is a regime map the experimenter reads off before building.

**An SRHT design rule.** When blind slow gain is the operative nuisance, do not run ordered Hadamard: its chronological coefficient ordering hands the object a temporal signature indistinguishable from drift. Randomize the carrier instead — via sign or pixel-permutation randomization of a Walsh-Hadamard design. The two ingredients play distinct roles and should not be compressed into one guarantee: the random *permutation* makes the temporal carrier exchangeable — the anchor mechanism of Sec. 5 — while the covariance-whitening guarantee is the *sign-randomization* statement applied after the permutation: sign flips whiten if and only if the realized (permuted)  $T^2$  is Walsh-flat (Sec. 7), an event the permutation makes likely once  $\min(\varepsilon^2 K_4, \varepsilon K_\infty) \geq C \log(K/\delta)$  (Fig. 6). Retain a positivity/offset margin, and choose the gain window  $W$  from the bias–variance tradeoff of the proposed windowed estimator. Under those checks, the proposed acquisition rule buys statistical self-calibration without surrendering orthogonal conditioning.

**A phase diagram as a falsifiable design map.** The master finite-noise identity (Theorem 1) and the finite- $N$  flip boundary  $\rho^*$  (Prop 3) render relMSE across DGI, SRHT, and pairwise-Hadamard as a chart over basis, drift, and noise (Fig. 5). It is worth noting that  $\rho_{\text{pair}}$  (adjacent-pair decorrelation, entering  $\rho^*$ ) and  $\rho_{\text{bw}}$  (bandwidth, entering the tall-design law) are distinct axes and must be labeled as such before any overlay (Remark, Sec. 8). The heuristic  $\rho^* \approx 2C_0(\text{pipeline})/(NK_{\text{eff}}s^2)$  appears as the leading-order limit of the full boundary — a prediction the diagram exposes to falsification, not a fitted curve.

**A cross-domain principle.** The exportable statement is basis-agnostic: *to self-calibrate a multiplicative nuisance, design the carrier so its distribution is stationary in the nuisance coordinate*. The following are offered as analogues and motivation, not as proven domains — each would require its own operator analysis of the diagonal-row map [14], [13]: blind gain and phase calibration, flat-field correction, coded-aperture and computational microscopy, and single-pixel spectroscopy all share the bilinear structure in which this reasoning applies. **Scope and limits.** The claims are theory and simulation only, with no hardware validation. Uniqueness near  $N = K + p$  is an

identifiability statement, not a conditioning guarantee; stability there is a separate random-matrix question. Random square unconstrained designs are *not* algebraically identifiable — any invertible design absorbs nonconstant gain into a redefined object (Theorem A). There is no universal DGI constant  $C_0$  or Hadamard factor  $D_H$ ; both must be measured per pipeline. The low-photon soft-log  $\psi_\alpha$  avoids the log-zero singularity but cannot manufacture information — zero photons carry zero Fisher information for log-intensity, an information limit, not an estimator defect.

**Experimental feasibility.** The theory issues sharp, falsifiable predictions for a bench with programmable drift: a drift-rate flip point  $\rho^*$  at which randomized carriers overtake pairwise Hadamard, and an SRHT-versus-ordered-Hadamard separation in blind gain-estimation error under stirred-scattering gain (Fig. 3). Testing these against a physical single-pixel system — with real read noise, Poisson statistics, and unmodeled gain jumps — is left to future hardware work, likely in collaboration with an optical-imaging group.

In conclusion, we have established an identifiability theory of blind gain–object separation for single-pixel acquisition: the square-design obstruction and the tall-design thresholds (Theorems A and A'), the stationarity anchor with its minimax-rate-optimal windowed estimator (Theorems B–D), the SRHT synthesis (Sec. 7), and the finite-noise reconstruction bridge with its flip-boundary phase diagram (Theorem 1, Prop 3). Taken together, these tools make gain observable not by measuring it directly, but by making the object stop pretending to be time. The proposed framework opens an avenue for self-calibrating acquisition design in real-world scenarios, e.g., single-pixel imaging through dynamic and complex scattering media.

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Appendix A supports Sec. 4 (Theorem A and Corollaries 1–2); Appendix B supports Sec. 4 (Theorem A', low-pass lemmas); Appendix C supports Sec. 5 (Propositions 1–2, condition (★), Theorem B); Appendix D supports Sec. 6 (Theorems C–D, Fisher geometry, low-photon); Appendix E supports Sec. 7 (SRHT Walsh condition, permutation bounds); Appendix F supports Sec. 8 (Theorem 1, specializations, Prop 3). Heuristic or genericity steps are labeled as such inside each proof; every appendix ends its theorems with a “What this does not claim” paragraph.

## APPENDIX A SQUARE-DESIGN NON-IDENTIFIABILITY

This appendix proves Theorem A of Sec. 4 together with its two corollaries, in the notation of the main text: noiseless bucket record  $R_n = a_n B_n$ , carrier  $B_n = (MT)_n$ , log-gain  $\ell = \log a$  (entrywise), drift class  $S$  a linear subspace of  $\mathbb{R}^N$  with  $\text{span}\{\mathbf{1}\} \subseteq S$  and  $\dim S = p$ . This appendix is the archival form of these results under the manuscript labels Theorem A, Corollary 1, Corollary 2.

### A. A.0 Standing assumptions

- **(A1) Model.** Noiseless record  $R_n = a_n(MT)_n$ ,  $n = 1, \dots, N$ , with  $N = K$  and  $M \in \mathbb{R}^{K \times K}$  **invertible** — Hadamard, Fourier, random square, or otherwise. Write  $c = MT$  for the carrier coefficient vector, so  $R = e^\ell \odot c$  ( $\odot$  = entrywise product).
- **(A2) Gain class.**  $a_n > 0$  for all  $n$ ; the admissible log-gains form (or contain a neighborhood inside) the linear subspace  $S \ni \mathbf{1}$ ,  $\dim S = p \geq 2$ ; write  $p - 1 = \dim(S/\text{span}\{\mathbf{1}\})$  for the nonconstant drift dimension.
- **(A3) Object class.**  $T \in \mathbb{R}^K$  unconstrained (Theorem A, Corollary 1); constrained classes are treated in Corollary 2.
- **(A4) Carrier support.** For the clean dimension statements we assume full carrier support,  $c_n = (MT)_n \neq 0$  for all  $n$ ; this holds for generic  $T$  (each  $\{c_n = 0\}$  is a hyperplane). Zero buckets are discussed in Remark A.2.
- **(A5) Gauge.** The global-scale gauge orbit of a pair  $(\ell, T)$  is  $\mathcal{G}(\ell, T) = \{(\ell + t\mathbf{1}, e^{-t}T) : t \in \mathbb{R}\}$ ; identifiability is always meant modulo  $\mathcal{G}$ .

### B. A.1 Theorem A (square non-identifiability, any invertible design)

**Theorem A.** Assume (A1)–(A5). For every  $s \in S$  the pair

$$\ell' = \ell + s, \quad c' = e^{-s} \odot c, \quad T'(s) = M^{-1}(e^{-s} \odot MT)$$

reproduces the bucket record exactly:  $e^{\ell'_n} c'_n = R_n$  for all  $n$ . If  $s$  is nonconstant on  $\text{supp}(MT)$  — under (A4), simply nonconstant — then  $(\ell', T'(s)) \notin \mathcal{G}(\ell, T)$  and  $T'(s) \neq \gamma T$  for every  $\gamma \in \mathbb{R}$ . The set of admissible pairs consistent with the record therefore contains a real-analytic family of dimension  $\dim S = p$ , of which the gauge orbit accounts for only  $\dim(S \cap \text{span}\{\mathbf{1}\}) = 1$ ; the non-gauge ambiguity dimension is at least  $p - 1 \geq 1$ . In particular, gain–object separation from

a square design is not identifiable modulo gauge whenever  $S$  contains a nonconstant element.

*Proof. (Data equality.)*  $e^{\ell'_n} c'_n = e^{\ell_n + s_n} e^{-s_n} c_n = e^{\ell_n} c_n = R_n$ , and  $T'(s) = M^{-1}c'$  is well defined because  $M$  is invertible. Both steps are exact; no smallness of  $s$  is used.

**(Distinctness of the object.)**  $T'(s) = \gamma T$  for some  $\gamma$  iff  $MT'(s) = \gamma MT$ , i.e.  $e^{-s_n} c_n = \gamma c_n$  for every  $n$ , i.e.  $e^{-s_n} = \gamma$  for every  $n \in \text{supp}(c)$ . Since  $x \mapsto e^{-x}$  is injective, this forces  $s$  constant on  $\text{supp}(c)$ . Contrapositively,  $s$  nonconstant on  $\text{supp}(MT)$  implies  $T'(s)$  is not proportional to  $T$ .

**(Distinctness modulo gauge.)**  $(\ell + s, T'(s)) \in \mathcal{G}(\ell, T)$  requires  $\ell + s = \ell + t\mathbf{1}$  in the gain coordinate, i.e.  $s = t\mathbf{1}$  (log is injective on positive gains). A nonconstant  $s$  therefore leaves the orbit; note that when  $s = t\mathbf{1}$  the construction reduces exactly to the gauge,  $T'(t\mathbf{1}) = e^{-t}T$ , so the family contains the orbit and nothing of the orbit is double-counted.

**(Dimension.)** The map  $s \mapsto (\ell + s, T'(s))$  is injective (read off the first coordinate) and real-analytic in  $s$ , hence its image is a  $p$ -dimensional analytically parametrized family of exact solutions. Intersecting with the gauge orbit uses up  $\dim(S \cap \text{span}\{\mathbf{1}\}) = 1$  dimension, leaving non-gauge ambiguity dimension  $\geq p - 1$ . ■

The display in the main text (Sec. 4) is the instance  $s = \bar{\ell}\mathbf{1} - \ell$ : the drift is absorbed in *one step* into the relabelled object  $T' = M^{-1} \text{diag}(a)MT$  (up to the gauge constant), leaving a constant gain. The hypothesis “ $a$  nonconstant on the support of  $MT$ ” in the main-text statement is exactly the distinctness condition established above.

a) *What this does not claim.*: Theorem A is a statement about the square case  $N = K$  and an unconstrained object; for  $N > K$  the construction fails unless  $e^{-s} \odot MT$  remains in  $\text{range}(M)$ , and the correct analysis is the tall-design collision count of Theorem A' (see Appendix B). It does not say the record is uninformative about the gain in a statistical model: under random illumination, the *relative* gain is consistently estimable (Theorem B; see Appendix C), a different — statistical, asymptotic — notion of identifiability which Theorem A neither contradicts nor implies. Finally, Theorem A is about uniqueness, not conditioning: it makes no claim about how well- or ill-posed any estimator is.

### C. A.2 Corollary 1 (slowness does not help)

**Corollary 1.** Assume (A1)–(A5).

(i) If the admissible log-gain class is the linear space  $S$  itself, the entire ambiguity family of Theorem A lies inside the class:  $\ell \in S$ ,  $s \in S \Rightarrow \ell + s \in S$ . In particular, choosing  $s = \bar{\ell}\mathbf{1} - \ell \in S$  produces a **constant-gain** exact explanation of the record with the static object  $T' = M^{-1}(e^{\bar{\ell} - \ell} \odot MT)$ : a slowly drifting gain over the true object is indistinguishable from no drift at all over a relabelled object.

(ii) If the admissible class is a slowness ball  $\mathcal{B}_L = \{\ell \in S : \|\ell\|_* \leq L\}$  for any seminorm  $\|\cdot\|_*$  vanishing on constants (e.g. a Hölder or low-pass-energy seminorm), and  $\ell$  is interior ( $\|\ell\|_* < L$ ), then every nonconstant  $s \in S$  with  $\|s\|_*$  small enough keeps  $\ell + s \in \mathcal{B}_L$ . The local non-gauge ambiguity dimension remains  $p - 1$ : slowness bounds the *size* of the ambiguity, never its *dimension*.

*Proof.* (i) is linearity of  $S$  plus Theorem A applied with the stated  $s$ ; nonconstancy of  $\ell$  on  $\text{supp}(c)$  makes the constant-gain alternative distinct modulo gauge. (ii) Subadditivity of the seminorm gives  $\|\ell + s\|_* \leq \|\ell\|_* + \|s\|_* \leq L$  once  $\|s\|_* \leq L - \|\ell\|_*$ ; the set of such  $s$  is a neighborhood of 0 in  $S$  and Theorem A applies to each nonconstant member. ■

a) *What this does not claim.*: Slowness is not useless in general — in the tall-design regime the whole point is that slow drift means small  $p$ , making the threshold  $N \geq K + p$  cheap to satisfy (Sec. 4). The corollary only asserts that at  $N = K$  a slowness prior *by itself* removes nothing from the ambiguity, because the ambiguity directions live in the slow class itself.

### D. A.3 Corollary 2 (nonnegativity is insufficient; known support zeros can restore local identifiability)

Throughout this corollary write  $H$  for the square invertible design ( $H = M$ ; the letter matches the fixed-basis usage, e.g. Hadamard), and define the gain-absorption operator

$$M_s := H^{-1} \text{diag}(e^{-s})H, \quad T'(s) = M_s T,$$

so that the ambiguity set at  $T$  under an object prior class  $\mathcal{T}$  is  $V_{\mathcal{T}}(T) = \{s \in S : M_s T \in \mathcal{T}\}$ , and identifiability modulo gauge is the assertion  $V_{\mathcal{T}}(T) = S \cap \text{span}\{\mathbf{1}\}$ .

**Corollary 2.** *Assume (A1)–(A5).*

(i) *(Positivity insufficient.)* Let  $\mathcal{T}_+ = \{T : T_j \geq 0 \ \forall j\}$  and let  $T$  have full support,  $T_j > 0$  for all  $j$ . Then there is  $r > 0$  such that every  $s \in S$  with  $\|s\|_\infty < r$  satisfies  $M_s T > 0$ ; explicitly, it suffices that

$$\|H^{-1}\|_2 \|HT\|_2 (e^r - 1) < \min_j T_j.$$

Hence  $V_{\mathcal{T}_+}(T)$  contains a full neighborhood of 0 in  $S$  and the local non-gauge ambiguity dimension is still  $p - 1$ .

(ii) *(Known support zeros; local criterion.)* Let  $\Omega \subset \{1, \dots, K\}$ ,  $|\Omega| = q$ , and  $\mathcal{T}_\Omega = \{T : T_{\Omega^c} = 0, T \geq 0\}$ , with the true  $T \in \mathcal{T}_\Omega$ ,  $T_\Omega > 0$ . Then local identifiability modulo gauge holds at  $T$  if and only if (at the first-order level)

$$\ker(P_{\Omega^c} H^{-1} \text{diag}(HT)|_S) = S \cap \text{span}\{\mathbf{1}\},$$

where  $P_{\Omega^c}$  is coordinate projection onto  $\Omega^c$ . The gauge direction always lies in the kernel; when the kernel is exactly the gauge, no ambiguity exists in a quantifiable ball: with  $\gamma$  the smallest singular value of the linearized map on a complement  $S_\perp$  of  $\text{span}\{\mathbf{1}\}$  in  $S$  and  $C_H$  a second-order remainder constant (below), every  $s$  with  $0 < \|s_\perp\|_2 < \gamma/C_H$  violates the support constraint.

(iii) *(Generic counts — dimension-counting arguments, labeled as such.)* The linearized map in (ii) sends the  $(p - 1)$ -dimensional space  $S/\text{span}\{\mathbf{1}\}$  into  $\mathbb{R}^{K-q}$ ; injectivity is possible only if  $K_0 := K - q \geq p - 1$ , i.e. **at least one known zero per nonconstant drift dimension**. For generic  $(H, T_\Omega)$  this necessary count is expected to be sufficient. For unknown support of size  $\leq q$ , dimension counting predicts the threshold  $K \geq 2q + (p - 1)$ , up to endpoint conventions.

*Proof.* (i)  $M_0 T = T$  and  $\|M_s T - T\|_2 = \|H^{-1}((e^{-s} - 1) \odot HT)\|_2 \leq \|H^{-1}\|_2 \|HT\|_2 \max_n |e^{-s_n} - 1| \leq$

$\|H^{-1}\|_2 \|HT\|_2 (e^{\|s\|_\infty} - 1)$ . If this is below  $\min_j T_j$  then  $M_s T > 0$  entrywise. Since the positive orthant is open and  $s \mapsto M_s T$  is continuous, positivity excludes nothing locally; Theorem A supplies the distinctness. ■ (i)

(ii) Define the support-constraint map  $F : S \rightarrow \mathbb{R}^{K-q}$ ,  $F(s) = P_{\Omega^c} M_s T$ . It is real-analytic,  $F(0) = P_{\Omega^c} T = 0$ , and its differential at 0 is

$$DF_0 u = -P_{\Omega^c} H^{-1} \text{diag}(HT) u,$$

by differentiating  $e^{-tu} = \mathbf{1} - tu + O(t^2)$  inside  $M_{tu}$ . The gauge is always in the kernel:  $H^{-1} \text{diag}(HT)\mathbf{1} = H^{-1}(HT) = T$  and  $P_{\Omega^c} T = 0$ ; consistently,  $F(t\mathbf{1}) = e^{-t} F(0) = 0$  exactly, for all  $t$ .

*Necessity of the criterion (first-order sense).* If the kernel strictly exceeds the gauge, some nonconstant  $u \in S$  satisfies the linearized support constraint. This is a first-order ambiguity direction: it makes  $T$  a critical point of the constrained model map and hence a Fisher-information null direction for the corresponding functional (cf. Remark A.1), so local *differential* identifiability fails. Whether the direction integrates to an exact ambiguity curve requires a further constant-rank or analyticity argument and can fail at second order; we claim only the first-order (differential) statement, which is what the criterion tests.

*Sufficiency in a ball.* Suppose the kernel equals the gauge and let  $\gamma = \sigma_{\min}(DF_0|_{S_\perp}) > 0$ . Entrywise Taylor with remainder,  $|e^{-x} - 1 + x| \leq \frac{x^2}{2} e^{|x|}$ , gives for  $\|s\|_\infty \leq 1$

$$\|F(s) - DF_0 s\|_2 \leq C_H \|s\|_2^2, \quad C_H := \frac{\epsilon}{2} \|H^{-1}\|_2 \|HT\|_2.$$

For gauge-fixed  $s = s_\perp \in S_\perp$  (any  $s$  may be reduced to this form using  $F(s + t\mathbf{1}) = e^{-t} F(s)$ , which rescales but never annihilates  $F$ ),

$$\|F(s_\perp)\|_2 \geq \gamma \|s_\perp\|_2 - C_H \|s_\perp\|_2^2 > 0$$

for  $0 < \|s_\perp\|_2 < \gamma/C_H$ .

A nonzero  $F(s)$  means  $M_s T \notin \mathcal{T}_\Omega$  (support constraint violated), so no non-gauge ambiguity exists in that ball. This is the quantitative inverse-function-theorem argument in explicit constants. ■ (ii)

(iii) *These are genericity/dimension-count arguments, not proofs, and we state them as such.* The necessity direction is rigorous: a linear map from a  $(p - 1)$ -dimensional space cannot be injective into  $\mathbb{R}^{K-q}$  if  $K - q < p - 1$ , so at least  $p - 1$  known zeros are required. The sufficiency direction is generic: the entries of  $P_{\Omega^c} H^{-1} \text{diag}(HT)|_{S_\perp}$  are polynomial in  $(H, T)$ , so full column rank holds off a proper algebraic subvariety *provided a single witness exists*; the dimension count  $K - q \geq p - 1$  makes a witness expected but, for a *fixed* named basis (e.g. Hadamard with a specific low-pass  $S$ ), the rank must be verified directly — the correct fixed-basis statement is rank-based, not dimensional. The unknown-support count  $K \geq 2q + (p - 1)$  arises from demanding that the  $q$ -sparse model avoid all of its gain-transformed copies  $M_s \Sigma_q$ , in analogy with the generic sparsity thresholds of Kech–Krahmer for generic bilinear maps; it is a heuristic prediction of the generic threshold, unverified for structured designs, and the uniform no-support-switching condition it summarizes is strictly stronger than (ii). ■ (heuristic scope)

a) *What this does not claim.*: Part (i) says nonnegativity fails to restore *identifiability*; it may still improve estimator conditioning and is routinely useful as regularization — no claim is made either way. Part (ii) is local: it excludes ambiguities in a ball around  $T$  and is silent about isolated global aliases, which can exist even with a nonsingular linearization; global uniqueness needs a separate compactness/non-vanishing argument outside the ball. The criterion is also stated at fixed, *known* support; for unknown support only the heuristic count of part (iii) is offered, and part (iii) makes no claim for any specific fixed basis without an explicit rank check.

#### E. A.4 Remarks

a) *Remark A.1 (Fisher-theoretic reading).*: Because the ambiguity of Theorem A is an explicit real-analytic curve  $t \mapsto (\ell + ts, T'(ts))$  along which the noiseless data are constant, every nonconstant  $s \in S/\text{span}\{\mathbf{1}\}$  yields an exact null direction of the (gauge-quotient) Fisher information in any regular dominated noise model built on this record: the Cramér–Rao bound is infinite for any functional that varies along the curve. Here the implication “explicit ambiguity curve  $\Rightarrow$  Fisher null direction” is exact, not merely first-order.

b) *Remark A.2 (zero buckets).*: If some  $c_n = (MT)_n = 0$ , two separate pathologies arise, which the statements above deliberately quarantine via (A4): the gain values  $a_n$  at zero-carrier frames are entirely unobservable (a trivial extra ambiguity, of dimension  $\dim(S \cap \{s : s = 0 \text{ on } \text{supp}(c)\})$  beyond the count of Theorem A), and log-domain processing of  $R_n = 0$  is undefined, an estimation-side issue treated by the calibrated soft-log machinery of the low-photon analysis (Sec. 6; see Appendix D), not an identifiability issue.

### APPENDIX B

#### TALL-DESIGN THRESHOLDS AND THE LOW-PASS DRIFT SPACE

This appendix proves the tall-design identifiability thresholds quoted as **Theorem A'** in Sec. 4 (local differential identifiability at  $N \geq K + p - 1$ ; generic exact finite-sample identifiability at  $N \geq K + p$ ; uniform identifiability at  $N \geq 2K + p - 1$ ), restated below as Theorem A'(i)–(iii), and the two lemmas that transfer these thresholds from *generic* drift subspaces to the *physical* Fourier low-pass drift space  $S_{\text{LP}}$ . Throughout we distinguish carefully between fully rigorous steps and steps that are standard algebraic-geometry *genericity* / *dimension-count* arguments; the latter are flagged where they occur.

#### A. B.0 Setting, notation, and the meaning of “generic”

The bucket record is  $R_n = a_n B_n$  with  $B = MT$ , i.e. in vector form

$$y = \text{diag}(e^\ell) MT, \quad M \in \mathbb{R}^{N \times K}, \quad N > K, \quad (B.1)$$

$$\ell \in S, \quad T \in \mathbb{R}^K \setminus \{0\},$$

where  $\ell = \log a$  is the log-gain trajectory, confined to the drift class  $S \subset \mathbb{R}^N$  with  $p = \dim S$ . Set

$$C = S \cap \text{span}\{\mathbf{1}\}, \quad c = \dim C \in \{0, 1\}, \quad r = p - c.$$

The integer  $r$  counts the *nonconstant* gain degrees of freedom. In the convention used throughout the paper,  $\mathbf{1} \in S$  (the constant direction is the unavoidable global-scale gauge  $(a, T) \mapsto (ca, T/c)$ ), so  $c = 1$  and  $r = p - 1$ . All identifiability statements are modulo this gauge. Via the bandwidth relation  $p \approx \rho_{\text{bw}} N$  of Sec. 2, every threshold below can be re-read as a condition on  $\rho_{\text{bw}}$ ; the pair-decorrelation parameter  $\rho_{\text{pair}}$  of Sec. 8 plays no role in this appendix.

a) *Genericity convention.*: “Generic  $M$ ” (and, where stated, “generic drift subspace”) means: outside a proper real semialgebraic subset of the relevant parameter space. Equivalently, the stated property holds almost surely when the excluded parameters are drawn from any distribution absolutely continuous with respect to Lebesgue measure. The standard proof route is to verify the property over  $\mathbb{C}$  on a Zariski-open set and restrict to the real points; we use this convention silently below. Note at the outset that a *deterministic structured* design — e.g. an ordered Hadamard matrix — is **not** covered by any genericity statement here and must be checked separately.

#### B. B.1 The collision equation

**Lemma B.1 (lifted collision equation).** *Two parameter pairs  $(\ell, T)$  and  $(\ell', T')$ , with  $\ell, \ell' \in S$  and  $T, T' \neq 0$ , produce identical noiseless records (B.1) if and only if*

$$MT = D_s MT', \quad s = \ell' - \ell \in S, \quad D_s = \text{diag}(e^s). \quad (B.2)$$

*The constant part of  $s$  is exactly the gauge (a relative rescaling of  $T$  vs.  $T'$ ); consequently all non-gauge aliases are described by the incidence relation*

$$[M, -D_s M] \begin{bmatrix} T \\ T' \end{bmatrix} = 0 \quad (B.3)$$

*with  $s$  nonconstant modulo  $C$ . Equivalently, writing  $U = \text{range}(M)$ , an alias exists iff  $MT \in U \cap D_s U$ .*

**Proof.**  $\text{diag}(e^\ell)$  is invertible, so equality of the two records is equivalent to (B.2). If  $s$  is constant,  $s = \sigma \mathbf{1}$ , then (B.2) reads  $MT = e^\sigma MT'$ ; since  $N > K$  and generic  $M$  has full column rank,  $T = e^\sigma T'$ , which is the gauge orbit. Conversely a nonconstant  $s \bmod C$  gives a genuinely distinct pair. Rewriting (B.2) as a nullspace condition gives (B.3). Finally, note that a kernel vector of  $[M, -D_s M]$  with  $T = 0$  forces  $D_s MT' = 0$ , hence  $T' = 0$  (both  $M$  and  $D_s M$  are injective for full-column-rank  $M$ ); so every nonzero kernel element has both blocks nonzero. ■

This diagonal structure is why the generic bilinear counts of Kech–Krahmer (arXiv:1603.07316) do not apply verbatim:  $y_n = a_n m_n^\top T$  is a *diagonal-row* measurement, not a generic rank-one measurement, and the ambiguity is governed by the intersection of the fixed  $K$ -plane  $U$  with its diagonally modulated copy  $D_s U$ .

#### C. B.2 The stratum-counting lemma

Let  $B = \exp(S)/\mathbb{R}_+$  be the nonconstant gain-ratio manifold (dimension  $r = p - 1$ ), parametrizing the classes  $b = e^s$

modulo positive scalars,  $s$  nonconstant. For a **fixed** design  $M$ , stratify  $B$  by the level sets of the rank function:

$$B = \bigsqcup_{\alpha} B_{\alpha}, \quad r_{\alpha} := \dim B_{\alpha},$$

$$\rho_{\alpha} := \text{rank}[M, -D_b M] \text{ constant for } b \in B_{\alpha}.$$

Such a finite stratification into locally closed strata exists for every  $M$ : after the substitution  $u = e^s$  the rank function is semialgebraic in  $u$ , its finitely many level sets are semi-algebraic, and finite semialgebraic stratification is standard (Bochnak–Coste–Roy, *Real Algebraic Geometry*, Ch. 9). Define the **projective collision dimension**

$$\Delta_{\alpha} := r_{\alpha} + 2K - 1 - \rho_{\alpha}. \quad (\text{B.4})$$

**Lemma B.2 (stratum counting).** *Fix  $M \in \mathbb{R}^{N \times K}$  of full column rank and stratify  $B$  as above. Consider the collision incidence variety*

$$Z_{\alpha} = \{(b, [T : T']) \in B_{\alpha} \times \mathbb{P}^{2K-1} : [M, -D_b M] \begin{bmatrix} T \\ T' \end{bmatrix} = 0\}.$$

Then:

- 1)  $Z_{\alpha} = \emptyset$  if and only if  $\rho_{\alpha} = 2K$ ; otherwise  $\dim Z_{\alpha} = \Delta_{\alpha}$  (and then automatically  $\Delta_{\alpha} \geq r_{\alpha} \geq 0$ ).
- 2) The “bad first-object” set contributed by stratum  $\alpha$  — the projection of  $Z_{\alpha}$  to  $[T] \in \mathbb{P}^{K-1}$  — has dimension at most  $\Delta_{\alpha}$ .
- 3) (Sufficiency, rigorous.) If  $\max_{\alpha} \Delta_{\alpha} < K - 1$ , where the maximum runs over strata with  $\rho_{\alpha} < 2K$ , then a generic object  $T$  has no non-gauge alias. (Necessity, genericity-flagged.) Conversely, if some stratum has  $\Delta_{\alpha} \geq K - 1$  **and** the projection  $Z_{\alpha} \rightarrow \mathbb{P}^{K-1}$  is dominant, a generic object has an alias.
- 4) Uniform identifiability (every nonzero object alias-free) holds if and only if  $\rho_{\alpha} = 2K$  for every stratum — equivalently, with the convention that the maximum in (3) runs over kernel-supporting strata (empty maximum  $= -\infty$ ), iff  $\max_{\alpha} \Delta_{\alpha} < 0$ .

**Proof.** (1) For fixed  $b \in B_{\alpha}$  the fiber of  $Z_{\alpha}$  is the projectivized nullspace of  $[M, -D_b M]$ , of projective dimension  $(2K - \rho_{\alpha}) - 1 = 2K - 1 - \rho_{\alpha}$ ; it is empty iff the nullspace is  $\{0\}$ , i.e. iff  $\rho_{\alpha} = 2K$  (possible only when  $N \geq 2K$ , since  $\rho_{\alpha} \leq \min(N, 2K)$ ). On a constant-rank stratum the nullspaces form a vector bundle over  $B_{\alpha}$  (constant-rank kernels vary continuously; locally trivial by choosing complementary minors), so the fiber dimension is constant and

$$\dim Z_{\alpha} = \dim B_{\alpha} + (2K - 1 - \rho_{\alpha}) = \Delta_{\alpha}.$$

When  $\rho_{\alpha} \leq 2K - 1$  we get  $\Delta_{\alpha} \geq r_{\alpha} \geq 0$ .

(2) The image of a semialgebraic set under the projection  $Z_{\alpha} \rightarrow \mathbb{P}^{K-1}$  has dimension at most  $\dim Z_{\alpha}$  (semialgebraic dimension is non-increasing under maps; Bochnak–Coste–Roy, Prop. 2.8.6).

(3) *Sufficiency*: the union over the finitely many strata of the bad projections is semialgebraic of dimension  $\leq \max_{\alpha} \Delta_{\alpha} < K - 1 = \dim \mathbb{P}^{K-1}$ , hence its complement is dense and full-measure. Every  $T$  in the complement has no alias with any

nonconstant ratio. This direction is fully rigorous. *Necessity*: if  $\Delta_{\alpha} \geq K - 1$  and the projection is dominant, the image fills  $\mathbb{P}^{K-1}$  up to lower-dimensional sets, so a generic  $T$  has an alias. **Status**: the dominance of the projection for generic  $M$  is a *standard algebraic-geometry genericity assertion* — failure of dominance is the vanishing of a Jacobian minor of the projection, hence a proper algebraic condition on  $M$  once a single example with full projection rank exists; this rests on the usual dimension/transversality count and does not exhibit the witness per stratum. We record the necessity direction as resting on this dominant-projection argument, in the same way as the generic bilinear identifiability literature (Kech–Krahmer; Li–Lee–Bresler).

(4) Uniform injectivity means (B.3) has no solution with nonconstant  $s$  and  $T, T' \neq 0$ , i.e. every fiber over every  $b \in B$  is empty. By (1) this happens iff  $\rho_{\alpha} = 2K$  on every stratum. Since strata with  $\rho_{\alpha} \leq 2K - 1$  have  $\Delta_{\alpha} \geq 0$ , the stated  $\max \Delta_{\alpha} < 0$  criterion is equivalent under the empty-maximum convention. This direction is deterministic and rigorous given the stratification. ■

a) *Remark (a boundary case).*: A looser phrasing of claim 1 — “ $Z_{\alpha}$  has dimension  $\Delta_{\alpha}$  if  $\Delta_{\alpha} \geq 0$ , empty if  $\Delta_{\alpha} < 0$ ” — would miss the case  $\rho_{\alpha} = 2K$ ,  $r_{\alpha} \geq 1$  (where  $\Delta_{\alpha} = r_{\alpha} - 1 \geq 0$  but  $Z_{\alpha} = \emptyset$ ). The corrected bookkeeping above — emptiness governed by  $\rho_{\alpha} = 2K$ , not by the sign of  $\Delta_{\alpha}$  — is what claims 3–4 actually use, and is how the uniform threshold at  $N \geq 2K + p - 1$  (which lives in the regime  $N > 2K$ ) is consistent with  $\rho_{\alpha} \leq \min(N, 2K)$ .

b) *What this does not claim.*: Lemma B.2 is a statement about *exact noiseless collisions* only: it says nothing about conditioning, noise robustness, or the behavior of any estimator. It does not evaluate the ranks  $\rho_{\alpha}$  — that is where genericity of  $M$  (Sec. B.3) or the low-pass lemmas (Sec. B.4) enter. It does not cover deterministic structured designs unless their stratum ranks are computed explicitly. The necessity half of claim 3 has genericity status as flagged.

#### D. B.3 Clean thresholds for generic drift families

On a *generic*  $p$ -dimensional log-gain subspace, the ratio manifold has a dense open stratum on which  $D_b$  has no forced coordinate equalities, and for generic  $M$  the rank on that stratum is maximal:

$$\text{rank}[M, -D_b M] = \min(N, 2K). \quad (\text{B.5})$$

**Status of (B.5)**: this is the *dominant-stratum hypothesis* — a genericity/dimension-count assertion (a single  $(M, b)$  witness with full rank makes the relevant minor a not-identically-zero polynomial, hence nonzero generically). For the physical low-pass space it is *proved at generic-pair level* — for each fixed nonconstant  $b$ , with the exceptional design set allowed to depend on  $b$  — in Sec. B.4–B.5 below; the uniform-over- $b$  version for a single generic  $M$  is a labeled dimension-count argument (Prop. B.7, level (ii)). That is precisely why the low-pass lemmas matter.

Under (B.5) the collision count takes a single unified form. If  $N < 2K$ , the dense stratum has  $\rho = N$  and, by Lemma B.2(1),

$$\Delta = r + 2K - 1 - N. \quad (\text{B.6})$$

If  $N \geq 2K$ , the dense stratum has  $\rho = 2K$  and empty fibers; collisions can then live only on the *rank-drop locus*  $\{b : \text{rank}[M, -D_b M] \leq 2K - 1\}$ . The corank-one determinantal variety in  $\mathbb{R}^{N \times 2K}$  has codimension  $(N - 2K + 1) \cdot 1$  (classical; Eagon–Northcott / Bruns–Vetter, *Determinantal Rings*), so a *transversally embedded*  $r$ -parameter family meets it in a set of dimension  $r - (N - 2K + 1)$ , over which the kernel fiber is 0-dimensional; the incidence dimension is again  $r - (N - 2K + 1) + 0 = r + 2K - 1 - N = \Delta$ . Thus (B.6) controls both regimes. **Status:** the transversality of the structured family  $b \mapsto [M, -D_b M]$  to the determinantal variety is again a genericity assertion of the same standard type, asserted rather than instance-verified.

**Theorem A'(i) (local differential threshold:  $N \geq K + p - 1$ ).** *Let  $M$  be generic,  $S$  a generic  $p$ -dimensional log-gain space with  $\mathbf{1} \in S$  ( $r = p - 1$ ), and  $T$  generic. After quotienting the global-scale gauge, the parameter manifold of pairs  $(\ell \bmod C, [T])$  has dimension  $(K - 1) + r = K + p - 2$ , and the projectivized data space has dimension  $N - 1$ . Local differential identifiability (injectivity of the differential of  $(\ell, T) \mapsto [\text{diag}(e^\ell)MT]$  at generic points) holds iff  $N - 1 \geq K + p - 2$ , i.e.  $N \geq K + p - 1$ .*

**Proof.** Necessity is the dimension inequality: an injective linear map from a  $(K + p - 2)$ -dimensional tangent space into an  $(N - 1)$ -dimensional one requires  $N - 1 \geq K + p - 2$ . For sufficiency, the differential has columns  $M \delta T$  (object directions) and  $\text{diag}(MT) \delta \ell$ ,  $\delta \ell \in S/C$  (gain directions), with the joint scale direction quotiented out. Full rank  $K + p - 2$  of this  $N \times (K + p - 1)$ -block-minus-gauge system is the nonvanishing of some minor — a polynomial condition in  $(M, S, T)$ . **Status:** the existence of a full-rank witness follows by “a generic rank calculation,” i.e. the standard specialization argument; the witness is not exhibited. We therefore record sufficiency as a *standard dimension-count genericity argument*, consistent with its use throughout this literature. ■

a) *What this does not claim:* Local injectivity of the differential does not imply global uniqueness; at the boundary  $N = K + p - 1$  the map is generically locally finite-to-one but **not** generically one-to-one (see Theorem A'(ii): there  $\Delta = K - 1$ , so aliases generically exist). No statement is made about the singular values of the Jacobian, i.e. about conditioning.

**Theorem A'(ii) (sharp generic-point threshold:  $N \geq K + p$ ).** *For generic tall  $M$  and a generic (transverse) nonconstant gain-ratio family of dimension  $r = p - 1 \geq 1$ , almost every pair  $(\ell, T)$  is identifiable modulo global scale iff  $N > K + r$ , i.e. iff*

$$N \geq K + p. \quad (\text{B.7})$$

*For  $K < N \leq K + p - 1$ , a generic object has a non-gauge algebraic alias: overdetermination alone is insufficient.*

**Proof.** By (B.6), the bad-object projection has dimension at most  $\Delta = r + 2K - 1 - N$  (Lemma B.2(2)). Generic-point identifiability holds when  $\Delta < K - 1$ , i.e.  $r + 2K - 1 - N < K - 1$ , i.e.  $N > K + r = K + p - 1$ ; integrality gives (B.7). This direction inherits only the rank hypothesis (B.5) — rigorous once (B.5) is established (for low-pass  $S$ , at generic-pair level

by Sec. B.5). For the converse,  $N \leq K + r$  gives  $\Delta \geq K - 1$ , and under the dominant-projection assertion (Lemma B.2(3), genericity-flagged) the bad set fills  $\mathbb{P}^{K-1}$ , so a generic object has an alias. ■

b) *What this does not claim:* “Generic” excludes an unquantified measure-zero exceptional set of objects; below the uniform threshold, exceptional objects with exact aliases *do* exist (the gap  $K + p \leq N < 2K + p - 1$  is real). The theorem says nothing about the conditioning of the inversion near  $N = K + p$  — uniqueness at the threshold is compatible with arbitrarily bad stability, which is a separate random-matrix question (Sec. 10). It does not apply to deterministic designs, and it does not say statistical mechanisms are useless above threshold — only that they are not *logically required* for identifiability.

**Theorem A'(iii) (sharp uniform threshold:  $N \geq 2K + p - 1$ ).** *In the setting of Theorem A'(ii), every nonzero object is identifiable modulo global scale iff  $N \geq 2K + r = 2K + p - 1$ .*

**Proof.** Uniform identifiability is emptiness of the whole collision incidence, i.e.  $\Delta < 0$  in the unified count:  $r + 2K - 1 - N < 0 \iff N \geq 2K + r$ . Concretely, for  $N \geq 2K + r$  the rank-drop locus has expected dimension  $r - (N - 2K + 1) \leq r - (r + 1) < 0$  in the  $r$ -dimensional family, so (under the determinantal-transversality assertion flagged above) the family misses it entirely and every  $b$  has trivial kernel,  $\rho = 2K$ ; Lemma B.2(4) concludes. Conversely for  $2K \leq N < 2K + r$  the drop locus has nonnegative expected dimension and (generically, same flag) is nonempty, producing an exact alias for some object; for  $N < 2K$  the kernel is nontrivial for *every*  $b$ , so uniform identifiability fails outright. ■

c) *What this does not claim:* Uniformity is over *objects*, not over *designs*:  $M$  and the drift family are still generic, and an adversarially chosen  $M$  can defeat the bound. No claim is made about recovery algorithms or noise.

#### E. B.4 The two low-pass lemmas

The clean thresholds above rest on the rank identity (B.5), asserted generically. The physical drift space is not generic — it is the Fourier low-pass space

$$S_{\text{LP}} = \text{span}\left\{\mathbf{1}, \cos \frac{2\pi q n}{N}, \sin \frac{2\pi q n}{N} : q = 1, \dots, m\right\},$$

$$p = \dim S_{\text{LP}} = 2m + 1, \quad (\text{B.8})$$

with  $r = p - 1 = 2m$  and the below-Nyquist condition  $p \leq N$ . The next two lemmas verify (B.5) for this space *without* any genericity assumption on  $S$ .

**Lemma B.5 (level-multiplicity bound; fully rigorous).** *Let  $s \in S_{\text{LP}}$  be nonconstant,  $p < N$ . Then no value is attained by  $(s_n)_{n=0}^{N-1}$  more than  $2m = p - 1$  times. Consequently the largest eigenvalue multiplicity of  $D_s = \text{diag}(e^{s_n})$  satisfies*

$$m_{\max}(s) \leq p - 1. \quad (\text{B.9})$$

**Proof.** Write  $z_n = e^{2\pi i n/N}$ ; the  $N$  points  $z_0, \dots, z_{N-1}$  are distinct. Every  $s \in S_{\text{LP}}$  has the Laurent representation  $s_n =$

$\sum_{q=-m}^m c_q z_n^q$  with conjugate symmetry  $c_{-q} = \overline{c_q}$  (real  $s$ ). Suppose  $s_n = c$  at more than  $2m$  distinct indices. Then the ordinary polynomial

$$P(z) = z^m \left( \sum_{q=-m}^m c_q z^q - c \right), \quad \deg P \leq 2m,$$

has more than  $2m$  distinct roots on the unit circle, hence  $P \equiv 0$  (fundamental theorem of algebra). Then  $s(z) \equiv c$  on the whole circle, so  $s$  is the constant sequence — contradiction. Since  $x \mapsto e^x$  is injective, the eigenvalue multiplicities of  $D_s$  equal the level multiplicities of  $s$ , giving (B.9). ■

**Lemma B.6 (generic rank of  $[M, -DM]$  with repeated diagonal values).** Let  $D \in \mathbb{R}^{N \times N}$  be a non-scalar diagonal matrix whose most-repeated diagonal value has multiplicity  $m_{\max}$ . Then for generic  $M \in \mathbb{R}^{N \times K}$ ,

$$\text{rank}[M, -DM] = \min(N, 2K, K + N - m_{\max}). \quad (\text{B.10})$$

**Proof.** Upper bound (deterministic — holds for every  $M$ ). The column space of  $[M, -DM]$  is  $U + DU$  with  $U = \text{range}(M)$ . Let  $\lambda$  be a most-repeated diagonal value and write  $D = \lambda I + Q$ , so  $Q$  is diagonal with exactly  $N - m_{\max}$  nonzero entries,  $\text{rank } Q = N - m_{\max}$ . Then  $U + DU = U + QU$ , hence

$$\dim(U + DU) \leq \dim U + \dim QU \leq K + \min(K, N - m_{\max}),$$

together with the trivial bounds  $\leq N$  and  $\leq 2K$ . This is the right-hand side of (B.10).

Lower bound (genericity via witness; status flagged). It suffices to exhibit a single  $M_0$  attaining the bound: then the corresponding maximal minor of  $[M, -DM]$  is a polynomial in the entries of  $M$  that is not identically zero, so it is nonzero off a proper algebraic set — the standard specialization argument. One constructs  $M_0$  in coordinates adapted to  $Q$ : order the coordinates so that the  $Q$ -active indices (those with  $d_i \neq \lambda$ ) come first, and build the  $K$  columns of  $M_0$  by pairing active coordinates with inactive ones (each pair  $u = e_{\text{act}} + e_{\text{inact}}$  contributes two independent directions to  $U + QU$ , since  $Qu = q e_{\text{act}}$ ) and, when inactive coordinates are exhausted, pairing active coordinates carrying distinct diagonal values (possible because no value has multiplicity exceeding  $m_{\max}$ ), until the count saturates at  $N, 2K$ , or  $K + (N - m_{\max})$ . **Status:** this is a construction sketch; the saturation bookkeeping across the finitely many cases is elementary but is not carried out line-by-line. We record the lower bound as a *standard witness-plus-specialization genericity argument*, not as an instance-checked computation. ■

a) *What these lemmas do not claim.*: Lemma B.5 is specific to the exact Fourier grid (B.8) below Nyquist; a different “low-pass” convention (e.g. a wavelet or spline drift class, or an off-grid Fourier family) can have larger level multiplicity and must be re-checked via (B.12) below. Lemma B.6 is a statement about *generic*  $M$  given  $D$ ; a deterministic design may realize strictly smaller rank (only the upper bound is universal).

### F. B.5 Transfer to the physical low-pass space

**Proposition B.7 (low-pass transfer: a three-level hierarchy).** Let  $S = S_{\text{LP}}$  as in (B.8),  $p = 2m + 1 < N$ , and suppose the

generic-point threshold  $N \geq K + p$  holds. The transfer of the clean thresholds of Theorem A'(i)–(iii),

$$\begin{aligned} \text{local: } N &\geq K + p - 1, & \text{generic: } N &\geq K + p, \\ \text{uniform: } N &\geq 2K + p - 1, \end{aligned}$$

to the physical low-pass drift space holds at three distinct levels, which we state — and use — separately.

(i) **Generic-pair statement (proven).** For each *fixed* non-constant  $s \in S_{\text{LP}}$ , a generic  $M$  — the exceptional design set may depend on  $s$  — satisfies

$$\text{rank}[M, -D_s M] = \min(N, 2K), \quad (\text{B.11})$$

i.e. the fixed pair  $(M, s)$  behaves as the dense stratum of Sec. B.3. Given (B.11) at a generic pair, the sufficiency directions of Theorem A'(ii)–(iii) are rigorous for that pair; the necessity directions retain the dominant-projection flag of Lemma B.2(3).

(ii) **Uniform over the drift family for a single generic  $M$  (dimension-count argument — not a theorem).** For fixed nonconstant  $s$ , let  $c$  denote the codimension in  $\mathbb{R}^{N \times K}$  of the bad-design set  $X_s = \{M : \text{rank}[M, -D_s M] < \min(N, 2K)\}$  (a proper algebraic subset by level (i), so  $c \geq 1$ ). The nonconstant gain-ratio family is  $(p - 1)$ -dimensional, so the union  $\bigcup_s X_s$  is expected to have codimension  $\geq c - (p - 1)$ , and a single generic  $M$  avoids the entire union provided  $c \geq p$ . If the codimension of the rank drop matched the estimate for a free  $N \times 2K$  matrix family — corank one is determinantal of codimension  $2K - N + 1$  in the regime  $K + p \leq N < 2K$  — then  $c = 2K - N + 1 \geq p$  gives the explicit sufficient window  $N \leq 2K - p + 1$ ; and in the regime  $N \geq 2K$  the drop codimension is  $N - 2K + 1$ , so  $c \geq p$  reads  $N \geq 2K + p - 1$  — recovering exactly the uniform threshold of Theorem A'(iii). The count is stated for the constrained family  $[M, -D_s M]$  — the two blocks share the same  $M$ , and  $D_s$  ranges over a structured exponential-trigonometric family — for which the free determinantal codimension is not automatic. We therefore label the whole of level (ii) a **dimension-count argument at heuristic level, not a theorem**. In particular, in the gap window  $2K - p + 1 < N < 2K + p - 1$  even this count fails to cover the union, and only the generic-pair statement (i) is available there.

(iii) **Empirical support (numerical, not proof).** The tall-design threshold scans at  $K = 64$  (1300 cells: five drift dimensions  $p \in \{3, 5, 9, 17, 33\}$ , 20 seeds per cell) and  $K = 128$  (1950 cells: same five  $p$ , 30 seeds per cell) — Sec. 9, Fig. 8 — find the local rank test transitioning from 0% to 100% pass **exactly** at  $N = K + p - 1$  for every  $p$  at both object sizes (3250 cells, no exceptions) — consistent with the uniform reading at the local-differential level, though a finite scan cannot prove it.

**Proof of (i).** Fix a nonconstant  $s \in S_{\text{LP}}$ . By Lemma B.5,  $m_{\max}(s) \leq p - 1$ . Apply Lemma B.6:

- If  $N < 2K$ : the threshold gives  $p - 1 \leq N - K - 1 < 2K - K - 1 < K$ , so  $K + N - m_{\max} \geq K + N - (p - 1) > N$ , and (B.10) yields  $\text{rank} = N = \min(N, 2K)$ .
- If  $N \geq 2K$ : the threshold gives  $p - 1 \leq N - K - 1 \leq N - K$ , so  $K + N - m_{\max} \geq 2K$ , and (B.10) yields  $\text{rank} = 2K = \min(N, 2K)$ .

This arithmetic is exact and rigorous, and for fixed  $s$  Lemma B.6's exceptional design set is a proper algebraic subset (depending on  $D_s$ ), which is exactly the generic-pair claim. Levels (ii) and (iii) carry the statuses stated above and are **not** part of what is proven. ■

a) *Other low-pass conventions.*: If the implemented drift class is not exactly (B.8), define its maximal nonconstant level multiplicity

$$L_S = \sup_{\substack{s \in S \\ s \text{ nonconstant}}} \max_c \#\{n : s_n = c\}. \quad (\text{B.12})$$

By the same arithmetic, the clean thresholds are guaranteed whenever

$$L_S \leq K \text{ (if } N < 2K) \quad \text{or} \quad L_S \leq N-K \text{ (if } N \geq 2K). \quad (\text{B.13})$$

If (B.13) fails, the clean  $N \geq K + p$  statement must **not** be used; instead apply the stratum formula of Lemma B.2 with  $\rho_\alpha = \min(N, 2K, K + N - m_\alpha)$ , where  $m_\alpha$  is the diagonal multiplicity on stratum  $\alpha$ : generic-point identifiability iff  $\max_\alpha(r_\alpha + 2K - 1 - \rho_\alpha) < K - 1$ , uniform iff  $< 0$  (kernel-supporting-strata convention). For the standard Fourier space (B.8), (B.13) holds automatically at  $N \geq K + p$  and no correction is needed.

b) *What this does not claim.*: The transfer is proven at the *generic-pair* level only: level (ii)'s uniform-over-drift statement is a labeled dimension-count argument — sufficient windows  $N \leq 2K - p + 1$  and  $N \geq 2K + p - 1$  under the free determinantal codimension estimate, with the gap window  $(2K - p + 1, 2K + p - 1)$  covered at generic-pair level only — and level (iii)'s exact empirical  $N = K + p - 1$  rank transition supports, but does not prove, the uniform reading at the local level. Beyond that, the transfer says nothing about the regime  $K < N < K + p$ , where tallness is insufficient *even generically* and the statistical stationarity anchor of Sec. 5 (Theorem B) or a support/sparsity prior is required — the low-pass structure does not rescue algebraic identifiability below threshold, and at  $p \approx N$  ( $\rho_{\text{bw}} \rightarrow 1$ ) blind algebraic separation is impossible for any  $N$ . It does not claim well-conditioned inversion at or near  $N = K + p$ . It does not cover deterministic designs (the ordered-Hadamard obstruction of Theorem A is unaffected). And it certifies only the exact grid-Fourier below-Nyquist model; any other drift-class implementation must pass the  $L_S$  check (B.12)–(B.13) before the clean thresholds are quoted.

## APPENDIX C

### CARRIER STATISTICS AND THE STATIONARITY ANCHOR

This appendix proves Proposition 1, formalizes condition (★) quantitatively, proves Proposition 2 (with counterexamples in both directions), and proves Theorem B. Throughout,  $R_n = a_n B_n$  is the bucket record,  $B_n = \langle I_n, T \rangle = \sum_{j=1}^K I_{n,j} T_j$  the carrier,  $\ell_n = \log a_n$  the log-gain,  $S$  the drift class with  $p = \dim S$ ,  $S_1 = \sum_j T_j$ ,  $S_2 = \|T\|_2^2$ ,  $K_{\text{eff}} = S_1^2/S_2$ ,  $K_\infty = S_2/\|T\|_\infty^2$ ,  $K_4 = S_2^2/\sum_j T_j^4$ , and  $W$  denotes the window length of the blind estimator  $\hat{g}_n = W^{-1} \sum_{k \in W_n} Y_k$  applied to  $Y_n = \log R_n$ . Asymptotic statements are for a triangular array indexed by  $(K, N)$ : the

object  $T = T^{(K)}$  and pattern law may vary with  $K$ , and limits are taken along  $K, N \rightarrow \infty$ .

### A. C.1 Proposition 1 — random-basis carrier statistics

**Proposition 1 (carrier moments, stationarity, CLT, Berry–Esseen).** *Assume:*

- **(P1)**  $T_j \geq 0$  for all  $j$ ,  $S_1 > 0$ ,  $S_2 > 0$ .
- **(P2)** The pattern entries  $\{I_{n,j}\}$  are i.i.d. across  $n$  and  $j$  with mean  $\mu_I > 0$  and variance  $\sigma_I^2 > 0$ ; the entries are nonnegative (so  $B_n \geq 0$  a.s., and  $B_n > 0$  a.s. whenever  $P(I = 0)^{|\text{supp } T|} = 0$ ).
- **(P3)** (for the CLT) Spikiness  $K_\infty \rightarrow \infty$  along the array, with the family  $\{(I - \mu_I)^2\}$  uniformly integrable (automatic for a fixed pattern law).
- **(P4)** (for Berry–Esseen)  $\nu_3 := \mathbb{E}|I - \mu_I|^3/\sigma_I^3 < \infty$ .

Then:

- (i) **Exact moments and stationarity.**  $\mathbb{E}B_n = \mu_I S_1$ ,  $\text{Var } B_n = \sigma_I^2 S_2$ , hence  $\text{CV}_B = (\sigma_I/\mu_I)/\sqrt{K_{\text{eff}}}$ . The sequence  $\{B_n\}_{n \geq 1}$  is i.i.d., hence strictly stationary and white.
- (ii) **CLT.** Under (P1)–(P3),  $(B_n - \mu_I S_1)/(\sigma_I \sqrt{S_2}) \Rightarrow \mathcal{N}(0, 1)$ .
- (iii) **Berry–Esseen.** Under (P4), for a universal constant  $C$ ,

$$\begin{aligned} \sup_x \left| P\left(\frac{B_n - \mu_I S_1}{\sigma_I \sqrt{S_2}} \leq x\right) - \Phi(x) \right| &\leq C \nu_3 \frac{\sum_j |T_j|^3}{S_2^{3/2}} \\ &\leq C \nu_3 K_\infty^{-1/2}. \end{aligned}$$

The correct CLT small parameter is  $K_\infty^{-1/2}$ , **not**  $K_{\text{eff}}^{-1}$ .

(iv) **Higher cumulants and the log carrier.** For  $r \geq 2$  the cumulants are  $\kappa_r(B_n) = \kappa_r(I) \sum_j T_j^r$ , so the standardized law of  $B_n$  depends on  $T$  through all ratios  $\sum_j T_j^r / S_2^{r/2}$ ,  $r \geq 3$ ; these vanish as  $K_\infty \rightarrow \infty$  but are not controlled by  $K_{\text{eff}}$  alone. Writing  $B_n = \mu_I S_1(1 + u_n)$  with  $\mathbb{E}u_n = 0$ ,  $\text{Var } u_n = (\sigma_I/\mu_I)^2/K_{\text{eff}}$ , a third-order Taylor expansion gives

$$\begin{aligned} \mathbb{E} \log B_n &= \log(\mu_I S_1) - \frac{1}{2} \text{Var}(u_n) + O(\mathbb{E}|u_n|^3), \\ \text{Var}(\log B_n) &= \frac{(\sigma_I/\mu_I)^2}{K_{\text{eff}}} + O(\mathbb{E}|u_n|^3). \end{aligned}$$

*Proof.* (i) Linearity of expectation and independence across  $j$  give the mean and (Bienaymé) the variance; independence across  $n$  gives the i.i.d. claim. (ii) Apply the Lindeberg–Feller CLT [Billingsley, *Probability and Measure*, Thm. 27.2] to the triangular array  $X_j = T_j(I_{n,j} - \mu_I)$  with  $s^2 = \sigma_I^2 S_2$ . Since  $|X_j| \leq \|T\|_\infty |I - \mu_I|$ , the event  $\{|X_j| > \varepsilon s\}$  is contained in  $\{|I - \mu_I| > \varepsilon \sigma_I \sqrt{K_\infty}\}$ , so the Lindeberg fraction is

$$\begin{aligned} &\frac{1}{\sigma_I^2 S_2} \sum_j T_j^2 \mathbb{E}[(I - \mu_I)^2 \mathbf{1}\{|X_j| > \varepsilon s\}] \\ &\leq \frac{1}{\sigma_I^2} \mathbb{E}[(I - \mu_I)^2 \mathbf{1}\{|I - \mu_I| > \varepsilon \sigma_I \sqrt{K_\infty}\}] \rightarrow 0 \end{aligned}$$

by uniform integrability as  $K_\infty \rightarrow \infty$ . (iii) Esseen's inequality for sums of independent, non-identically distributed variables [Petrov, *Sums of Independent Random Variables*, Ch. V] gives the bound with third-moment ratio  $\nu_3 \sum_j |T_j|^3 / S_2^{3/2}$ , and  $\sum_j |T_j|^3 \leq \|T\|_\infty S_2$  yields the  $K_\infty^{-1/2}$  form. (iv) Cumulant homogeneity ( $\kappa_r(cX) = c^r \kappa_r(X)$ ) and additivity

over independent summands give the cumulant formula. The log expansion is Taylor's theorem with Lagrange remainder applied to  $\log(1+u)$  on the event  $\{|u_n| \leq \frac{1}{2}\}$ . ■

**Status of the log expansion (flagged).** Step (iv) is a **controlled moment expansion, not a distribution-free theorem**: making the  $O(\mathbb{E}|u_n|^3)$  remainder rigorous, and ensuring integrability of  $\log B_n$  near  $B_n = 0$ , requires a high-probability bound  $P(|u_n| > \frac{1}{2})$  decaying fast enough — e.g. sub-exponential pattern entries plus  $K_\infty$ -driven concentration, or a strictly positive pattern floor  $I \geq I_{\min} > 0$ . This step is a moment calculation; it enters only through the variance scale  $\sigma_{\text{abs}}^2 = \sigma_{\text{LR}}^2 \asymp 1/K_{\text{eff}}$  (per-frame, summable-absolute, and long-run variances all coincide for the i.i.d. carrier) entering rate constants, never in identifiability statements. Zero and near-zero buckets are excluded here; the calibrated soft-log  $\psi_\alpha(c) = \log(c+\alpha)$  of Appendix D removes this restriction with a photon-dependent sensitivity factor.

a) *What this does not claim.*: The proposition does not say the carrier law is determined by  $(\mu_B, \sigma_B)$  — that reduction holds only at the Gaussian-approximation level; the exact law feels every weighted cumulant of  $T$ . For gain recovery this residual object dependence enters  $m_T = \mathbb{E} \log B_n$  only as a *time-constant* scalar, absorbed by the global-scale gauge. It also does not assert cross-row decorrelation or window-energy whitening for structured (SRHT) carriers — those require the Walsh-spectrum/ $K_4$  conditions of Appendix E, where  $K_{\text{eff}}$  alone is again insufficient.

### B. C.2 The quantitative stationarity condition (★)

**Definition (★)( $W, \epsilon, \delta$ ).** Let  $\{B_n\}_{n \leq N}$  be the carrier and  $W_n$  length- $W$  windows covering  $\{1, \dots, N\}$ . The carrier satisfies the *quantitative stationarity anchor* with tolerance  $\epsilon$  and confidence  $1 - \delta$  if there exists a scalar  $m_T \in \mathbb{R}$  (allowed to depend on  $T$ , not on  $n$ ) such that

$$P\left(\max_{1 \leq n \leq N} \left| W^{-1} \sum_{k \in W_n} \log B_k - m_T \right| \leq \epsilon\right) \geq 1 - \delta.$$

We say (★) **holds asymptotically** if for every  $\delta > 0$  there are windows  $W = W_N \rightarrow \infty$ ,  $W/N \rightarrow 0$ , with  $\epsilon = \epsilon_{W,N,\delta} \rightarrow 0$ . The scalar  $m_T$  is exactly the quantity absorbed by the global-scale gauge  $a \mapsto ca$ ,  $T \mapsto T/c$  (which shifts  $m_T \mapsto m_T - \log c$ ,  $\ell \mapsto \ell + \log c$ , leaving  $Y$  invariant).

### C. C.3 Proposition 2 — (★) is a window-estimator criterion, not a universal one

**Proposition 2.** Assume the drift is slow in the sense of hypothesis (B3) below, with  $C_\beta L (W/N)^\beta \rightarrow 0$ . Then:

(a) **Equivalence for the windowed corrector.** The blind windowed estimator  $\hat{g}_n = W^{-1} \sum_{k \in W_n} Y_k$  is uniformly consistent for  $\ell_n + m_T$  (in probability, uniformly over  $n$ ) **if and only if** (★) holds asymptotically.

(b) (★) **is not necessary for identifiability.** There are designs violating (★) that are nonetheless identifiable: a tall generic design with  $N \geq K + p$  is exactly algebraically identifiable modulo scale (Theorem A'; see Appendix B) with no stationarity whatsoever — including deterministic ordered designs; likewise,  $K_0 \geq p - 1$  known support zeros restore local identifiability under ordered Hadamard (see Appendix A).

(c) (★) **is not sufficient for algebraic identifiability.** There are settings where (★) holds yet exact separation of  $(a, T)$  is impossible: (c1) any square random design  $N = K$  — Proposition 1 gives (★), yet Theorem A applies to every invertible realization, so a nonconstant  $s \in S$  still maps  $(\ell, T)$  to an exact alias; (c2) a uniform random re-ordering  $\pi$  of the deterministic ordered Hadamard record makes (★) hold without changing the coefficient multiset — hence without adding any algebraic information. Quantitatively, Serfling's inequality for sampling without replacement [Serfling, Ann. Statist. 2 (1974) 39] gives, for the finite population  $g_k = \log B_k$  with variance  $\tau_g^2$  and range  $R_g$ ,

$$P\left(\left|W^{-1} \sum_{i=n}^{n+W-1} g_{\pi(i)} - \bar{g}\right| \geq t\right) \leq 2 \exp\left[-\frac{Wt^2}{2\tau_g^2 + \frac{2}{3}R_g t}\right],$$

so a union bound over the  $\leq N$  windows yields (★)( $W, \epsilon, \delta$ ) whenever  $\log N \ll W$  (and  $W$  remains below the drift coherence time). This device fails when some  $g_k$  are undefined (zero buckets) or the population log-variance/range is uncontrolled, and it does **not** deliver the small carrier variance  $\sigma_{\text{LR}}^2 \asymp 1/K_{\text{eff}}$  of genuinely random nonnegative patterns.

*Proof.* (a) Decompose exactly

$$\begin{aligned} \hat{g}_n - (\ell_n + m_T) &= \underbrace{\left(W^{-1} \sum_{k \in W_n} \ell_k - \ell_n\right)}_{\text{drift bias}} \\ &\quad + \underbrace{\left(W^{-1} \sum_{k \in W_n} \log B_k - m_T\right)}_{\text{carrier deviation}}. \end{aligned}$$

The drift bias is  $\leq C_\beta L (W/N)^\beta \rightarrow 0$  deterministically by (B3). Hence  $\max_n |\hat{g}_n - (\ell_n + m_T)| \rightarrow 0$  in probability iff the maximal carrier deviation does — which is verbatim the asymptotic form of (★). Both implications are immediate from this identity. (b) Theorem A' (see Appendix B) is proved without any distributional assumption on the carrier order; the support-zero criterion is Corollary 2 (see Appendix A). (c1) Theorem A (see Appendix A) holds for *any* invertible  $M$ , in particular a random draw, conditionally on the realization; (★), a statement about the temporal law, is unaffected. (c2) Serfling's bound plus the union over windows; the multiset  $\{g_k\}$ , hence every order-independent (algebraic) functional of the data, is invariant under  $\pi$ . ■

a) *What this does not claim.*: Part (a) is an equivalence for one estimator family (windowed local averages of  $\log R_n$ ; the same proof covers kernel weights of matching order). It is not a characterization of identifiability by all mechanisms — (b) and (c) are precisely the two failure modes of that over-reading. Nor does (★) by itself bound the *rate*: rates require the variance model of Theorem B, and reconstruction quality after gain correction additionally requires the corrected linear inverse problem to be well posed (see Appendix F).

### D. C.4 Theorem B — statistical relative-gain identifiability

**Theorem B.** Assume:

- **(B1) Signal model.**  $Y_n = \log R_n = \ell_n + m_T + z_n$ ,  $n = 1, \dots, N$ , where  $m_T = \mathbb{E} \log B_n$  is a time-constant scalar (finite by  $B_n > 0$  a.s. and  $\mathbb{E} |\log B_n| < \infty$ ), and  $z_n = \log B_n - m_T$ .
- **(B2) Carrier noise.**  $\{z_n\}$  is centered and strictly stationary, and either (B2a)  $\alpha$ -mixing with  $\mathbb{E} |z_0|^{2+\eta} < \infty$  and  $\sum_{h \geq 1} \alpha(h)^{\eta/(2+\eta)} < \infty$ , which by Davydov's covariance inequality [Davydov, Theory Probab. Appl. 13 (1968) 691] gives absolutely summable autocovariances — write  $\gamma(h) = \text{Cov}(z_0, z_h)$  and  $\sigma_{\text{abs}}^2 := \sum_{h \in \mathbb{Z}} |\gamma(h)| < \infty$  for the **summable-autocovariance constant**, the quantity entering every non-asymptotic variance bound below — and hence also a finite **signed long-run variance**  $\sigma_{\text{LR}}^2 := \gamma(0) + 2 \sum_{h \geq 1} \gamma(h) (\leq \sigma_{\text{abs}}^2)$ , which is the asymptotic variance of window means,  $W \text{Var}(W^{-1} \sum_{k \in W_n} z_k) \rightarrow \sigma_{\text{LR}}^2$ ; the two constants play distinct roles and must not be conflated; or (B2b), for high-probability bounds,  $\beta$ -mixing with  $\beta(k) \leq \beta_0 \exp[-(k/b)^\kappa]$  and  $\|z_n\|_{\psi_1} \leq M$ .
- **(B3) Slow drift (normalized time).**  $\ell \in S$  embeds in a normalized-time Hölder ball:  $|\ell_j - \ell_k| \leq L(|j - k|/N)^\beta$ ,  $\beta \in (0, 2]$  — equivalently the per-frame constant  $L_a = L N^{-\beta}$  of Appendix D — matched to the window order so the windowed bias obeys  $|W^{-1} \sum_{k \in W_n} \ell_k - \ell_n| \leq C_\beta L (W/N)^\beta$  (for  $\beta \in (1, 2]$  this requires centered symmetric windows so the first-order bias cancels).
- **(B4) Window asymptotics.** Both  $W = W_N \rightarrow \infty$  and  $W/N \rightarrow 0$ ; for fixed  $L$  the windowed drift bias  $C_\beta L (W/N)^\beta \rightarrow 0$  then holds automatically (for a triangular array with drift constant  $L = L_N$  growing with  $N$ , require additionally  $L (W/N)^\beta \rightarrow 0$ ).

Then:

(i) **Pointwise consistency with rate.** Under (B2a), the non-asymptotic bound  $\mathbb{E} [\hat{g}_n - (\ell_n + m_T)]^2 \leq C(L^2(W/N)^{2\beta} + \sigma_{\text{abs}}^2/W) \rightarrow 0$  holds, the windowed bias being  $O(L(W/N)^\beta)$ ; the asymptotic variance of the window mean is  $\sigma_{\text{LR}}^2/W$  (signed long-run variance, (B2a)). Under (B2b), with block length  $q = \lceil b[\log(4W/\delta)]^{1/\kappa} \rceil$ , with probability  $\geq 1 - \delta$ ,

$$|\hat{g}_n - (\ell_n + m_T)| \leq C_\beta L (W/N)^\beta + C \left[ \sqrt{\frac{\nu_q^2 q \log(4/\delta)}{W}} + \frac{M q \log(4/\delta)}{W} \right],$$

where  $\nu_q^2$  is a block-variance proxy bounded by a multiple of  $\sigma_{\text{abs}}^2$ ; uniformity over  $n \leq N$  follows with  $\delta \mapsto \delta/N$ .

(ii) **Centered-gain recovery.**  $\hat{\ell}_n^c := \hat{g}_n - N^{-1} \sum_m \hat{g}_m$  satisfies  $\mathbb{E} [\hat{\ell}_n^c - (\ell_n - \bar{\ell})]^2 \leq 4C(L^2(W/N)^{2\beta} + \sigma_{\text{abs}}^2/W)$ ; hence the relative gain  $\ell_n - \bar{\ell}$  (equivalently  $a_n$  up to one multiplicative constant) is identifiable and consistently estimable.

(iii) **Gauge.** The scalar  $m_T + \bar{\ell}$  is not identifiable: for every  $c > 0$ , the substitution  $a \mapsto ca$ ,  $T \mapsto T/c$  leaves the law of  $\{Y_n\}$  exactly invariant while shifting  $\ell \mapsto \ell + \log c$ ,  $m_T \mapsto m_T - \log c$ . This is precisely the constant direction  $\text{span}\{1\} \subset S$  of Theorem A, and it is the only ambiguity remaining.

*Proof.* (i) Bias–variance split as in Proposition 2(a). Bias: (B3). Variance: for a stationary sequence,  $\text{Var}(W^{-1} \sum_{k \in W_n} z_k) = W^{-1} \sum_{|h| < W} (1 - |h|/W) \gamma(h) \leq W^{-1} \sum_h |\gamma(h)| =$

$\sigma_{\text{abs}}^2/W$  — the non-asymptotic bound, with the summable-autocovariance constant; as  $W \rightarrow \infty$  the same expression converges to  $\sigma_{\text{LR}}^2/W (1 + o(1))$  with the *signed* long-run variance, which is the correct asymptotic constant and can be strictly smaller. Finiteness of both is (B2a)/Davydov;  $(x + y)^2 \leq 2x^2 + 2y^2$  combines bias and variance. For (B2b), partition the window into alternating blocks of length  $q$ ; the exponential  $\beta$ -mixing coupling replaces blocks by independent copies up to total-variation cost  $W\beta(q)/q \leq \delta/2$  by the choice of  $q$ , and Bernstein's inequality for independent sub-exponential block sums [Merlevède–Peligrad–Rio, Probab. Theory Relat. Fields 151 (2011) 435, and the standard blocking scheme] gives the display. (ii)  $\hat{\ell}_n^c - (\ell_n - \bar{\ell}) = [\hat{g}_n - (\ell_n + m_T)] - N^{-1} \sum_m [\hat{g}_m - (\ell_m + m_T)]$ ; Jensen bounds the second term's second moment by the maximal pointwise one, and  $(x + y)^2 \leq 2x^2 + 2y^2$  gives the factor 4. (iii) Direct computation:  $B_n(T/c) = B_n(T)/c$ , so  $Y_n$  is unchanged term by term; conversely, consistency in (ii) implies identifiability of the estimand  $\ell_n - \bar{\ell}$  (two parameter configurations inducing the same law of  $\{Y_n\}$  would force the same limit), so the ambiguity is exactly one dimensional. ■

a) *What this does not claim.* Theorem B is a **statistical, asymptotic** statement about *relative-gain* recovery in a triangular array; it is not exact finite-sample algebraic identifiability of  $(a, T)$  — Theorem A still holds verbatim for a square random design, and the two results are complementary, not in tension. It presupposes  $B_n > 0$  and integrable logs: in photon-counting records with empty bins the theorem must be replaced by its calibrated soft-log variant (see Appendix D), where  $\psi_\alpha(c) = \log(c + \alpha)$ , the calibration curve  $m_\alpha$  is inverted, and the same bias–variance structure holds with sensitivity  $\kappa_\alpha$  and photon-starved variance  $\sim 1/(W\lambda)$ . The constants  $\sigma_{\text{abs}}^2$  and  $\sigma_{\text{LR}}^2$  are model inputs (they coincide for the i.i.d. carrier, where  $\gamma(h) = 0$  for  $h \neq 0$ ); identifying them with  $(\sigma_I/\mu_I)^2/K_{\text{eff}}$  invokes the flagged expansion of Proposition 1(iv). Finally, consistent gain correction does not by itself guarantee good reconstruction — the corrected inverse problem must be independently well posed, and the propagation of the residual gain variance  $v$  through a basis with leverage  $B_L$ , floor  $C_0/N$ , and the  $\rho_{\text{pair}}$  flip boundary is the subject of Appendix F.

## APPENDIX D

### ESTIMATION RATES, MINIMAX LOWER BOUNDS, AND FISHER GEOMETRY

This appendix proves the rate and optimality statements of Sec. 6: the high-probability windowed-estimator bound (Theorem C's engine), its minimax optimality at the level of rates (Theorem D), the quotient-Fisher geometry underlying the identifiability dichotomy, and the low-photon soft-log rate. Notation is locked to the main text:  $R_n = a_n B_n$ ,  $\ell = \log a \in S$  with  $\dim S = p$  and  $\text{span}\{1\} \subset S$ ,  $Y_n = \log R_n = \ell_n + m_T + z_n$  with  $m_T = \mathbb{E} \log B_n$  a time-independent scalar, windows  $W$ , and the spread functionals  $K_{\text{eff}}$ ,  $K_\infty$  as in Appendix C. The Hölder smoothness order is written  $\beta \in (0, 2]$  with seminorm  $L_a$  (the drift constant of Theorem C); the symbol  $\alpha$  is reserved for the soft-log offset  $\psi_\alpha(c) = \log(c + \alpha)$ .

Throughout, the *discrete Hölder ball*  $H_\beta(L_a)$  is defined operationally:  $\ell$  belongs to  $H_\beta(L_a)$  relative to a window family  $\{W_n\}$  of order  $\beta$  if the deterministic smoothing bias obeys

$$\left| W^{-1} \sum_{k \in W_n} \ell_k - \ell_n \right| \leq C_\beta L_a W^\beta \quad \text{for all } n. \quad (\text{D.1})$$

For  $\beta \in (0, 1]$  a one-sided moving average of  $\ell$  with modulus  $|\ell_k - \ell_n| \leq L_a |k - n|^\beta$  satisfies (D.1). For  $\beta \in (1, 2]$ , (D.1) requires a *centered symmetric* window (or a kernel of order  $\lfloor \beta \rfloor$ ), so that the first-order bias cancels and only curvature  $L_2$  survives; a one-sided window does not attain  $\beta > 1$ .

**A. D.1 Lemma D.1 (windowed gain-estimation rate under mixing)**

a) *Statement.*: Assume:

- 1) (*Model*)  $Y_n = \ell_n + m_T + z_n$ , with  $\{z_n\}$  strictly stationary and  $\mathbb{E}z_n = 0$ .
- 2) (*Mixing*)  $\{z_n\}$  is  $\beta$ -mixing with  $\beta(k) \leq \beta_0 \exp[-(k/b)^\kappa]$  for some  $\beta_0, b > 0, \kappa > 0$ .
- 3) (*Tails*)  $\|z_n\|_{\psi_1} \leq M$  (sub-exponential).
- 4) (*Smoothness*)  $\ell \in H_\beta(L_a)$  in the sense (D.1), with the window order matched to  $\beta$  as above.

Let  $\hat{g}_n = W^{-1} \sum_{k \in W_n} Y_k$  and set  $q = \lceil b[\log(4W/\delta)]^{1/\kappa} \rceil$ . Then for each fixed  $n$  and  $\delta \in (0, 1)$ , with probability at least  $1 - \delta$ ,

$$\begin{aligned} |\hat{g}_n - (\ell_n + m_T)| &\leq C_\beta L_a W^\beta \\ &\quad + C \left[ \sqrt{\frac{\nu_q^2 q \log(4/\delta)}{W}} + \frac{M q \log(4/\delta)}{W} \right], \end{aligned} \quad (\text{D.2})$$

where  $\nu_q^2 = \text{Var}(q^{-1/2} \sum_{i=1}^q z_i)$  is the block-variance proxy; when the autocovariances are absolutely summable,  $\nu_q^2 \leq C' \sigma_{\text{abs}}^2$  with the summable-autocovariance constant  $\sigma_{\text{abs}}^2 = \sum_h |\gamma(h)|$  of Appendix C (hypothesis (B2a)) — the signed long-run variance  $\sigma_{\text{LR}}^2 \leq \sigma_{\text{abs}}^2$  of Appendix C is the asymptotic variance of window means and plays no role in these non-asymptotic bounds. Uniformity over  $n = 1, \dots, N$  follows by replacing  $\delta$  with  $\delta/N$  (a  $\log N$  inflation inside the brackets).

(*Expectation version, weaker hypotheses.*) If instead  $\{z_n\}$  is only  $\alpha$ -mixing with  $\sum_{h \geq 1} \alpha(h)^{\eta/(2+\eta)} < \infty$  and  $\|z_0\|_{2+\eta} < \infty$  for some  $\eta > 0$ , then

$$\mathbb{E}[\hat{g}_n - (\ell_n + m_T)]^2 \leq C L_a^2 W^{2\beta} + C \sigma_{\text{abs}}^2 / W. \quad (\text{D.3})$$

b) *Proof.*: *Bias.*  $\hat{g}_n - (\ell_n + m_T) = [W^{-1} \sum_k \ell_k - \ell_n] + W^{-1} \sum_k z_k$ ; the first bracket is bounded by  $C_\beta L_a W^\beta$  by (D.1).

*Fluctuation, high probability.* Partition the window  $W_n$  into  $m = \lceil W/q \rceil$  consecutive blocks of length  $q$  and separate them into odd- and even-indexed families. By Berbee's coupling lemma [Berbee 1979; see Rio, *Asymptotic Theory of Weakly Dependent Random Processes*, 2017], on an enlarged probability space each family can be replaced by a sequence of *independent* blocks with the same marginals, at total-variation cost at most  $m\beta(q) \leq \beta_0 W \exp[-(q/b)^\kappa]$ . The choice  $q = \lceil b[\log(4W/\delta)]^{1/\kappa} \rceil$  makes this cost at most  $\delta/2$  (adjusting  $\beta_0$  into  $C$ ). On the coupling event, each family

is a sum of independent block sums  $S_j = \sum_{i \in \text{block } j} z_i$  with  $\|S_j\|_{\psi_1} \leq CqM$  (triangle inequality for the  $\psi_1$  norm) and  $\text{Var}(S_j) = q\nu_q^2$ . Bernstein's inequality for sums of independent sub-exponential variables [e.g. Vershynin, *High-Dimensional Probability*, Thm. 2.8.1] applied to each family, with a union bound over the two families (whence the  $4/\delta$ ), yields the bracketed term in (D.2) with the characteristic Gaussian-plus-linear tail: a  $\sqrt{\nu_q^2 q \log(4/\delta)/W}$  moderate-deviation term and an  $Mq \log(4/\delta)/W$  heavy-tail correction. When  $\sum_h |\text{Cov}(z_0, z_h)| < \infty$ ,  $\nu_q^2 = \text{Var}(z_0) + 2 \sum_{h=1}^{q-1} (1 - h/q) \text{Cov}(z_0, z_h) \leq C' \sigma_{\text{abs}}^2$ .

*Fluctuation, expectation.* Davydov's covariance inequality [Davydov 1968] gives  $|\text{Cov}(z_0, z_h)| \leq 8\alpha(h)^{\eta/(2+\eta)} \|z_0\|_{2+\eta}^2$ ; the summability hypothesis makes the autocovariance series absolutely summable, so  $\text{Var}(W^{-1} \sum_{k \in W_n} z_k) \leq \bar{\sigma}^2/W$  with  $\bar{\sigma}^2 \leq C\sigma_{\text{abs}}^2$ . Combining with the squared bias via  $(x+y)^2 \leq 2x^2 + 2y^2$  gives (D.3). ■

**Corollary D.2 (optimized rate = Theorem C's high-photon case).** *Minimizing the right side of (D.3) over  $W$  gives*

$$W^* \asymp (\sigma_{\text{abs}}^2 / L_a^2)^{1/(2\beta+1)}, \quad \text{MSE}^* \asymp L_a^{2/(2\beta+1)} \sigma_{\text{abs}}^{4\beta/(2\beta+1)}, \quad (\text{D.4})$$

*valid when  $W^* \in [1, N]$  and  $W^*$  is shorter than the gain coherence time; otherwise the boundary window  $W = 1$  or  $W = N$  applies. For  $\beta = 1$  with per-frame drift scale  $L_1 \asymp s\rho_{\text{pair}}$  (adjacent-frame decorrelation convention — not  $\rho_{\text{bw}}$ ), this is  $\text{MSE}^* \asymp \sigma_{\text{abs}}^{4/3} (s\rho_{\text{pair}})^{2/3}$ ; for random carriers the carrier is white,  $\sigma_{\text{abs}}^2 = \sigma_{\text{LR}}^2 \approx (\sigma_I / \mu_I)^2 / K_{\text{eff}}$  under the spikiness condition  $K_\infty \rightarrow \infty$  of Prop. 1.*

c) *What this does not claim.*: The constants  $C, C_\beta$  are not tracked and depend on the mixing constants  $(\beta_0, b, \kappa)$  and  $M$ . The bound is for the *relative* gain  $\ell_n + m_T$ ; the scalar  $m_T + \text{gauge}(\ell)$  is not estimable (Theorem B). Nothing is claimed for  $\beta > 2$  (would need local polynomials), for non-stationary carriers, or about *adaptive* window selection when  $(\beta, L_a)$  are unknown. The step  $\nu_q^2 \lesssim \sigma_{\text{abs}}^2$  requires summable autocovariances, which the stated mixing plus moment hypotheses supply.

**B. D.2 Theorem D (minimax lower bounds; rate level, constants open)**

a) *Statement.*: (*Hölder / Assouad–two-point.*) Consider the i.i.d. Gaussian submodel  $z_n \sim \mathcal{N}(0, \sigma^2)$  — a member of the model class of Lemma D.1. Then for every  $n$  interior to the design,

$$\inf_{\hat{\ell}} \sup_{\ell \in H_\beta(L_a)} \mathbb{E}(\hat{\ell}_n - \ell_n)^2 \geq c_\beta L_a^{2/(2\beta+1)} \sigma^{4\beta/(2\beta+1)}, \quad (\text{D.5})$$

the infimum over all measurable estimators of the gauge-fixed log-gain. In normalized continuous time ( $N$  samples on  $[0, 1]$ , bandwidth  $h = W/N$ ) the equivalent form is  $\inf \sup \mathbb{E}(\hat{\ell}(t) - \ell(t))^2 \geq c(\sigma^2/N)^{2\beta/(2\beta+1)} L^{2/(2\beta+1)}$ .

(*Sobolev / van Trees–Pinsker.*) In the Gaussian sequence formulation — expand  $\ell(t) = \sum_j \theta_j \varphi_j(t)$  in a Fourier basis

and impose the Sobolev ellipsoid  $\sum_j j^{2\beta} \theta_j^2 \leq L^2$  — the integrated risk obeys

$$\inf_{\hat{\ell}} \sup_{\|\ell\|_{H^\beta} \leq L} \frac{1}{N} \sum_n \mathbb{E}(\hat{\ell}_n - \ell_n)^2 \geq c_\beta L^{2/(2\beta+1)} (\sigma^2/N)^{2\beta/(2\beta+1)}, \quad (\text{D.6})$$

with the gauge direction (the constant mode, confounded with  $m_T$ ) excluded from the risk. Consequently the windowed estimator of Lemma D.1 is **minimax-rate optimal** over  $H_\beta(L_a)$  for  $\beta \leq 1$  with one-sided windows and for  $\beta \leq 2$  with centered symmetric windows. **Sharp minimax constants are open** (this is Theorem D of the main text).

*b) Proof.: Two-point (pointwise).* Fix  $n_0$  and let  $\phi$  be a fixed  $C^\infty$  bump,  $\text{supp } \phi \subset [-1, 1]$ ,  $\phi(0) = 1$ ,  $\|\phi\|_{H_\beta} \leq 1$ . Take the hypotheses  $\ell^{(0)} \equiv 0$  and  $\ell_k^{(1)} = L_a W_0^\beta \phi((k - n_0)/W_0)$ ; both lie in  $H_\beta(L_a)$  (up to an absolute constant absorbed into  $c_\beta$ ). Since the noise is i.i.d. Gaussian, the Kullback–Leibler divergence between the two data laws is  $\text{KL} = \frac{1}{2\sigma^2} \sum_k (\ell_k^{(1)})^2 \leq C L_a^2 W_0^{2\beta+1} / \sigma^2$ . Choosing  $W_0 \asymp (\sigma^2/L_a^2)^{1/(2\beta+1)}$  makes  $\text{KL} \leq c < \infty$ , and the two hypotheses are separated at  $n_0$  by  $|\ell_{n_0}^{(1)} - \ell_{n_0}^{(0)}| = L_a W_0^\beta$ . Le Cam’s two-point lemma [Tsybakov, *Introduction to Nonparametric Estimation*, 2009, §2.4] then gives (D.5). The nuisance scalar  $m_T$  only enlarges the model, so the bound holds a fortiori; the bumps can be taken mean-centered over the record so that the two hypotheses differ in the gauge-invariant component  $\ell - \bar{\ell}$  itself. For the integrated version, tile  $[1, N]$  with  $\sim N/W_0$  disjoint bumps carrying independent signs and apply Assouad’s lemma [Tsybakov 2009, Thm. 2.12].

*van Trees / Pinsker.* In the sequence model  $y_j = \theta_j + (\sigma/\sqrt{N})\epsilon_j$  (exact for equispaced discrete-Fourier regression with white Gaussian noise), place independent priors  $\theta_j \sim \mathcal{N}(0, \tau_j^2)$  truncated inside the ellipsoid. The van Trees inequality [Gill–Levit 1995] bounds the Bayes risk of each coordinate by the reciprocal of (data Fisher + prior Fisher):  $\text{BayesRisk} \geq \sum_j (N/\sigma^2 + \tau_j^{-2})^{-1}$ , the constant mode omitted (gauge). Maximizing the right side over prior spectra supported in the ellipsoid is Pinsker’s calculation [Pinsker 1980] and yields the scaling (D.6). Since minimax risk dominates Bayes risk, (D.6) follows. ■

*c) What this does not claim.:* (i) Constants: neither  $c_\beta$  nor a Pinsker-type sharp constant is claimed for our model; the exact Pinsker constant transfers only where the acquisition is genuinely the Gaussian sequence model — for mixing, non-Gaussian carriers the statement is at rate level only (a Brown–Low-type asymptotic-equivalence argument would be needed for more, and is not attempted). (ii) The lower bound is proven inside the i.i.d. Gaussian *submodel*; this suffices for minimaxity of the full class (sup over the class dominates sup over the submodel, while the upper bound (D.3) holds over the full class), but no lower bound is claimed *within* every fixed mixing law. (iii) No adaptivity claim: rate optimality is for known  $(\beta, L_a)$ .

*C. D.3 Proposition D.4 (quotient Fisher geometry: singularity  $\Leftrightarrow$  the Theorem-A ambiguity)*

*a) Statement.:* (a) *Linear-Gaussian window model.* Let  $\ell = U\theta$  with  $U \in \mathbb{R}^{N \times p'}$ , and observe  $Y = U\theta + m\mathbf{1} + z$ ,  $z \sim \mathcal{N}(0, \Sigma)$ ,  $\Sigma \succ 0$  known,  $m \in \mathbb{R}$  the unknown carrier log-mean (confounded with global gain scale). The Fisher information for  $\theta$  after eliminating the nuisance  $m$  is

$$I_\theta = U^\top \Sigma^{-1/2} P_\perp \Sigma^{-1/2} U, \quad P_\perp = I - \frac{\Sigma^{-1/2} \mathbf{1} \mathbf{1}^\top \Sigma^{-1/2}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}. \quad (\text{D.7})$$

For a linear functional  $L\theta$ : if  $\text{row}(L) \subseteq \text{range}(I_\theta)$ , every unbiased estimator obeys  $\text{Cov}(L\hat{\theta}) \succeq L I_\theta^+ L^\top$  (Moore–Penrose CRB for singular information [Rao–Mitra; Stoica–Marzetta 2001]); if  $L$  has a nonzero component along  $\ker I_\theta$ , **no unbiased estimator of  $L\theta$  exists and the CRB is infinite** — the functional is not locally estimable.

*(b) Quotient statement.* Let  $\Theta$  be the parameter manifold,  $G$  the global-scale group  $(a, T) \mapsto (ca, T/c)$ , and  $P_\theta$  a dominated model differentiable in quadratic mean. Define the quotient score operator  $S_\theta : T_{[\theta]}(\Theta/G) \rightarrow L_0^2(P_\theta)$  and  $I_\theta = S_\theta^* S_\theta$ . Then:  $I_\theta \succ 0$  on  $T_{[\theta]}(\Theta/G)$  **iff**  $[\theta] \mapsto P_\theta$  is an immersion, i.e. iff the model is *locally differentially identifiable* modulo scale. Any  $C^1$  ambiguity curve  $[\theta(t)]$  with  $P_{\theta(t)} = P_{\theta(0)}$  satisfies  $S_\theta \theta'(0) = 0$ , hence produces a Fisher null direction. For the Theorem-A ambiguity this implication is *exact*: the explicit data-preserving path  $\ell' = \ell + ts$ ,  $T' = M^{-1}(c \odot e^{-ts})$ ,  $s \in S \setminus \text{span}\{1\}$ , is a genuine ambiguity curve for a square invertible design, so the Fisher matrix in the full  $(\ell, T)$  parameterization is singular along its tangent, and the CRB is  $+\infty$  for every functional that varies along it.

*(c) Local-window corollary.* For a window over which  $\ell$  is approximately constant, the Gaussian CRB reduces to  $\text{Var}(\hat{\ell}_n) \geq \sigma_z^2 / W_{\text{eff}}$  with  $W_{\text{eff}} = W / (1 + 2 \sum_h \rho_z(h))$  under weak dependence (equivalently,  $\sigma_z^2$  replaced by the signed long-run variance  $\sigma_{\text{LR}}^2$ , which is the exact asymptotic constant here) — the Fisher-side counterpart of the  $\sigma_{\text{abs}}^2 / W$  upper bound in (D.3).

*b) Proof.:* (a) The joint Fisher matrix of  $(\theta, m)$  in the Gaussian model is the block matrix  $\begin{pmatrix} U^\top \Sigma^{-1} U & U^\top \Sigma^{-1} \mathbf{1} \\ \mathbf{1}^\top \Sigma^{-1} U & \mathbf{1}^\top \Sigma^{-1} \mathbf{1} \end{pmatrix}$ . Nuisance elimination is the Schur complement with respect to the  $m$ -block [Lehmann–Casella, *Theory of Point Estimation*, §6], which is exactly (D.7). The pseudoinverse CRB for singular FIM is Stoica–Marzetta (2001). The infinite-CRB claim is cleanest without CRB machinery: a direction  $u \in \ker I_\theta$  that is tangent to an exact ambiguity curve leaves the data law invariant, so any estimator has *identical* distribution under  $\theta$  and  $\theta + tu$ ; unbiasedness for a functional with  $Lu \neq 0$  would force  $L\theta = L(\theta + tu)$ , a contradiction — hence no unbiased finite-variance estimator exists.

(b) Differentiability in quadratic mean makes  $S_\theta$  well defined and  $I_\theta = S_\theta^* S_\theta$  the pullback metric [van der Vaart, *Asymptotic Statistics*, Ch. 7].  $I_\theta \succ 0$  iff  $S_\theta$  is injective iff the differential of  $[\theta] \mapsto \sqrt{dP_\theta}$  is injective, i.e. an immersion. The ambiguity-curve implication is the chain rule:  $t \mapsto P_{\theta(t)}$  constant  $\Rightarrow$  its  $L^2$  derivative  $S_\theta \theta'(0)$  vanishes. For Theorem A the curve is explicit (see Appendix A), so no genericity is needed. ■

c) *What this does not claim.*: The converse direction — Fisher singularity implying an actual *ambiguity manifold* — requires constant-rank (or analytic) regularity so that the rank defect integrates (rank theorem); Fisher singularity by itself only defeats *local differential* identifiability. Nonsingular Fisher does not exclude isolated *global* aliases. Part (a) assumes Gaussian noise with known  $\Sigma$ ; part (c) is a Gaussian-model computation used as an order-of-magnitude floor, not a bound proven for the actual log-carrier distribution. No efficiency claim is made for the windowed estimator beyond rate (constants open, Theorem D).

D. D.4 Theorem C (low-photon soft-log rate and the  $1/(W\bar{\lambda})$  law)

a) *Statement.*: Let counts follow the calibrated model  $C_n \mid B_n, \ell_n \sim \text{Pois}(\lambda_n)$ ,  $\lambda_n = \Lambda_0 e^{\ell_n} B_n + d$ , with  $d \geq 0$  dark counts and  $\{B_n\}$  the stationary random carrier. Fix  $\alpha > 0$ , let  $\psi_\alpha(c) = \log(c + \alpha)$ , define the calibration curve  $m_\alpha(\theta) = \mathbb{E}[\psi_\alpha(C)]$  for  $C \mid B \sim \text{Pois}(\Lambda_0 e^\theta B + d)$ , and the estimator  $\hat{\theta}_n = m_\alpha^{-1}(W^{-1} \sum_{k \in W_n} \psi_\alpha(C_k))$  (global-scale gauge imposed). Assume:

- 1)  $\{\psi_\alpha(C_n)\}$  is stationary and  $\beta$ -mixing after removing the slow gain, with long-run variance  $\sigma_{\alpha, \text{LR}}^2$ ;
- 2)  $\ell \in H_\beta(L_a)$ ,  $\beta \in (0, 2]$ , in the sense (D.1);
- 3) on the operating gain range,  $0 < \kappa_{\min} \leq m'_\alpha(\theta) \leq \kappa_{\max} < \infty$ .

Then

$$\mathbb{E}(\hat{\theta}_n - \ell_n - \text{gauge})^2 \leq C \left( \frac{\kappa_{\max}}{\kappa_{\min}} \right)^2 L_a^2 W^{2\beta} + C \frac{\sigma_{\alpha, \text{LR}}^2}{\kappa_{\min}^2 W}, \quad (\text{D.8})$$

and optimizing over  $W$ ,

$$\text{MSE}_\alpha^* \leq C \kappa_{\min}^{-2} [\kappa_{\max} L_a]^{2/(2\beta+1)} \sigma_{\alpha, \text{LR}}^{4\beta/(2\beta+1)} \quad (\text{D.9})$$

— Theorem C of the main text. In the photon-starved regime  $\bar{\lambda} \ll 1$  the variance term of (D.8) becomes  $C/(W\bar{\lambda})$ , which is minimax-order sharp: the Poisson Fisher information for the log-intensity parameter is  $I(\theta) = \lambda$ , so  $W$  counts carry information  $\asymp W\bar{\lambda}$  and no estimator improves on the  $1/(W\bar{\lambda})$  order (van Trees). Zeros only reduce Fisher information; they never make the estimator diverge.

b) *Proof.*: *Sensitivity and variance of the soft-log.* The Poisson derivative identity  $\frac{d}{d\lambda} \mathbb{E}f(C) = \mathbb{E}[f(C+1) - f(C)]$  (differentiate  $\sum_c e^{-\lambda} \lambda^c f(c)/c!$  term by term) gives

$$\kappa_\alpha(\lambda) := \frac{d}{d \log \lambda} \mathbb{E} \log(C + \alpha) = \lambda \mathbb{E} \log \frac{C + 1 + \alpha}{C + \alpha},$$

with the expansions  $\kappa_\alpha(\lambda) = 1 + O(1/\lambda)$  for  $\lambda \gg 1$  and  $\kappa_\alpha(\lambda) = \lambda \log(1 + 1/\alpha) + O(\lambda^2)$  for  $\lambda \ll 1$ . The Poisson Poincaré inequality [Klaassen 1985; Bobkov–Ledoux 1998],  $\text{Var} f(C) \leq \lambda \mathbb{E}[(f(C+1) - f(C))^2]$ , yields  $\text{Var}[\psi_\alpha(C)] = O(1/\lambda)$  at high  $\lambda$  and  $= \lambda \log^2(1 + 1/\alpha) + O(\lambda^2)$  at low  $\lambda$ . These verify that hypothesis 3 holds with  $\kappa_{\min} \asymp c_\alpha \bar{\lambda}$  in the starved regime and  $\kappa_{\min} \approx \kappa_{\max} \approx 1$  at high counts.

*Rate.* Write  $Y_k = \psi_\alpha(C_k)$  and decompose

$$W^{-1} \sum_{k \in W_n} Y_k - m_\alpha(\ell_n) = \left[ W^{-1} \sum_k m_\alpha(\ell_k) - m_\alpha(\ell_n) \right] + \left[ W^{-1} \sum_k (Y_k - m_\alpha(\ell_k)) \right].$$

Since  $m_\alpha$  is  $\kappa_{\max}$ -Lipschitz on the gain range,  $m_\alpha \circ \ell$  inherits the discrete Hölder modulus scaled by  $\kappa_{\max}$ , so the first bracket is  $\leq C \kappa_{\max} L_a W^\beta$  by (D.1). The second bracket is a window average of centered stationary  $\beta$ -mixing variables; the expectation bound of Lemma D.1 controls its variance by  $C \sigma_{\alpha, \text{LR}}^2 / W$ . Because  $m'_\alpha \geq \kappa_{\min} > 0$ ,  $m_\alpha$  is strictly increasing and  $m_\alpha^{-1}$  is  $1/\kappa_{\min}$ -Lipschitz (inverse function theorem); composing and squaring gives (D.8), and the standard bias-variance optimization gives (D.9).

*Photon-starved limit.* Substituting  $\kappa_{\min} \asymp c_\alpha \bar{\lambda}$  and  $\sigma_{\alpha, \text{LR}}^2 \asymp c_\alpha^2 \bar{\lambda}$  into the variance term of (D.8) gives  $\sigma_{\alpha, \text{LR}}^2 / (\kappa_{\min}^2 W) \asymp 1/(W\bar{\lambda})$ . **This substitution is a leading-order moment calculation:** it identifies the long-run variance with the single-count variance (neglecting residual temporal correlation of the carrier at low counts) and uses the small- $\lambda$  expansions above; the *order* is what is claimed, not the constant. Sharpness at rate level: for  $\theta = \log$ -intensity,  $I(\theta) = (\partial_\theta \lambda)^2 / \lambda = \lambda$ ; independence across frames gives total information  $\asymp W\bar{\lambda}$ , and the van Trees inequality converts this into a  $c/(W\bar{\lambda})$  lower bound on the local MSE of any estimator. ■

**Shrinkage-bias caveat (load-bearing, per Sec. 6).** For  $\bar{\lambda} < 1$  the estimator is *bias-dominated*: the mass at  $C = 0$  pulls  $W^{-1} \sum \psi_\alpha$  toward  $\log \alpha$ , and the amplification factor  $(\kappa_{\max}/\kappa_{\min})^2$  in the bias term of (D.8) blows up as  $\kappa_{\min} \propto \bar{\lambda} \rightarrow 0$ . Empirically the MSE can then sit *below* the Fisher reference curve — that is bias, not super-efficiency, and it does not contradict the lower bound (which constrains variance-limited, locally unbiased behavior). The  $1/(W\bar{\lambda})$  law is asserted only in the variance-limited regime above the  $\bar{\lambda} \sim 1$  crossover; below it, honest accounting reports the bias floor. Recovering the rate there requires the offset-design, Anscombe ( $2\sqrt{C} + 3/8$ ), or full Poisson-mixture MLE variants; for the mixture MLE the claim  $\text{MSE} \approx 1/(WJ(\theta))$  with  $J \sim \mathbb{E}\lambda$  at low photons is standard parametric asymptotics under regularity of the mixture likelihood and is *not* re-proven here.

c) *What this does not claim.*: No absolute gain scale is estimated (gauge remains). The calibration curve  $m_\alpha$  is assumed known (flat-field) or jointly calibrated up to scale; miscalibration adds a bias not covered by (D.8). Constants in (D.8)–(D.9) depend on the mixing constants of  $\psi_\alpha(C_n)$  and are not tracked. As  $\Phi \rightarrow 0$  with no offset and no reference, Fisher information for the gain vanishes — an information limit no estimator evades (Sec. 10). Finally, (D.9) is an upper bound whose minimax matching is at rate level only, inherited from Theorem D.

## APPENDIX E

### SRHT WHITENING: EXACT WALSH CONDITION AND PERMUTATION BOUNDS

This appendix proves the claims of Sec. 7: the exact covariance identity  $\text{Cov}_D(Z_g, Z_h) = \hat{w}(g+h)$ , the necessary-and-sufficient Walsh-flatness conditions for sign-only whitening (pairwise, spectral-window, and window-average forms),

the Bernstein–Serfling permutation bound and its union-bound corollary, the permutation-alone carrier-variance identity with its flat-object counterexample, and the Hanson–Wright window-energy bound.

#### A. E.0 Setting and standing assumptions

- **(E1)**  $K = 2^m$  and pixels are indexed by the Walsh group  $G = \mathbb{F}_2^m$ , with characters  $\chi_g(j) = (-1)^{g \cdot j}$ , unnormalized Hadamard matrix  $H_{g,j} = \chi_g(j)$ , and orthonormal carrier  $U = K^{-1/2}H$ . For every  $g \neq 0$ ,  $\chi_g$  is balanced: exactly  $K/2$  entries equal  $+1$ .
- **(E2)** The object  $T \in \mathbb{R}^K$  is deterministic and nonzero. Write  $w_j = T_j^2$ ;  $S_1$ ,  $S_2$ , and the spread functionals  $K_{\text{eff}}$ ,  $K_\infty$ ,  $K_4$  are as in Appendix C. The Walsh transform of  $w$  is  $\hat{w}(h) = \sum_j \chi_h(j) w_j$ , so  $\hat{w}(0) = S_2$ .
- **(E3)** The SRHT coefficient vector is  $x = UDPT$ , where  $D = \text{diag}(\eta_1, \dots, \eta_K)$  with i.i.d. Rademacher signs  $\eta_j$ , and  $P$  is a uniformly random pixel permutation independent of  $D$ . The normalized carrier is  $Z_g = \sqrt{K} x_g = \sum_j \chi_g(j) \eta_j (PT)_j$ . Physical nonnegative patterns are the offset form  $B_g = \mu S_1 + \sigma Z_g$  with positivity margin  $\mu S_1 > \sigma |Z_g|$ ; the bucket record is  $R_n = a_n B_n$  with  $\ell = \log a$  as in the main text. A *window* is a subset  $A \subset G$  of  $W = |A|$  rows.

Permutation leaves  $S_1$ ,  $S_2$ ,  $K_{\text{eff}}$ ,  $K_4$ ,  $K_\infty$  invariant; it only reorders  $w$ , i.e. replaces  $\hat{w}$  by  $\hat{w}^P$  with  $w_j^P = T_{Pj}^2$ .

#### B. E.1 The exact covariance identity

**Lemma E.1.** *Under (E1)–(E3) with  $P$  fixed (in particular  $P = \text{id}$ ),  $\mathbb{E}_D Z_g = 0$  and for all  $g, h \in G$*

$$\text{Cov}_D(Z_g, Z_h) = \sum_j \chi_{g+h}(j) (PT)_j^2 = \widehat{w^P}(g+h). \quad (\text{E.1})$$

**Proof.**  $\mathbb{E} \eta_j = 0$  gives the mean. Since  $\mathbb{E} \eta_i \eta_j = \delta_{ij}$ ,

$$\begin{aligned} \mathbb{E}_D Z_g Z_h &= \sum_{i,j} \chi_g(i) \chi_h(j) (PT)_i (PT)_j \mathbb{E} \eta_i \eta_j \\ &= \sum_j \chi_g(j) \chi_h(j) (PT)_j^2, \end{aligned}$$

and  $\chi_g \chi_h = \chi_{g+h}$  by the character property of the Walsh group. ■

Consequently the covariance of any window  $A$  is the pattern matrix  $\Gamma_A(g, h) = \widehat{w^P}(g+h)$ , and, normalized by the common marginal variance  $S_2$ ,  $\Gamma_A/S_2 = I_A + R_A$  with off-diagonal entries  $R_A(g, h) = \widehat{w^P}(g+h)/S_2$ ,  $g \neq h$ . The entire gain–object coupling of sign-randomized acquisition is therefore carried by the non-DC Walsh spectrum of the squared (permuted) object — nothing else.

#### C. E.2 Sign-only whitening: necessary and sufficient conditions

**Theorem E.2 (exact Walsh condition).** *Under (E1)–(E3) with no permutation ( $P = \text{id}$ ):*

- 1) (exact pairwise)  $\text{Cov}_D(Z_g, Z_h) = S_2 \delta_{g,h}$  for all  $g, h$  **iff**  $\hat{w}(q) = 0$  for every  $q \neq 0$ , equivalently **iff**  $T_j^2 = S_2/K$  for all  $j$  ( $T^2$  flat).
- 2) (pairwise  $\varepsilon$ -whitening)  $|\text{Cov}_D(Z_g, Z_h)| \leq \varepsilon S_2$  for all  $g \neq h$  **iff**  $\max_{q \neq 0} |\hat{w}(q)| \leq \varepsilon S_2$ .
- 3) (spectral window) For a window class  $\mathcal{A}$ ,  $\|\Gamma_A/S_2 - I_A\|_{\text{op}} \leq \varepsilon$  for all  $A \in \mathcal{A}$  **iff**  $\sup_{A \in \mathcal{A}} \|R_A\|_{\text{op}} \leq \varepsilon$ .
- 4) (window average) With  $\bar{Z}_A = W^{-1} \sum_{g \in A} Z_g$  and multiplicity  $m_A(q) = \#\{(g, h) \in A \times A : g \neq h, g+h=q\}$ , the variance  $\text{Var}(\bar{Z}_A)$  lies in  $[(1 \mp \varepsilon) S_2/W]$  for every  $A \in \mathcal{A}$  **iff**  $\sup_{A \in \mathcal{A}} |\sum_{q \neq 0} m_A(q) \hat{w}(q)| \leq \varepsilon W S_2$ .

**Proof.** (1) If whitening holds, then for any  $q \neq 0$  pick  $g, h$  with  $g+h=q$  (always possible in  $G$ ); (E.1) forces  $\hat{w}(q) = 0$ . Conversely, vanishing non-DC coefficients make (E.1) equal  $S_2 \delta_{g,h}$ . The flatness equivalence is Walsh inversion: the transform is invertible, so  $\hat{w}$  supported on  $\{0\}$  iff  $w$  is constant, i.e.  $T_j^2 = S_2/K$ . (2) Immediate from (E.1) after dividing by  $S_2$ . (3) The normalized window covariance is  $I_A + R_A$ ; the statement is definitional. (4) Direct expansion:

$$\text{Var}(\bar{Z}_A) = W^{-2} \sum_{g,h \in A} \hat{w}(g+h) = \frac{S_2}{W} + W^{-2} \sum_{q \neq 0} m_A(q) \hat{w}(q),$$

and the two-sided requirement is exactly the stated inequality. ■

Two standard reductions relate the forms. If  $\mathcal{A}$  contains all two-point windows, (3) implies (2), and (2) is the sharpest scalar condition. In the converse direction, since  $R_A$  has zero diagonal, Gershgorin’s disc theorem gives  $\|R_A\|_{\text{op}} \leq (W-1) \max_{q \neq 0} |\hat{w}(q)|/S_2$ , so pairwise condition (2) at tolerance  $\varepsilon/(W_{\text{max}}-1)$  implies spectral condition (3) for all windows with  $W \leq W_{\text{max}}$ . This Gershgorin step is a (generally loose) sufficient bound, not an equivalence.

The obstruction is sharp: if  $T^2$  is aligned with the positive support of a single character  $\chi_q$ , then  $|\hat{w}(q)| = S_2$  and the row pair  $(g, h)$  with  $g+h=q$  is *perfectly* correlated under sign randomization. Signs cannot remove this; it is a property of the object’s pixel ordering.

a) *What this does not claim.*: Theorem E.2 characterizes second-moment decorrelation over the sign draw for a *fixed* object and ordering; it does not assert Gaussianity of  $Z_g$  (a Rademacher sum, whose normal approximation requires  $K_\infty \rightarrow \infty$  with Berry–Esseen error  $\lesssim K_\infty^{-1/2}$ ), independence of rows, or anything about noise, gain drift, or estimation error — those enter through Secs. 5–6 and Appendix F. It is also a statement about the design/object pair, not a high-probability event: no randomness beyond  $D$  is involved.

#### D. E.3 Random permutation as a probabilistic Walsh-flattener

By Lemma E.1, Theorem E.2 holds verbatim for a real-ized permutation  $P$  with  $w$  replaced by  $w^P$ : the *deterministic* necessary-and-sufficient condition remains Walsh-flatness, now of the permuted squared object. A random  $P$  makes that condition likely.

**Theorem E.3 (Bernstein–Serfling permutation bound).** *Under (E1)–(E3), there exist universal constants  $c, C > 0$  such that for every fixed  $q \neq 0$  and  $\varepsilon > 0$ ,*

$$\Pr(|\widehat{w^P}(q)| \geq \varepsilon S_2) \leq 2 \exp[-c \min(\varepsilon^2 K_4, \varepsilon K_\infty)], \quad (\text{E.2})$$

and by the union bound over the  $K - 1$  nonzero frequencies,

$$\Pr\left(\max_{q \neq 0} |\widehat{w^P}(q)| \geq \varepsilon S_2\right) \leq 2(K - 1) \exp[-c \min(\varepsilon^2 K_4, \varepsilon K_\infty)]. \quad (\text{E.3})$$

In particular, pairwise  $\varepsilon$ -whitening (Theorem E.2(2) for  $w^P$ ) holds with probability at least  $1 - \delta$  whenever

$$\min(\varepsilon^2 K_4, \varepsilon K_\infty) \geq C \log(K/\delta). \quad (\text{E.4})$$

**Proof.** Fix  $q \neq 0$  and let  $A_q = \{j : \chi_q(j) = +1\}$ , so  $|A_q| = K/2$  by (E1). Then

$$\widehat{w^P}(q) = \sum_{j \in A_q} w_j^P - \sum_{j \notin A_q} w_j^P = 2 \sum_{j \in A_q} w_j^P - S_2,$$

a centered sum of  $n = K/2$  draws without replacement from the finite population  $\{w_1, \dots, w_K\}$  with mean  $\mu_w = S_2/K$ . Its exact variance is  $n\sigma_w^2 \frac{K-n}{K-1} \leq \frac{K}{4}\sigma_w^2 \leq \frac{1}{4} \sum_j w_j^2 = \frac{S_2^2}{4K_4}$ , where  $\sigma_w^2 = K^{-1} \sum_j (w_j - \mu_w)^2$ ; and each summand satisfies  $0 \leq w_j \leq \|T\|_\infty^2 = S_2/K_\infty$ . The Bernstein inequality for sampling without replacement (Serfling 1974; the Bernstein–Serfling form of Bardenet–Maillard 2015) then gives, for  $t > 0$ ,

$$\Pr(|\widehat{w^P}(q)| \geq t) \leq 2 \exp\left[-c \min\left(\frac{t^2}{S_2^2/K_4}, \frac{t}{S_2/K_\infty}\right)\right].$$

Setting  $t = \varepsilon S_2$  yields (E.2); (E.3) is the union bound; (E.4) follows by requiring the right side of (E.3) to be at most  $\delta$  and absorbing the factor  $2(K - 1)$  into  $C \log(K/\delta)$ . ■

a) *What this does not claim.*: (E.2)–(E.4) are sufficient high-probability conditions with unspecified universal constants; they are not the deterministic criterion for a realized acquisition — that remains Walsh-flatness of  $w^P$  (Theorem E.2), which can and should be checked numerically for the ordering actually used. For spectral whitening of all windows up to size  $W_{\max}$ , the simple route substitutes  $\varepsilon \rightarrow \varepsilon/(W_{\max} - 1)$  in (E.4) via Gershgorin; sharper window-size dependence via matrix-Bernstein or Hanson–Wright arguments (Sec. E.5) is a concentration refinement, again sufficient rather than necessary. Both branches of the min collapse when energy concentrates on few pixels ( $K_4, K_\infty$  small): permutation cannot flatten a spike.

**E. E.4 Permutation alone: exact carrier variance and the flat-object counterexample**

**Theorem E.4 (permutation-alone variance).** Under (E1)–(E2), with no signs and a uniformly random permutation  $P$ , the carrier  $Z_g(P) = \sum_j \chi_g(j) T_{Pj}$  of any non-DC row  $g \neq 0$  satisfies  $\mathbb{E}_P Z_g = 0$  and

$$\text{Var}_P Z_g = \frac{K S_2 - S_1^2}{K - 1} = S_2 \frac{K - K_{\text{eff}}}{K - 1}. \quad (\text{E.5})$$

Consequently, for offset physical patterns  $B_g = \mu S_1 + \sigma Z_g(P)$  with positivity margin,

$$\frac{\text{Var}_P B_g}{(\mathbb{E}_P B_g)^2} = \left(\frac{\sigma}{\mu}\right)^2 \frac{K - K_{\text{eff}}}{(K - 1) K_{\text{eff}}} \leq \left(\frac{\sigma}{\mu}\right)^2 \frac{1}{K_{\text{eff}}}. \quad (\text{E.6})$$

No matching lower bound of order  $1/K_{\text{eff}}$  holds uniformly over objects: for flat  $T$  ( $T_j$  constant),  $K_{\text{eff}} = K$  while  $Z_g(P) = 0$  for *every* permutation and every  $g \neq 0$ .

**Proof.** For  $g \neq 0$ ,  $\sum_i \chi_g(i) = 0$ , so  $\mathbb{E}_P Z_g = (\sum_i \chi_g(i)) \mathbb{E}_P T_{P1} = 0$ . Exchangeability of  $(T_{P1}, \dots, T_{PK})$  gives  $\mathbb{E}_P T_{Pi}^2 = S_2/K$  and, for  $i \neq j$ ,  $\mathbb{E}_P T_{Pi} T_{Pj} = (S_1^2 - S_2)/(K(K - 1))$ . Hence

$$\mathbb{E}_P Z_g^2 = S_2 + \frac{S_1^2 - S_2}{K(K - 1)} \sum_{i \neq j} \chi_g(i) \chi_g(j),$$

and  $\sum_{i \neq j} \chi_g(i) \chi_g(j) = (\sum_i \chi_g(i))^2 - K = -K$ , giving (E.5); the second form follows from  $K_{\text{eff}} = S_1^2/S_2$ . For (E.6),  $\mathbb{E}_P B_g = \mu S_1$  and  $\text{Var}_P B_g = \sigma^2 \text{Var}_P Z_g$ ; divide. The counterexample: if  $T_j \equiv t$ , every non-DC Walsh sum of a constant vector vanishes identically, while  $S_1^2/S_2 = K$ . Note (E.5) also exhibits the finite-population correction — the relative variance is  $(\sigma/\mu)^2 (1/K_{\text{eff}} - 1/K) \cdot K/(K - 1)$ . ■

a) *What this does not claim.*: Theorem E.4 does not say permutation-alone whitens: it provides finite-population stationarity (exchangeability of the carrier along the acquisition, the (★) mechanism of Sec. 5) and an  $O(1/K_{\text{eff}})$  upper variance scale, but it is not equivalent to i.i.d. Rademacher or nonnegative random-pattern carrier noise, and it guarantees no  $\Theta(1/K_{\text{eff}})$  excitation — a flat object has exactly zero. Two-sided coefficientwise control requires the sign flip and the spectral-spread condition (E.4) together (the “both” cell of the Fig. 6 ablation). An earlier claim (R1/v3) that permutation-alone “cannot attain variance  $\sim 1/K_{\text{eff}}$ ” was too strong as an upper-scale statement and is corrected by (E.6); it is correct only in the two-sided reading.

#### F. E.5 Hanson–Wright window-energy bound

Whitening of window energy — the quantity the window estimator of Sec. 6 actually averages — admits a direct quadratic-form bound. For a window  $A$  of  $W$  rows let  $P_A$  select those rows and set

$$Q_A = \|P_A U D P T\|_2^2 = \eta^\top M \eta, \\ M = \text{diag}(P T) U_A^\top U_A \text{diag}(P T),$$

with  $\eta$  the Rademacher sign vector and  $U_A$  the selected rows of  $U$ .

**Proposition E.5 (window-energy concentration).** Condition on any realized  $P$ . Then  $\mathbb{E}_D Q_A = (W/K) S_2$ , and there is a universal  $c > 0$  with

$$\Pr_D\left(|Q_A - \frac{W}{K} S_2| \geq \varepsilon \frac{W}{K} S_2\right) \leq 2 \exp\left[-c \min\left(\varepsilon^2 \frac{W K_\infty}{K}, \varepsilon \frac{W K_\infty}{K}\right)\right]. \quad (\text{E.7})$$

Over a specified family of  $M$  windows, uniform relative energy whitening at tolerance  $\varepsilon \leq 1$  holds with probability  $\geq 1 - \delta$  once

$$\frac{W K_\infty}{K} \geq C \varepsilon^{-2} \log(M/\delta). \quad (\text{E.8})$$

**Proof.** Since  $U$  has orthonormal rows,  $U_A U_A^\top = I_W$ , so  $U_A^\top U_A$  is an orthogonal projector and  $(U_A^\top U_A)_{jj} =$

$\sum_{g \in A} U_{gj}^2 = W/K$  (Hadamard entries are  $\pm K^{-1/2}$ ). Hence  $M \succeq 0$ , and  $\mathbb{E}_D Q_A = \text{tr } M = \sum_j (PT)_j^2 (W/K) = (W/K)S_2$ . Because  $\eta_j^2 = 1$ , the diagonal of  $M$  contributes deterministically and  $Q_A - \mathbb{E}Q_A$  is a pure off-diagonal Rademacher chaos, to which the Hanson–Wright inequality applies (Hanson–Wright 1971; sub-Gaussian form of Rudelson–Vershynin 2013):

$$\Pr(|Q_A - \text{tr } M| \geq t) \leq 2 \exp \left[ -c \min \left( \frac{t^2}{\|M\|_F^2}, \frac{t}{\|M\|_{\text{op}}} \right) \right].$$

The proxies:  $\|M\|_{\text{op}} \leq \|\text{diag}(PT)\|_{\text{op}}^2 \|U_A^\top U_A\|_{\text{op}} \leq \|T\|_\infty^2 = S_2/K_\infty$ , and for PSD  $M$ ,  $\|M\|_F^2 \leq \|M\|_{\text{op}} \text{tr } M \leq (W/K)S_2\|T\|_\infty^2$ . Substituting  $t = \varepsilon(W/K)S_2$  gives  $t^2/\|M\|_F^2 \geq \varepsilon^2 W K_\infty / K$  and  $t/\|M\|_{\text{op}} \geq \varepsilon W K_\infty / K$ , i.e. (E.7); for  $\varepsilon \leq 1$  the quadratic branch is active. A union bound over  $M$  windows gives (E.8). ■

For dense objects of bounded dynamic range,  $K_\infty \asymp K_4 \asymp K_{\text{eff}} \asymp K$  and (E.8) reduces to the mild  $W \gtrsim \varepsilon^{-2} \log M$  — the window only needs logarithmic length. For an object supported on  $m$  comparable pixels,  $K_\infty \asymp m$  and the condition hardens to  $Wm/K \gtrsim \varepsilon^{-2} \log M$ : this is precisely where  $K_{\text{eff}}$  alone is the wrong functional and the three spread parameters separate (Sec. 6).

*a) What this does not claim.*: Proposition E.5 controls window *energy* over the sign draw for a pre-specified finite window family; it is a sufficient concentration tool, not the deterministic necessary-and-sufficient condition (which remains the spectral condition of Theorem E.2(3) applied to  $w^P$ ), and it does not cover all  $\binom{K}{W}$  windows simultaneously without paying for the larger union. The constants  $c, C$  are universal but not optimized, and no claim is made that (E.8) is tight in its  $W$ - or  $K_\infty$ -dependence — Hanson–Wright and matrix-Bernstein refinements can improve window-family dependence for structured  $\mathcal{A}$ , but they do not change the deterministic criterion. Finally, all of Appendix E concerns the carrier statistics of the *design*; the translation of whitening into gain-estimation rates ( $1/W$  per window) and into image error (leverage  $B_L = 1$  for full SRHT inversion) is carried out in Appendix D and Appendix F respectively.

## APPENDIX F

### THE MASTER FINITE-NOISE IDENTITY AND THE FLIP BOUNDARY

This appendix proves Theorem 1 (the master finite-noise relMSE identity) and Prop 3 (the finite- $N$  flip boundary) of Sec. 8, together with the noise plug-ins and the three basis specializations quoted there. Throughout,  $T \in \mathbb{R}^K$  is the object, with  $S_1, S_2, K_{\text{eff}}$  as in Appendix C;  $A \in \mathbb{R}^{N \times K}$  is the design,  $b = AT$  the ideal bucket vector; the raw bucket is  $R_n = a_n B_n$  with gain  $a_n = e^{\ell_n}$ ,  $\ell \in S$  (the drift class of Sec. 4,  $p = \dim S$ ). After the pipeline applies a gain estimate  $\hat{a}_n = e^{\hat{\ell}_n}$ , the corrected bucket is

$$z_n = R_n / \hat{a}_n = (1 + \delta_n) b_n + \xi_n, \quad \delta_n = \frac{a_n}{\hat{a}_n} - 1 = e^{\ell_n - \hat{\ell}_n} - 1 \quad (\text{F.1})$$

where  $\delta_n$  is the *residual* multiplicative gain error and  $\xi_n$  is additive bucket-domain noise after correction, with  $\mathbb{E} \delta = m_\delta$ ,

$\text{Cov}(\delta) = V_\delta$ ,  $\mathbb{E} \xi = 0$ ,  $\text{Cov}(\xi) = \Sigma_\xi$ , and  $\delta \perp \xi$  (uncorrelated,  $C_{\delta\xi} = \text{Cov}(\delta, \xi) = 0$  — a hypothesis, revisited in the cross-term note below). The reconstruction is  $\hat{T} = Lz$  for any fixed linear operator  $L \in \mathbb{R}^{K \times N}$  (exact inverse, pseudoinverse, DGI correlator, or a regularized estimator linearized around a fixed design).

#### A. F.1 The master identity

**Theorem 1 (exact second-moment bridge).** *Assume model (F.1) with  $A, T, L$  fixed (all expectations conditional on them),  $\delta$  and  $\xi$  uncorrelated, and second moments  $m_\delta, V_\delta, \Sigma_\xi$  finite. Then, exactly and with no Gaussian or small-noise approximation,*

$$\frac{\mathbb{E} \|\hat{T} - T\|_2^2}{S_2} = \underbrace{\frac{\|(LA - I)T + L \text{diag}(b) m_\delta\|_2^2}{S_2}}_{\text{bias}} + \underbrace{\frac{\text{tr}(L \text{diag}(b) V_\delta \text{diag}(b) L^\top)}{S_2}}_{\text{residual gain}} + \underbrace{\frac{\text{tr}(L \Sigma_\xi L^\top)}{S_2}}_{\text{additive noise}}. \quad (\text{F.2})$$

*(a) If moreover  $\mathbb{E} \delta_n = 0$  and the  $\delta_n$  are uncorrelated with common variance  $v$ , the residual-gain term equals  $v B_L(A, T)$  with the **leverage***

$$B_L(A, T) = \frac{1}{S_2} \sum_{n=1}^N b_n^2 \|Le_n\|_2^2. \quad (\text{F.3})$$

*(b) If instead the residual gains are stationary with  $\text{Cov}(\delta_n, \delta_m) = v r_\delta(n - m)$ , the residual-gain term equals*

$$R_{\text{gain}} = \frac{v}{S_2} \sum_{n,m} r_\delta(n - m) b_n b_m \langle Le_n, Le_m \rangle. \quad (\text{F.4})$$

**Proof.** Write  $z = b + \text{diag}(b) \delta + \xi$ , so  $\hat{T} - T = (LA - I)T + L \text{diag}(b) \delta + L\xi$ . Decompose  $\delta = m_\delta + (\delta - m_\delta)$ . The mean of  $\hat{T} - T$  is  $(LA - I)T + L \text{diag}(b) m_\delta$ ; its squared norm is the bias term. By the bias–variance decomposition  $\mathbb{E} \|X\|^2 = \|\mathbb{E} X\|^2 + \text{tr } \text{Cov}(X)$  and the uncorrelatedness of  $\delta$  and  $\xi$ , the covariance of  $\hat{T}$  is  $L \text{diag}(b) V_\delta \text{diag}(b) L^\top + L \Sigma_\xi L^\top$ ; taking traces and dividing by  $S_2$  gives (F.2). For (a),  $V_\delta = vI$  and  $\text{tr}(L \text{diag}(b) \text{diag}(b) L^\top) = \sum_n b_n^2 \|Le_n\|_2^2$ , giving (F.3). Case (b) is the coordinate expansion of the same trace with  $(V_\delta)_{nm} = v r_\delta(n - m)$ . ■

*a) What this does not claim.*: (F.2) requires the uncorrelatedness hypothesis  $C_{\delta\xi} := \text{Cov}(\delta, \xi) = 0$ : without it, the same bias–variance decomposition produces exactly one additional term,  $(2/S_2) \text{tr}[L \text{diag}(b) C_{\delta\xi} L^\top]$ , which vanishes under the stated hypothesis but *not* automatically when the gain estimate  $\hat{a}$  is computed from the same noisy record that carries  $\xi$  (same-record dependence — the generic situation for a blind pipeline, and one the injected-residual protocol of Sec. 9 deliberately does not probe). Beyond that, (F.2) is a second-moment accounting identity for a *linear* reconstructor

with the design held fixed; it does not bound nonlinear or data-adaptive  $L$ , and for random-pattern pipelines the constants below are obtained by additionally averaging over the pattern draw. It says nothing about how small  $v$  can be made — that is the estimation-rate question of Sec. 6. Finally, the scalar collapse  $v B_L$  *requires* uncorrelated residuals: for smooth (coherent) residual gain — the generic output of a windowed blind estimator — only the matrix form (F.4) is correct. In particular, the simple summary “orthogonal inversion gives  $v$ ” holds only for independent coefficient-wise residuals under exact inversion.

### B. F.2 Noise plug-ins

**Proposition F.2 (read and Poisson noise, exact).** (a) Gaussian read. If detector-domain noise  $\eta_n \sim \mathcal{N}(0, \sigma_{\text{read},n}^2)$  (independent across frames) precedes gain correction, then  $\xi_n = \eta_n/\hat{a}_n$  and, exactly,  $\text{Var}(\xi_n) = \sigma_{\text{read},n}^2/\hat{a}_n^2 = (\sigma_{\text{read},n}^2/a_n^2)(a_n/\hat{a}_n)^2$ ; for calibrated gains  $\hat{a}_n \approx a_n$  the factor  $(a_n/\hat{a}_n)^2 \rightarrow 1$  and  $\text{Var}(\xi_n) = \tau_{G,n}^2 := \sigma_{\text{read},n}^2/a_n^2$ . (b) Poisson. If counts obey  $C_n | T, a \sim \text{Pois}(\Phi_n)$  with  $\Phi_n = \gamma_n a_n b_n$  and  $z_n = C_n/(\gamma_n \hat{a}_n)$ , then exactly

$$\mathbb{E}[z_n | T, a] = \frac{a_n}{\hat{a}_n} b_n, \quad \text{Var}(z_n | T, a) = \frac{b_n^2}{\Phi_n} \left( \frac{a_n}{\hat{a}_n} \right)^2. \quad (\text{F.5})$$

**Proof.** (a) is scaling of a Gaussian. (b): a Poisson variable has mean = variance =  $\Phi_n$ ; dividing by  $\gamma_n \hat{a}_n$  scales the mean by  $(\gamma_n \hat{a}_n)^{-1}$  and the variance by  $(\gamma_n \hat{a}_n)^{-2}$ , and  $\Phi_n/(\gamma_n \hat{a}_n) = (a_n/\hat{a}_n) b_n$ ,  $\Phi_n/(\gamma_n \hat{a}_n)^2 = (b_n^2/\Phi_n)(a_n/\hat{a}_n)^2$ . ■

Hence — and only after the post-calibration simplification  $\hat{a}_n \rightarrow a_n$ , which sets the exact  $(a_n/\hat{a}_n)^2$  factors of (a) and (F.5) to one —  $\Sigma_\xi = \text{diag}(\tau_{G,n}^2 + b_n^2/\Phi_n)$  plugs into (F.2), and the Poisson term is **exact at all photon counts**, including  $\Phi_n \ll 1$ ; no log or Gaussian approximation enters the reconstruction layer. Under a total budget  $\Phi_n = \omega_n \Phi_{\text{tot}}$ ,  $R_{\text{Pois}} = (S_2 \Phi_{\text{tot}})^{-1} \sum_n (b_n^2/\omega_n) \|L e_n\|^2$ .

a) *What this does not claim.*: Exactness holds *conditionally on the gains*; the difficulty of low photons lives entirely in estimating  $\hat{a}_n$  (the calibrated soft-log analysis; see Appendix D), not in propagating noise through  $L$ . The optimal allocation  $\omega_n \propto |b_n| \|L e_n\|$  requires knowing  $T$  and is not claimed to be achievable.

### C. F.3 Specializations

a) *F.3.1 Orthogonal / full-SRHT inversion* ( $B_L = 1$ ).: If  $A = U$  is orthonormal ( $N = K$ , signed coefficients available) and  $L = U^\top$ , then  $LA = I$  and  $\|L e_n\| = 1$ , so  $B_L = (1/S_2) \sum_n b_n^2 = 1$  by Parseval. With independent residuals,

$$\text{relMSE}_{\text{orth}} = v + \frac{1}{S_2} \sum_n (\tau_{G,n}^2 + b_n^2/\Phi_n). \quad (\text{F.6})$$

Exact orthogonal inversion transmits independent coefficient-wise residual gain at unit leverage. A full SRHT acquisition  $A = P_{\text{row}} U D P_{\text{col}}$ ,  $L = A^\top$  has identically the same propagation; SRHT’s role is *not* a better inverse but temporal stationarization of the carrier so that the blind estimator can shrink  $v$  (Sec. 5/7).

### b) F.3.2 Pairwise Hadamard: exact perturbation law.

Let  $c_k = h_k^\top T$ ,  $h_k \in \{\pm 1\}^K$ , with physical masks  $B_k^\pm = (S_1 \pm c_k)/2$  and reconstruction  $T = K^{-1} H^\top c$ , so coefficient errors  $e_k$  give  $\text{relMSE} = (K S_2)^{-1} \sum_k \mathbb{E} e_k^2$ . If the paired masks carry gains  $a_k^+$  and  $a_k^- = a_k^+(1 + \Delta_k)$ , the pairwise-normalized estimator satisfies the **exact** algebraic law

$$\hat{c}_k - c_k = -\frac{S_1 \Delta_k (1 - x_k^2)}{2 + \Delta_k (1 - x_k)}, \quad x_k = c_k/S_1, \quad (\text{F.7})$$

whence, exactly,  $R_{\text{pair,gain}} = \frac{S_1^2}{K S_2} \sum_k \mathbb{E} \left[ \frac{\Delta_k^2 (1 - x_k^2)^2}{(2 + \Delta_k (1 - x_k))^2} \right]$ . For small mismatch with  $\Delta_k$  independent of the coefficient index,

$$R_{\text{pair,gain}} = \frac{K_{\text{eff}}}{4} D_H(T) \text{Var}(\Delta) + O(K_{\text{eff}} \mathbb{E} |\Delta|^3), \quad D_H(T) = \frac{1}{K} \sum_k (1 - x_k^2)^2, \quad (\text{F.8})$$

by Taylor expansion of (F.7) in  $\Delta_k$ . For most non-DC Hadamard coefficients of a nonnegative object  $|x_k| \ll 1$ , so  $D_H \approx 1$ ;  $D_H$  is nevertheless a *pipeline constant to be measured*, not assumed. The  $K_{\text{eff}}$  amplification is structural: pairwise normalization converts a differential gain error into a coefficient error of order  $S_1$ , and Parseval then divides by  $S_2$ . Noise terms: read noise gives exactly  $R_{\text{pair,read}} = 2\sigma_{\text{read}}^2/S_2$ ; for Poisson counts with pair budget  $\Phi_{\text{pair}}$ , using  $B_k^+ + B_k^- = S_1$ , the coefficient shot variance is  $\text{Var}(e_k | c_k) \approx S_1^2/\Phi_{\text{pair},k}$  — an approximation, exact only up to the  $c_k$ -dependent split between the two buckets — giving  $R_{\text{pair,Pois}} = K_{\text{eff}}/\Phi_{\text{pair}}$  (equal-pair budget) or  $(K_{\text{eff}} + 1)/(2\Phi_{\text{mask}})$  (equal-mask budget); these differ only by allocation convention.

c) *What F.3.2 does not claim.*: (F.7)–(F.8) presuppose a stable DC/ $S_1$  normalization; at low photons the denominator bucket sum can be noisy or zero, and a robust implementation needs offset/clipping or a likelihood-ratio estimator. The exact law (F.7) is algebra; only the moment step (F.8) invokes small  $\Delta$  and index-independence.

d) *F.3.3 Random patterns / DGI: the one-sample constant*  $C_0$ .: Let  $I_n = \mu \mathbf{1} + \sigma x_n$  with iid coordinates,  $\mathbb{E} x = 0$ ,  $\mathbb{E} x^2 = 1$ ,  $\mathbb{E} x^3 = \gamma_3$ ,  $\mathbb{E} x^4 = \beta_4$ , and the ideal correlator  $\hat{T}_{\text{DGI}} = (N\sigma)^{-1} \sum_n x_n z_n$ . Define the one-sample vector  $Z = \sigma^{-1} x (\mu S_1 + \sigma x^\top T)$ ; then  $\mathbb{E} Z = T$  and, with known gain and no detector noise,

$$\frac{\mathbb{E} \|\hat{T}_{\text{DGI}} - T\|^2}{S_2} = \frac{C_0}{N}, \quad C_0 := \frac{\mathbb{E} \|Z - T\|^2}{S_2}, \quad (\text{F.9})$$

which is the implementation-independent *definition* of  $C_0$ . **The evaluation of  $C_0$  for iid patterns is a moment calculation:** expanding  $\mathbb{E} \|Z\|^2$  with  $Z = (\mu S_1/\sigma) x + x x^\top T$  gives the three terms  $K(\mu/\sigma)^2 S_1^2$ ,  $2(\mu/\sigma) \gamma_3 S_1^2$ , and  $(K + \beta_4 - 1) S_2$ ; subtracting  $\|\mathbb{E} Z\|^2 = S_2$ ,

$$C_0 = K + \beta_4 - 2 + K_{\text{eff}} [K(\mu/\sigma)^2 + 2\gamma_3(\mu/\sigma)] \xrightarrow{\mu=0, \gamma_3=0} K + \beta_4 - 2. \quad (\text{F.10})$$

The  $K_{\text{eff}}K(\mu/\sigma)^2$  term is the familiar nonnegative-background penalty. Residual gain, read, and Poisson terms (equal-count convention  $\Phi_n = \Phi_{\text{frame}}$ ):

$$\text{relMSE}_{\text{DGI}} = \frac{C_0}{N} + \frac{(1+C_0)v}{N} + \frac{K\tau_G^2}{N\sigma^2 S_2} + \frac{1+C_0}{N\Phi_{\text{frame}}}. \quad (\text{F.11})$$

The gain term follows from Theorem 1(a) with the *leverage*  $\times N$  framing: for the correlator,  $\|Le_n\|^2 = \|x_n\|^2/(N^2\sigma^2)$ , so averaging over pattern draws,  $N\mathbb{E}B_L = \mathbb{E}\|Z\|^2/S_2 = 1+C_0$  — i.e.,  $R_{\text{DGI,gain}} = (1+C_0)v/N$ , and in residual-injection experiments the measurable slope is  $NB_L \rightarrow 1+C_0$ . **Status (labeled).** This pattern-averaging step treats the residuals  $\delta$  as independent of the pattern draw — exact for injected residuals (the Fig. 4 protocol), a heuristic decoupling for blind pipelines, where  $\hat{a}$  is computed from the same record the patterns generated. Correlated residuals multiply  $v$  by  $\chi\delta = 1+2\sum_{h\geq 1}\text{corr}(\delta_0, \delta_h)$  — a heuristic long-run-variance correction valid only when correlations are weak enough that pattern cross-terms still average; the rigorous object remains (F.4).

*e) What F.3.3 does not claim.:* There is **no universal**  $C_0$ : (F.10) is exact only for the ideal known- $(\mu, \sigma)$  correlator with iid coordinates. Mean subtraction, reference normalization, and photon allocation all change the constant (not the  $C_0/N$  structure), so  $C_0$  must be measured from the pipeline via definition (F.9). The flux-limited convention  $\Phi_n = \gamma b_n$  replaces the last term of (F.11) by  $(N\sigma^2 S_2)^{-1}\mathbb{E}[\|x\|^2 b_n/\gamma]$ .

#### D. F.4 Prop 3: the finite- $N$ flip boundary

*a) Drift model and assumptions.:* Let  $s^2$  be the variance scale of the log-gain process  $\ell$  and let  $\rho_{\text{pair}}$  be the **adjacent-pair decorrelation parameter**, meaning the paired-mask gain increment obeys  $\text{Var}(\Delta_k) \approx \text{Var}(\ell_k^- - \ell_k^+) = 2s^2 r(\rho_{\text{pair}})$  with  $r(\rho) = \rho + O(\rho^2)$  small- $\rho$ ; for an Ornstein–Uhlenbeck log-gain,  $r(\rho) = 1 - e^{-\rho}$ . Assume the small-mismatch regime of (F.8) (so  $R_{\text{pair,gain}}(\rho_{\text{pair}}) \approx \frac{1}{2}K_{\text{eff}}D_H(T)s^2 r(\rho_{\text{pair}})$ ) and the DGI decomposition (F.11) with blind residual variance  $v_{\text{blind}}(\rho_{\text{pair}}, N)$  — the window-optimized output of the Sec.-6 rate theorem,  $v_{\text{blind}} \lesssim \min_{W \in [1, N]} [\sigma_{\text{eff}}^2/W + C(s\rho_{\text{pair}}W)^2] \lesssim C\sigma_{\text{eff}}^{4/3}(s\rho_{\text{pair}})^{2/3}$  when the optimizer is interior.

**Prop 3 (implicit flip boundary).** Define  $\rho^* = \inf\{\rho_{\text{pair}} \geq 0 : R_{\text{pair}}(\rho_{\text{pair}}) \geq R_{\text{rand}}(\rho_{\text{pair}}, N)\}$ , where  $R_{\text{pair}} = R_{\text{pair,gain}} + R_{\text{pair,read}} + R_{\text{pair,Pois}}$  and  $R_{\text{rand}}$  is (F.11) with  $v = v_{\text{blind}}(\rho_{\text{pair}}, N)$ . **Assume single crossing:** since  $v_{\text{blind}}(\rho_{\text{pair}}, N)$  itself varies with  $\rho_{\text{pair}}$ , both sides of the comparison move with  $\rho_{\text{pair}}$ , and we assume the difference  $R_{\text{pair}} - R_{\text{rand}}$  changes sign exactly once on the range considered — existence and uniqueness of the crossing is an assumption at the same heuristic level as the small-drift expansion, not a proven property. Under the small-drift expansion and this assumption,  $\rho^*$  solves the implicit equation

$$r(\rho^*) = \frac{2}{K_{\text{eff}}D_H(T)s^2} \left[ \frac{C_0}{N} + \frac{(1+C_0)v_{\text{blind}}(\rho^*, N)}{N} + R_{\text{DGI,noise}} - R_{\text{pair,noise}} \right]. \quad (\text{F.12})$$

For the OU convention this is explicit:  $\rho^* = -\log(1-Q)$  with  $Q = 2\Delta_R/(K_{\text{eff}}D_H s^2)$ ,  $\Delta_R$  the bracket of (F.12). If  $Q \leq 0$ , random/DGI is already worse at  $\rho = 0$  (sampling/noise floor dominates); if  $Q \geq 1$ , pairwise never flips within the OU small-to-moderate range.

**Proof.** At the boundary,  $R_{\text{pair}}(\rho^*) = R_{\text{rand}}(\rho^*, N)$ . Substituting  $R_{\text{pair,gain}} = \frac{1}{2}K_{\text{eff}}D_H s^2 r(\rho^*)$  (from F.8 with  $\text{Var}\Delta = 2s^2 r$ ) and (F.11), and solving for  $r(\rho^*)$ , gives (F.12); monotonicity of  $r$  makes the crossing the infimum. The OU form inverts  $r(\rho) = 1 - e^{-\rho}$ . Note (F.12) is a fixed-point equation because  $v_{\text{blind}}$  is evaluated at  $\rho^*$ ; the single-crossing assumption stated in the proposition is what makes the crossing exist and be unique. The monotonicity intuition —  $R_{\text{pair,gain}}$  increasing in  $\rho_{\text{pair}}$  while the right side varies slowly ( $v_{\text{blind}} \propto \rho_{\text{pair}}^{2/3}$ ) — motivates the assumption but does not prove it. ■

*b) Leading-order heuristic (labeled as such).:* If  $r(\rho) = \rho$ ,  $D_H = 1$ , noise differences are negligible,  $v_{\text{blind}}$  is negligible (or absorbed into  $C_0$ ), and  $\rho$  means adjacent-pair decorrelation, (F.12) collapses to the engineering rule

$$\rho^* \approx \frac{2C_0}{N K_{\text{eff}} s^2}, \quad (\text{F.13})$$

with first corrections  $\rho^* \approx \frac{2(C_0 + (1+C_0)v_{\text{blind}})/[N K_{\text{eff}} D_H s^2] + 2(R_{\text{DGI,noise}} - R_{\text{pair,noise}})/(K_{\text{eff}} D_H s^2)}{r(\rho^*)}$  and the nonlinear replacement  $\rho \mapsto r^{-1}(\cdot)$  when adjacent increments are not infinitesimal. (F.13) is a *leading-order heuristic*, not a theorem; the theorem is (F.12).

*c)  $\rho$ -convention warning.:* Throughout,  $\rho_{\text{pair}}$  (adjacent-pair decorrelation) is the variable of (F.12)–(F.13); it is **not** the normalized bandwidth  $\rho_{\text{bw}}$  of the tall-design thresholds ( $p \approx \rho_{\text{bw}}N$ , Sec. 4). If a bandwidth parameter with adjacent-increment variance  $\sim 2s^2\rho_{\text{bw}}/N$  is substituted, the explicit  $1/N$  in (F.13) cancels and the boundary reads  $\rho_{\text{bw}}^* \approx 2C_0/(K_{\text{eff}}s^2)$  — same physics, different symbol. Every phase-diagram axis must declare its convention.

*d) What Prop 3 does not claim.:* The boundary compares two *specific* pipelines (pairwise-normalized Hadamard vs. blind random+DGI with the ideal correlator); it does not rank SRHT-paired pipelines, externally calibrated systems, or prior-regularized reconstructions, each of which shifts  $\Delta_R$ . It inherits every assumption of its ingredients: small mismatch (F.8), a measured pipeline  $C_0$  and  $D_H$ , a stationary drift with a declared  $r(\rho)$ , and a  $v_{\text{blind}}$  from the window rate theorem whose constants are implementation-specific. It is a mean-square crossing, not a guarantee about any single realization; the single-crossing property itself is assumed, not derived; and at  $\rho_{\text{pair}}$  beyond the small-drift expansion only the defining infimum — not (F.12) — is meaningful.